When Export Controls Backfire: Evidence from 2019 Korea-Japan Trade Dispute*

Younghoon Kim†

August 2026

Abstract

I show that export controls can "backfire," increasing productivity and exports in the target country. This productivity response mitigates the negative welfare impact in the target country and exacerbates it in the imposing country. I leverage Korea's response to the 2019 Korea-Japan Trade Dispute. In 2019, Japan announced export controls on South Korea for national strategic items, leaving enforcement to the discretion of Japanese officials. Although no export restrictions were imposed in practice, the threat alone triggered substantial changes in Korea's imports and exports. Imports from Japan declined substantially, irrespective of whether items were directly targeted by the announcement. However, imports from Japan decreased disproportionately in sectors where Japan had been the leading supplier to Korea. Surprisingly, Korea's exports in these same sectors expanded while their prices declined. Domestic production in these sectors also expanded broadly with capital deepening, pointing to scale economies. Motivated by these empirical findings, I structurally estimate the strength of scale economies. Finally, I quantify the welfare effects, showing that scale economies mitigate the loss in Korea and exacerbate the loss in Japan.

JEL Classification: F12, F13, F14, F51

Keywords: Trade Policy, Export Controls, Scale Economies

*I am indebted to Jonathan Vogel, Pablo Fajgelbaum, and Oleg Itskhoki for their invaluable guidance and support. I am also grateful to Ariel Burstein and Jinyong Hahn for their guidance. I thank Yan Bai, Jeonghwan Choi, Mark Colas, Diego Comin, Daisoon Kim, Ryan Kim, Yasutaka Koike, Andrei Levchenko, Nuno Limão, Isabela Manelici, Yuan Mei, Michael Peters, Stephen Redding, and Esteban Rossi-Hansberg for helpful comments and suggestions.

†National University of Singapore (email: kimyounghoon@nus.edu.sg).

1 Introduction

Export controls have recently emerged as a pivotal tool in geopolitics. For instance, since October 2022, the US has required multinational firms to obtain prior authorization before exporting advanced semiconductors and related manufacturing equipment to China. Some argue that such measures could be detrimental to the target country due to restricted access to crucial intermediate goods. Others, however, view them as an opportunity for the target country to bolster domestic production. The feasibility of the latter depends on whether production exhibits increasing returns to scale, allowing the industry to become more productive as it scales up to meet the increased demand for domestically produced goods.

I examine this issue in the context of Korea's response to Japan's 2019 announcement of export controls. I show that Japan's export controls backfire in the sense that they increase Korea's productivity and exports, thereby mitigating the welfare loss in Korea while exacerbating it in Japan. The key mechanism driving this effect is scale economies in production. By shifting demand toward domestic goods, foreign export controls lead to an increase in domestic production. In the presence of scale economies, this increased output enhances productivity, which in turn drives down prices and boosts exports. My empirical findings provide direct support for this mechanism. Korean producers' output and exports increased, accompanied by a decrease in prices, precisely in sectors where Korea had been most reliant on Japan. Motivated by these empirical findings, I structurally estimate the strength of scale economies, leveraging demand-side variation across sectors in exposure to Korea's substitution away from Japan. The results indicate that they are substantial: a 1% increase in a sector's total sales leads to a 0.164% increase in productivity on average. The welfare implications of scale economies are quantitatively significant. Compared to a scenario without them, their presence offsets 20% of the welfare loss in Korea while amplifying the welfare loss in Japan by 11%.

In late 2018, tensions between Korea and Japan emerged following a Korean Supreme Court ruling that ordered Japanese corporations to compensate Korean victims forced into labor during World War II, a decision Japan strenuously contested. In July 2019, the Japanese government responded to this decision by announcing potential export controls on three items essential to Korea's semiconductor and display industries. In September 2019, Japan removed Korea from its "White List" of preferential trade partners, exposing all national strategic items exported to Korea to potential export controls. Therefore, Japanese firms exporting these national strategic items to Korea were required to obtain government approval for each shipment. The approval decision was uncertain, as the

Japanese government was legally authorized to delay, deny, or restrict shipments at any time. Despite this possibility, the Japanese government did not enforce any export restrictions on these items.

Although the controls were never enforced in practice, I demonstrate that the announcement itself led to a significant decline in Korea's imports from Japan. First, Korea's overall imports from Japan declined substantially, both relative to the pre-event period and to imports from other countries.[1] Second, the extent of this decline was similar between targeted and non-targeted goods.[2] This suggests that Korean firms anticipated broader restrictions on Japanese goods, not only the items directly targeted by the announcement. Third, Korea's imports from Japan decreased disproportionately in goods for which Japan had been the leading supplier to Korea (*Japan-dominant* goods).[3] This indicates that what governed firms' response was not whether a good was directly targeted, but the degree of pre-event dependence on Japan. Fourth, despite the substantial decrease in imports from Japan, Korea's imports of these Japan-dominant goods from the rest of the world did not increase.

More surprisingly, Korea's exports of these same Japan-dominant goods to the world increased following the announcement.[4] Given the import patterns above, these exports should, if anything, have declined, since domestic production would likely be redirected to the domestic market to satisfy the demand previously met by imports from Japan. Even more surprisingly, their prices declined. Here, too, the standard prediction is the opposite: with an upward-sloping supply curve, increased domestic demand should have raised prices. Import substitution alone can explain neither outcome.

One plausible reconciliation is scale economies, under which an expansion in domestic production raises productivity and lowers prices. To examine this interpretation, I turn to Korea's production side. Korean producer data reveal a broad-based expansion with capital deepening, precisely in these Japan-dominant sectors over the post-event period. These responses, combined with additional pieces of evidence, are difficult to reconcile with alternative mechanisms, such as markup cutting. Taken together, the patterns point to scale economies in production as the underlying mechanism.

To demonstrate how foreign export controls can lead to an increase in exports ac-

[1]Note that the Korean government did not impose any import restrictions on Japanese goods during the post-event period. I return to the government's policy responses in Section 5.1.

[2]Targeted goods here are the Japanese national strategic items exposed to potential export controls by the "White List" removal. Section 3 provides the formal definition.

[3]These goods are concentrated mainly in machinery and equipment, chemicals, non-metallic mineral products, and rubber and plastic products. See Appendix A.5 for details.

[4]In fact, Korea was already exporting 97 percent of these goods during the pre-event period. This capability is also supported by the two countries' similar export structures. See Appendix A.7 for details.

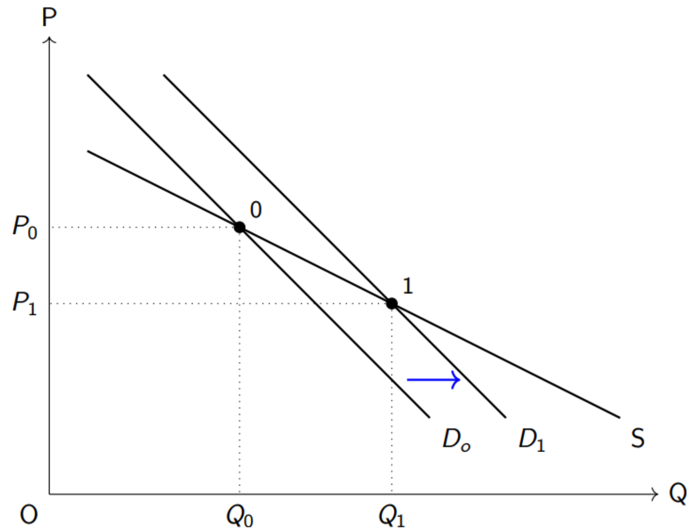


Figure 1: Downward Supply Curve

Notes: Supply curves in a partial equilibrium model can slope downward in the presence of scale economies. Within this framework, foreign export controls directly shift the demand curve for domestic goods to the right (from D_0 to D_1). This, in turn, leads to an increase in quantity (from Q_0 to Q_1) and a decrease in price (from P_0 to P_1) along the downward-sloping supply curve. This framework rationalizes the empirical findings in Section 3.

companied by a price decline in the target country, I present an Armington trade model featuring external economies of scale, following Kucheryavyy et al. (2023) and Bartelme et al. (2025). The key mechanism of the model is the one suggested by the empirical patterns: an increase in output enhances productivity, which in turn lowers prices and boosts sales. To provide intuition for this, supply curves in a partial equilibrium model can slope downward in the presence of scale economies, as shown in Figure 1. Along this downward-sloping supply curve, an increase in demand leads to a decrease in price and an increase in quantity. Within this framework, the threat of Japan's export controls acted as a positive shock to the demand for domestically produced goods, resulting in an increase in Korean producers' output, an increase in export quantities, and a decrease in export prices—exactly as observed in my empirical analysis.

Based on this structural model, I estimate the strength of scale economies by projecting changes in origin-destination-sector exports on changes in origin-sector total sales, pooling the variation across sectors. A challenge is the potential endogeneity of total sales, which could be driven by unobserved supply-side shocks. To address this, I employ an instrumental variable strategy that uses purely demand-side variation. For the Korea-only analysis, the instrument is the interaction between Japan's pre-event share of Korea's imports and the trade elasticity. This instrument is designed to capture the exogenous demand shock from the 2019 Korea-Japan Trade Dispute by leveraging the increase in demand for domestically produced goods that arose from Korea's substitution away from Japan. The results provide strong evidence for scale economies in Korea. The estimate of the scale elasticity is 0.164, indicating that a 1% increase in a sector's total sales raises its productivity by 0.164% on average.

The analysis is then extended to Korea's major import sources, providing a cross-country test for the presence of scale economies. These countries could have expanded their sales to Korea by replacing Japanese goods; with scale economies, this would boost their exports to all other destinations. The identification strategy is the same as in the Korea-only analysis, with the instrument augmented by a third component: Korea's pre-event share of the origin's exports, which captures its exposure to the demand shock in the Korean market. The estimated scale elasticity is 0.206, consistent with the Korea-only estimate. Both estimates fall well within the range of sector-level elasticities reported by Bartelme et al. (2025).

The final part of the paper quantifies the welfare effects of the 2019 Korea-Japan Trade Dispute, using the structural model with scale elasticities consistent with these estimates. The core exercise is to compare the welfare outcomes for Korea and Japan across two scenarios: one without scale economies and one with scale economies. The analysis shows that the presence of scale economies substantially alters the welfare outcomes: it mitigates the negative welfare impact of the export controls in the target country (Korea) while exacerbating it in the imposing country (Japan). Quantitatively, compared to the scenario without scale economies, their presence offsets 20% of the welfare decline in Korea and amplifies the decline in Japan by 11%. The model also reproduces the untargeted moments documented in my empirical analysis. These findings demonstrate how export controls can backfire: by increasing productivity in the target country through scale economies, such policies can lead to unintended and self-harming consequences for the imposing country.

Related Literature. This paper contributes to several strands of literature. First, it relates to a growing literature on the use of trade policy as a geoeconomic instrument. Clayton et al. (2026) provide a framework for geoeconomics, in which a hegemon wields power by coordinating threats across economic relationships. A number of studies assess the economic effects of substantial tariff increases resulting from the US-China Trade War (e.g., Amiti et al. (2019); Fajgelbaum et al. (2020)).[5] Another set of studies exam-

[5] Fajgelbaum and Khandelwal (2022) provide a comprehensive review of the economic consequences of the US-China Trade War.

ines economic sanctions (Morgan et al. (2023)), most recently those imposed on Russia following its invasion of Ukraine (Moll et al. (2023); Itskhoki and Ribakova (2024); Itskhoki and Mukhin (2026); Mayer et al. (2026)). Fan et al. (2026) analyze the economic consequences of US national-security export controls. This paper's primary contribution is to provide novel evidence that export controls backfire by increasing productivity in the target country, which in turn mitigates the welfare loss for the target country and exacerbates the loss for the imposing country. To the best of my knowledge, this is the first study to document such unintended, self-harming outcomes in the context of a contemporary trade dispute between two major, technologically advanced economies.

Second, this paper builds on the literature on economies of scale (Antweiler and Trefler (2002); Kucheryavyy et al. (2023); Lashkaripour and Lugovskyy (2023); Bartelme et al. (2025)). Kucheryavyy et al. (2023) develop a well-behaved multi-industry trade model with industry-level external economies of scale. Bartelme et al. (2025) propose an empirical strategy that estimates sector-level scale economies by exploiting changes in sector size driven by long-run variation in domestic demand. Building on these frameworks, this paper makes two contributions: it estimates the scale elasticity by leveraging a demand shift induced by a trade dispute, using data at a more disaggregated 6-digit level; and it provides direct empirical evidence for this mechanism by documenting a simultaneous decrease in export prices and an increase in export quantities at the product level.

Third, this paper contributes to the literature examining the effects of trade policy in the presence of scale economies. This literature demonstrates that trade policy can have significant, and sometimes unexpected, effects on production and exports. Juhàsz (2018) finds that temporary trade protection during the Napoleonic Wars fostered the adoption of mechanized cotton spinning in France. Breinlich et al. (2026) show that trade normalization with China reduced US export growth in affected sectors. Fajgelbaum et al. (2024) highlight that third countries increased their exports to the rest of the world in products affected by the higher tariffs of the US-China Trade War. This paper contributes to this literature by identifying the specific mechanism through which export controls can paradoxically boost exports in the target country. The target country experiences an expansion in domestic production, absorbing the demand previously met by imports from the imposing country; this expansion raises productivity through scale economies. This effect is most pronounced in sectors where the imposing country had been the dominant supplier.

Fourth, this paper is directly connected to the literature on the home-market effect, a key prediction of trade models with scale economies (Krugman (1980); Head and

Ries (2001); Davis and Weinstein (2003); Costinot et al. (2019)). This effect predicts that a larger domestic market for a good leads to increased exports of that same good. This paper provides a direct empirical demonstration of this channel. Specifically, it shows how an exogenous trade policy shock that reduces imports can generate a home-market effect by increasing demand for domestic goods, a process that in turn raises productivity, lowers prices, and ultimately expands exports.

Fifth, this paper extends the literature on trade policy uncertainty (Handley and Limão (2015); Pierce and Schott (2016); Handley and Limão (2017); Crowley et al. (2018); Alessandria et al. (2024)). The analysis reveals that the threat of export controls—even without actual enforcement—can act as a catalyst for import substitution. Thus, this paper demonstrates that trade policy uncertainty can have unintended, and even beneficial, consequences, as this substitution spurs a positive supply-side response.

Lastly, this paper adds to the literature on the 2019 Korea-Japan Trade Dispute. Hayakawa et al. (2023) and Makioka and Zhang (2024) empirically study the first phase of the export controls—announced in July 2019—on the three chemicals. Kim (2024) examines the short-run dynamics of Korea's imports of these chemicals. Kim and Kim (2022) quantify the effect of the subsequent Korean boycott of Japanese products in the beer market. This paper analyzes the dispute across the full range of goods, as the controls were extended beyond the three chemicals. It shows that Korea's import decline was governed not by targeted status but by pre-event dependence on Japan. It then traces Korea's domestic production and exports, identifying scale economies as the underlying mechanism, and quantifies the welfare effects in general equilibrium.

Outline. The remainder of the paper is organized as follows. Section 2 provides the background on Japan's 2019 announcement of export controls on Korea. Section 3 presents empirical findings regarding changes in Korea's imports, exports, and domestic production following the event. Section 4 introduces a trade model featuring external economies of scale. Section 5 estimates the strength of scale economies, first for Korea and then for its major import sources. Section 6 quantifies the welfare effects of the trade dispute on Korea and Japan. Finally, Section 7 concludes the paper.

2 Background

Decades-long disputes between South Korea and Japan stem from the issue of reparations for Korean forced laborers who worked under Japanese colonial rule during World War II. In late 2018, political tensions between the two countries escalated once more, following a ruling by the Korean Supreme Court that ordered Japanese corporations to compensate Korean victims of this forced labor. Japan vehemently opposed the Korean court's decision, arguing that the 1965 Korea-Japan treaty had already settled such claims for indemnification for forced labor.

In the midst of heightened tensions, the Japanese government abruptly announced in July 2019 that it would tighten export procedures for the shipment of three chemicals—photoresist, fluorinated polyimide, and hydrogen fluoride—to South Korea, citing national security. The new measures required Japanese firms exporting these chemicals to Korea to obtain government authorization for each individual shipment. In response, the Korean government immediately expressed concerns regarding the announcement, as Korean firms importing these chemicals from Japan were confronted with uncertainty over whether their future imports would be permitted. Moreover, these chemicals were essential inputs for two of Korea's key industries—semiconductors and display panels—and Japan was the leading supplier of these chemicals to Korea.

In September 2019, Japan removed South Korea, its third-largest trading partner, from its "White List" of 27 countries, leading South Korea to lose its status as a preferential trade partner. The exclusion from the "White List" served, in essence, as an extension of the export controls: exports of national strategic items to Korea were no longer eligible for a "General Comprehensive Permit," a simplified export process, and instead became subject to individual export approval on a case-by-case basis. This shift granted the Japanese authorities the legal discretion to delay, deny, or restrict shipments of these items to Korea at any time. Approximately 21 percent of all goods exported from Japan to Korea were directly exposed to potential export controls due to this exclusion.[6]

In Korea, the dispute quickly became a matter of national concern: even though it began only in the second half of 2019, it was mentioned in more news articles than any other topic that year.[7] In retaliation, the Korean government took a similar action, removing Japan from its own list of preferential trade partners.[8] The Korean government also announced that it would not renew the General Security of Military Information Agreement with Japan, a military agreement that was critically important to Japan and even to the United States in their efforts against North Korea. It further filed a complaint

[6]Of the 4,207 HS-6 items Korea imported from Japan during the pre-event period (July 2018–June 2019), 886 were classified as Japanese national strategic items by the Korea Security Agency of Trade and Industry, which distributed the list to Korean firms. See Appendix A.4 for details.

[7]The ranking is from the News Based Statistics Search of Statistics Korea. In the second half of 2019 alone, more than 7,000 news articles mentioned the dispute; see Kim (2024) for the underlying series. The Korean President, Prime Minister, and Deputy Prime Minister each addressed the nation on the issue.

[8]While symmetric in form, this measure applied to the opposite direction of trade: it tightened the approval procedures for Korea's exports of strategic items to Japan.

with the WTO regarding Japan's announcement of export controls. At home, the government unveiled longer-term plans to promote domestic production of key materials and components.[9] The escalating measures on both sides intensified bilateral tensions, culminating in a trade dispute between the two countries.

Despite these tensions, for nearly four years, the Japanese government did not implement any actual export restrictions, suggesting the policy was primarily a coercive tactic to influence the resolution of the historical dispute in Japan's favor. In March 2023, the leaders of South Korea and Japan held their first summit in four years, agreeing to normalize trade relations to their pre-dispute status. In June 2023, the Japanese government officially lifted the export controls and reinstated South Korea to its "White List," restoring the country's status as a preferential trade partner.

3 Empirical Facts

In this section, I report empirical facts regarding changes in Korea's imports and exports following the announcement of Japan's 2019 export controls. The key finding is that Korea's imports decreased disproportionately in goods for which Japan had been the leading supplier to Korea (henceforth, *Japan-dominant* goods), rather than in goods directly targeted by the export controls. More surprisingly, Korea's exports of these same *Japan-dominant* goods increased, accompanied by a decline in export prices (Fact 5). This unexpected pattern leads me to examine Korea's domestic production, where I document a broad-based expansion with capital deepening in *Japan-dominant* sectors. Combined with additional pieces of evidence, the patterns point to scale economies in production as the underlying mechanism, motivating the model in Section 4.

Data. The empirical analysis in this section uses Korea's trade data at the 6-digit level of the Harmonized System (HS) from the Korea Customs Service. The data cover 251 countries and 5,807 goods at the HS-6 level over 11 years, from July 2014 to June 2025. Throughout this section, 'good k' refers to the HS-6 level. The original monthly data are aggregated to the quarterly level. The analysis restricts the sample to intermediate and capital goods, which account for 91.6 percent of Korea's imports from Japan in 2018 (the pre-event period) by value, as well as 95.6 and 93.7 percent of *Targeted* and *Japan-dominant* goods, respectively.[10] Appendix A.1 confirms that the main findings are robust

[9] The implications of these plans for the analysis are discussed in Section 5.1.

[10] By the number of HS-6 products, intermediate and capital goods account for 93.7 and 88.8 percent of *Targeted* and *Japan-dominant* goods, respectively. See Appendix A.3 for the full composition statistics.

to including consumer goods, with the exception of Fact 2 (discussed below).

Empirical Strategy. I document the empirical facts in this section using two approaches: a difference-in-differences specification and an event study that traces the time path of the same comparison in one-year windows. The exact specification is reported in each fact below. The main estimator is ordinary least squares (OLS), chosen for consistency with Fact 5, where the dependent variables are export price and quantity. The nature of these variables makes Poisson pseudo-maximum likelihood (PPML) estimation unsuitable for Fact 5. Price cannot, in principle, be coded as zero in the absence of trade; moreover, since it is defined as a unit value (value/quantity), the price is undefined when quantity is missing despite a recorded value. Appendix A.1 confirms that the main OLS findings are robust to PPML estimation for Facts 1 to 4.

Fact 1: Korea's imports from Japan substantially decreased following the event, both relative to the pre-event period and to imports from other countries.

To investigate the effect of Japan's announcement of export controls on Korea's imports, I use a difference-in-differences specification:

$$\log m_{i,k,t}^{kor} = \alpha_{i,k} + \alpha_{k,t} + \beta\Big(Japan_i\Big) \times \Big(Post_t\Big) + \varepsilon_{i,k,t},$$

where $m_{i,k,t}^{kor}$ denotes the value of Korea's imports of good k from origin i at time t. $\alpha_{i,k}$ and $\alpha_{k,t}$ denote origin-good and good-time fixed effects, respectively. $Japan_i$ is a country indicator variable for Japan and $Post_t$ is a time indicator for the post-event period, which spans from the third quarter of 2019 to the second quarter of 2025. Standard errors are clustered at origin i and good k levels. The coefficient β is estimated to be -0.238, as shown in Column (1) of Table 1. This indicates that Korea's imports from Japan decreased by 21.2 percent, both relative to the pre-event period and to imports from other countries.[11] Time-varying coefficients β_t, estimated using an event study, are also plotted in Panel (A) of Figure 2, along with 95 percent confidence intervals for the estimates. This event-study analysis confirms the absence of a pre-trend. This decline is robust to PPML estimation and to an alternative sample (including consumer goods). The same pattern also holds for Japan's exports to Korea, confirming the finding from the mirror perspective. See Appendix A.1 and Appendix A.2 for details.

[11] Note that the Korean government did not impose any import restrictions on Japanese goods during the post-event period. I return to the government's policy responses in Section 5.1.

Fact 2: The extent of the decline in Korea's imports from Japan was similar between targeted and non-targeted goods.

To examine whether the substantial decrease in imports from Japan is limited to goods directly targeted by Japan's export controls, I use a triple difference-in-differences specification:

$$\log m_{i,k,t}^{kor} = \alpha_{i,k} + \alpha_{k,t} + \alpha_{i,t} + \beta\Big(Japan_i\Big) \times \Big(Post_t\Big) \times \Big(Targeted_k\Big) + \varepsilon_{i,k,t},$$

where $\alpha_{i,t}$ denotes origin-time fixed effects and $Targeted_k$ is an indicator variable for goods that were directly targeted by the export controls.[12] The remaining terms and clustering level are as previously described. The coefficient β is estimated to be -0.011, essentially indistinguishable from zero, as shown in Column (2) of Table 1. This indicates that the extent of the decline in *Non-targeted* goods from Japan is not statistically different from that of *Targeted* goods. Moreover, time-varying coefficients β_t, estimated from an event study, further support this finding, as shown in Panel (B) of Figure 2. This symmetric decline indicates that Korean firms anticipated broader restrictions on Japanese goods, not only on the items directly targeted by the announcement, reflecting a widespread perception of Japan as an unreliable source across industries. The result is robust to PPML estimation within the intermediate and capital goods sample.[13] The same absence of significant difference between *Targeted* and *Non-targeted* goods also holds for Japan's exports to Korea, confirming the finding from the mirror perspective. See Appendix A.1 and Appendix A.2 for details.

Fact 3: Korea's imports from Japan decreased disproportionately in goods for which Japan had been the leading supplier to Korea.

To explore whether the decrease was disproportionately larger in sectors where Korea had been most reliant on Japan, I use a triple difference-in-differences specification:

$$\log m_{i,k,t}^{kor} = \alpha_{i,k} + \alpha_{k,t} + \alpha_{i,t} + \beta\Big(Japan_i\Big) \times \Big(Post_t\Big) \times \Big(Dominant_k\Big) + \varepsilon_{i,k,t},$$

[12] *Targeted* goods are defined as the items on a list compiled by the Korea Security Agency of Trade and Industry (KOSTI). This list identifies Japanese national strategic items at the HSK-10 level that became subject to individual export approval after Japan removed Korea from its "White List" of countries. See Appendix A.4 for details.

[13] In the sample including consumer goods, the comparison between *Targeted* and *Non-targeted* goods depends on the choice of estimator due to zero values among *Non-targeted* consumer goods. Restricting to intermediate and capital goods removes this divergence: OLS and PPML yield consistent results. See Appendix A.1 for details.

where $Dominant_k$ is an indicator variable for *Japan-dominant* goods, defined as goods for which Japan was the leading supplier during the pre-event period (July 2018 to June 2019).[14] The remaining terms and clustering level are as previously described. The coefficient β is estimated to be -0.119, as shown in Column (3) of Table 1. This indicates that Korea's imports of *Japan-dominant* goods declined approximately 11.2 percent more than *Non-dominant* goods. This also implies that the relevant margin governing the magnitude of firms' response was not whether a good was directly targeted by the export controls (Fact 2), but rather the degree of pre-event supply dependence on Japan.[15] Firms more reliant on Japan faced greater exposure to potential disruption, prompting larger preemptive reductions. The time-varying coefficients β_t, plotted in Panel (C) of Figure 2, show a dramatic structural break following the announcement. β_t trended upward toward zero during the pre-event period but turned sharply downward thereafter, with the magnitude continuing to widen through the end of the sample. This decline is robust to PPML estimation and to an alternative sample (including consumer goods). The same pattern also holds for Japan's exports to Korea, confirming the finding from the mirror perspective. See Appendix A.1 and Appendix A.2 for details.

Fact 4: Korea's imports from the rest of the world in goods for which Japan had been the leading supplier to Korea did not increase.

Given the decline in imports from Japan in *Japan-dominant* goods (Fact 3), I examine whether Korea's imports of these goods from the rest of the world increased to compensate, using the following difference-in-differences specification:

$$\log m_{i,k,t}^{kor} = \alpha_{i,k} + \alpha_{i,t} + \beta\Big(Dominant_k\Big) \times \Big(Post_t\Big) + \varepsilon_{i,k,t}.$$

The specification details, including clustering level, are as previously described. Japan is excluded from the set of origin countries to focus on the change in Korea's imports from the rest of the world. The coefficient β is estimated to be -0.028, essentially indistinguishable from zero, as shown in Column (4) of Table 1. This indicates that Korea's imports from the rest of the world in *Japan-dominant* goods did not increase, both relative to the pre-event period and to *Non-dominant* goods. This also suggests that the decline in imports from Japan was not offset by an increase from the rest of the world. Moreover,

[14] These goods are concentrated mainly in machinery and equipment, chemicals, non-metallic mineral products, and rubber and plastic products. Appendix A.5 lists representative *Japan-dominant* sectors at the aggregated 5-digit level.

[15] *Targeted* and *Japan-dominant* goods are largely distinct. See Appendix A.6 for details.

time-varying coefficients β_t, estimated using an event study, further support this finding, as plotted in Panel (D) of Figure 2. The result is robust to PPML estimation and to an alternative sample (including consumer goods). See Appendix A.1 for details.

Fact 5: Korea's export quantities of goods for which Japan had been the leading supplier to Korea increased, while their prices decreased.

Given the findings from Fact 3 and Fact 4, one might expect Korea's exports of *Japan-dominant* goods to decline, since domestic production in these goods would likely be redirected to the domestic market to satisfy the demand previously met by imports from Japan. To test this hypothesis, I use the following difference-in-differences specification:

$$\log x_{k,t}^{kor} = \alpha_k + \alpha_{K(k),t} + \beta\Big(Dominant_k\Big) \times \Big(Post_t\Big) + \varepsilon_{k,t},$$

where $x_{k,t}^{kor}$ is Korea's export quantity of good k at time t. α_k and $\alpha_{K(k),t}$ denote good and sector-time fixed effects, respectively, where $K(k)$ is the KSIC-3 sector to which good k belongs.[16] In place of simple time fixed effects (α_t), the sector-time fixed effects $\alpha_{K(k),t}$ absorb sector-specific shocks over time, isolating the differential response of *Japan-dominant* goods within each sector.[17] Standard errors are clustered at good k level. The remaining terms are as previously described. Contrary to the hypothesis, the coefficient β is estimated to be 0.124, as documented in Column (5) of Table 1. This indicates that Korea's export quantities of *Japan-dominant* goods increased by approximately 13.2%, both relative to *Non-dominant* goods and to the pre-event period. This rapid expansion is plausible given that Korea was already exporting 97 percent of these *Japan-dominant* goods during the pre-event period, and that Korea and Japan share similar export structures. See Appendix A.7 for details.

Furthermore, a standard economic prediction would be that Korea's export prices of *Japan-dominant* goods would rise, driven by increased domestic demand and an upward-sloping supply curve. To test this hypothesis, the dependent variable $\log x_{k,t}^{kor}$ is now defined as the log of the export unit value (value divided by quantity). All other terms of the specification remain the same. The new coefficient β is estimated to be -0.047, as shown in Column (6) of Table 1. This result indicates that, contrary to the hypothesis, Korea's export prices of *Japan-dominant* goods declined by approximately 4.6%, both rel-

[16]KSIC refers to the Korean Standard Industrial Classification, the Korean adaptation of the United Nations' International Standard Industrial Classification (ISIC).

[17]The fixed effects absorb sector-level confounders such as policy shocks (e.g., government support including subsidies) and input-output linkage effects. The KSIC-3 level is chosen as the most disaggregated level at which input-output tables are constructed and government spending by sector is reported.

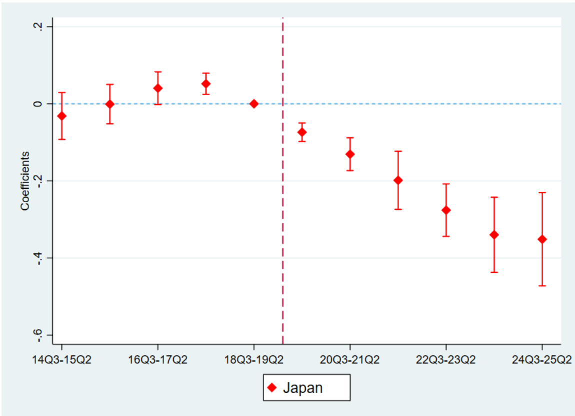
Table 1: Empirical Facts (Difference-in-Differences)

	Fact 1 (1) $\ln m$	Fact 2 (2) $\ln m$	Fact 3 (3) $\ln m$	Fact 4 (4) $\ln m$	Fact 5-1 (5) $\ln x^q$	Fact 5-2 (6) $\ln x^p$
JP $\times$ Post	-0.238^{***}					
	(0.044)					
JP $\times$ Post $\times$ Targeted		-0.011				
		(0.022)				
JP $\times$ Post $\times$ Dominant			-0.119^{***}			
			(0.028)			
Dominant $\times$ Post				-0.028	0.124^{**}	-0.047^{**}
				(0.021)	(0.049)	(0.024)
Origin $\times$ Good FE	Yes	Yes	Yes	Yes		
Good $\times$ Time FE	Yes	Yes	Yes			
Origin $\times$ Time FE		Yes	Yes	Yes		
Good FE					Yes	Yes
Sector $\times$ Time FE					Yes	Yes
No. of Goods	4,012	4,012	4,012	4,213	3,601	3,601
No. of Targeted Goods		946				
No. of Dominant Goods			594	591	579	579
Observations	2,143,891	2,142,683	2,142,683	2,028,141	139,719	139,719
R-squared	0.81	0.81	0.81	0.78	0.89	0.86

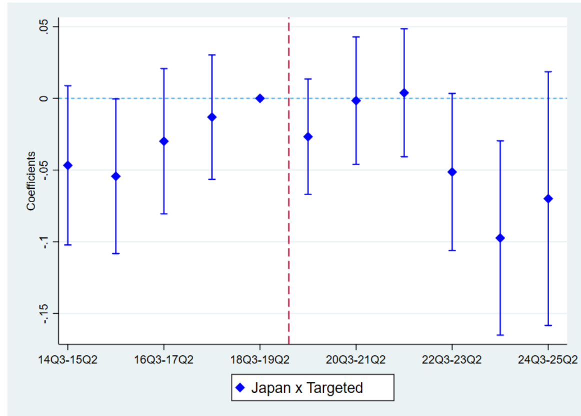
Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

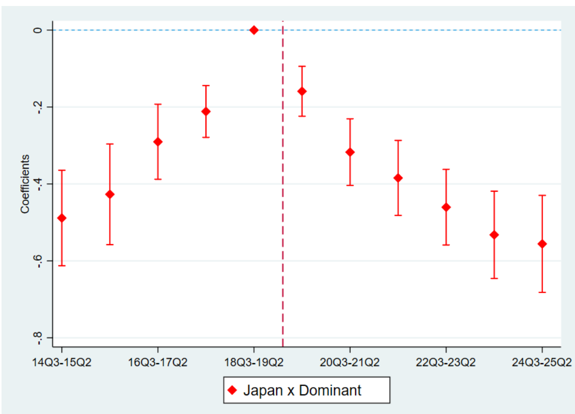
Notes: Table 1 presents the estimation results for the difference-in-differences specifications on Korea's imports and exports from Section 3. Columns (1) through (4) correspond to Facts 1 through 4, and Columns (5) and (6) both correspond to Fact 5. The dependent variable is the log of Korea's import value in Columns (1) through (4), the log of Korea's export quantity in Column (5), and the log of Korea's export price in Column (6). The sample is restricted to intermediate and capital goods. All specifications are estimated by ordinary least squares (OLS) at the HS-6 product level.
Source: Korea Customs Service (2014–2025).



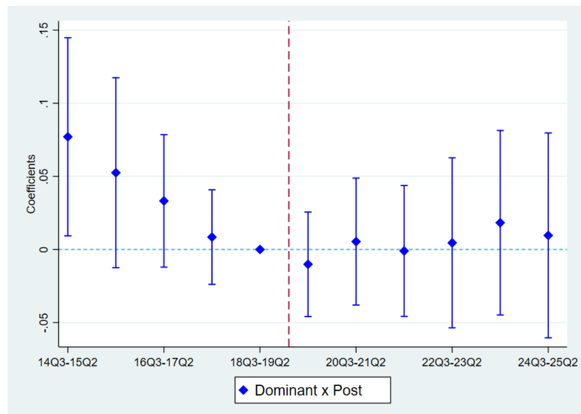
(A) Fact 1: Imports from Japan



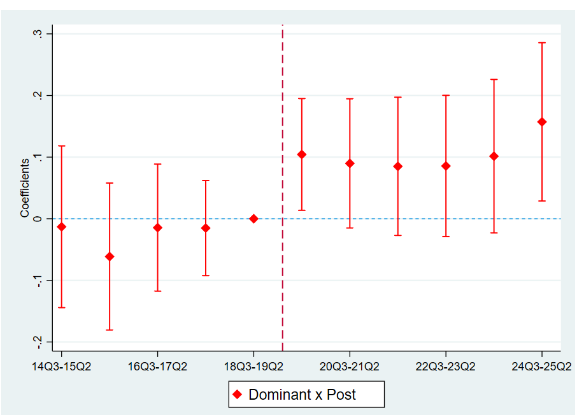
(B) Fact 2: Targeted vs. Non-Targeted



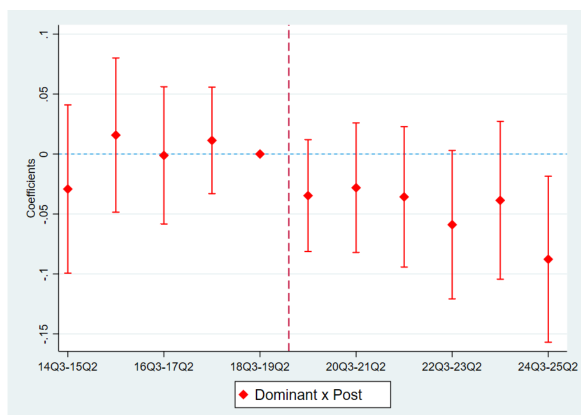
(C) Fact 3: Dominant vs. Non-Dominant



(D) Fact 4: Imports from RoW



(E) Fact 5-1: Korea's Export Quantity



(F) Fact 5-2: Korea's Export Price

Figure 2: Empirical Facts (Event Study)

Notes: Figure 2 illustrates the estimation results for the event-study specifications on Korea's imports and exports from Section 3. Panels (A) through (D) correspond to Facts 1 through 4, and Panels (E) and (F) both correspond to Fact 5. The sample is restricted to intermediate and capital goods. All specifications are estimated by ordinary least squares (OLS) at the HS-6 product level, using quarterly data for the period from Q3 2014 to Q2 2025.
Source: Korea Customs Service (2014–2025).

ative to *Non-dominant* goods and to the pre-event period. Combined with the increase in quantities, this is consistent with productivity gains in the production of *Japan-dominant* goods after the event. Moreover, this pattern is consistent with scale economies, as illustrated by the downward-sloping supply curve in Figure 1. An event study further confirms these two contrasting patterns: the time-varying coefficients β_t for export quantities and prices show the rise in quantities and the decline in prices following the event, as presented in Panels (E) and (F) of Figure 2. At monthly frequency, quantities rise first while prices decline with a lag, further supporting the scale economies interpretation. The result is robust to an alternative sample (including consumer goods). See Appendix A.1 and Appendix A.8 for details.

The pattern documented in Fact 5—a simultaneous rise in Korean export quantities and decline in export prices for *Japan-dominant* goods—is, at first glance, a puzzle. If the domestic production that replaced Japan's goods were simply absorbed by import substitution, neither outcome should have arisen. One plausible reconciliation is scale economies, under which an expansion in domestic production raises productivity and lowers prices. To examine whether this interpretation aligns with the production-side response, I turn to evidence from Korean producer data.

Production Response. The data come from the Korean Mining and Manufacturing Survey by Statistics Korea. The data are annual, covering 476 sectors at the 5-digit level of the Korean Standard Industrial Classification (KSIC), the Korean adaptation of the United Nations' International Standard Industrial Classification (ISIC). The sample period spans 11 years, from 2014 to 2024. I estimate the following difference-in-differences specification on each of the eight sector-level dependent variables $Y_{s,t}$ by ordinary least squares (OLS)[18]:

$$\log Y_{s,t} = \alpha_s + \alpha_{S(s),t} + \beta\Big(Dominant_s\Big) \times \Big(Post_t\Big) + \varepsilon_{s,t},$$

where s indexes a 5-digit KSIC sector, t indexes a year, and $S(s)$ denotes the 3-digit KSIC sector to which the 5-digit sector s belongs. α_s and $\alpha_{S(s),t}$ denote 5-digit sector and 3-digit-sector-year fixed effects, respectively. In place of simple year fixed effects (α_t), $\alpha_{S(s),t}$ absorb 3-digit-sector-specific shocks over time, isolating the differential response of *Japan-dominant* 5-digit sectors within each 3-digit sector. $Dominant_s$ identifies sectors

[18] Unlike the import and export analyses at the HS-6 product level in Facts 1 through 5, zero observations are rare in the annual production data at the KSIC-5 sector level used here, making the distinction between OLS and PPML inconsequential.

in which Japan was the leading supplier to Korea during the pre-event period (2018).[19] $Post_t$ is a time indicator for the post-event period, which spans from 2019 to 2024. The eight dependent variables $Y_{s,t}$ are: gross output, the year-end book value of machinery and equipment, intermediate inputs, employment, the wage bill, the year-end book value of tangible fixed assets, annual investment in tangible fixed assets, and the number of establishments.

Panel A of Table 2 reports the estimates from the specification with $\alpha_{S(s),t}$. Column (1) shows that gross output in *Japan-dominant* sectors rises by approximately 6.4% over the post-event period, even after controlling for time-varying shocks at the 3-digit-sector level, such as government support and input-output linkage effects. Given that gross output is measured in nominal terms, the price decline documented in Fact 5 suggests an even larger real expansion. The machinery and equipment stock, intermediate inputs, and employment all rise significantly, with the machinery stock showing the largest expansion, as shown in Columns (2) through (4). This pattern suggests that capital deepening is a key driver of the output expansion. Panel B of Table 2 reports the pooled estimates from the specification with α_t instead of $\alpha_{S(s),t}$, in which *Japan-dominant* sectors are compared with *Non-dominant* sectors across all 3-digit sectors rather than within each. The same pattern documented in Panel A holds, and all eight outcomes are now statistically significant, indicating a broad-based expansion in *Japan-dominant* sectors. In particular, the stock of tangible fixed assets and annual investment rise significantly, reinforcing the evidence of broader capital accumulation across *Japan-dominant* sectors, as shown in Columns (6) and (7). The number of establishments also increases, suggesting extensive-margin entry into these sectors, as shown in Column (8).

Additional Evidence. Beyond the production-level estimates, several additional pieces of evidence are consistent with the scale economies interpretation. First, the *Japan-dominant* sectors are predominantly capital- and R&D-intensive industries, in which scale economies are a natural feature. Representative examples include semiconductor manufacturing machinery, sensitized materials and related chemicals, carbon fibers, and optical lenses. Second, Korea was already exporting 97 percent of these *Japan-dominant* goods during the pre-event period, indicating that the necessary production

[19] Sectors in which Japan was the leading supplier to Korea are identified by aggregating Korea's imports from the HS-6 to the KSIC-5 level using a concordance provided by Statistics Korea. A single KSIC-5 sector typically corresponds to multiple HS-6 goods. These sectors are concentrated mainly in machinery and equipment, chemicals, non-metallic mineral products, and rubber and plastic products. Appendix A.5 lists representative *Japan-dominant* sectors and reports their distribution across KSIC-2 industries.

Table 2: Production Response in Korea (Difference-in-Differences)

	(1) *output*	(2) *machinery*	(3) *inputs*	(4) *emp*	(5) *wagebill*	(6) *assets*	(7) *invest*	(8) *estab*
Panel A								
Dominant×Post	0.062**	0.095**	0.063**	0.044**	0.033	0.029	0.066	0.015
	(0.031)	(0.042)	(0.030)	(0.019)	(0.022)	(0.032)	(0.059)	(0.015)
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
KSIC-3 × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of Sectors	457	455	457	455	455	455	455	457
Dominant Sectors	63	63	63	63	63	63	63	63
Observations	4,941	4,497	4,951	4,944	4,944	4,497	4,488	5,009
R-squared	0.98	0.96	0.97	0.98	0.98	0.97	0.92	0.99
Panel B								
Dominant×Post	0.069**	0.106***	0.081***	0.084***	0.063***	0.067**	0.102**	0.031**
	(0.027)	(0.037)	(0.027)	(0.018)	(0.020)	(0.028)	(0.051)	(0.015)
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of Sectors	473	472	473	472	472	472	472	476
Dominant Sectors	63	63	63	63	63	63	63	63
Observations	5,105	4,646	5,115	5,110	5,110	4,646	4,637	5,218
R-squared	0.97	0.96	0.97	0.98	0.97	0.96	0.90	0.98

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Table 2 presents the estimation results for the difference-in-differences specifications on production-side responses from Section 3. Panel A reports the within-3-digit-sector estimates with 5-digit sector (α_s) and 3-digit sector-year ($\alpha_{S(s),t}$) fixed effects, while Panel B reports the pooled estimates with 5-digit sector (α_s) and year (α_t) fixed effects. The eight dependent variables are: *output* = gross output; *machinery* = year-end book value of machinery and equipment; *inputs* = major intermediate inputs; *emp* = number of regular workers; *wagebill* = annual wage bill of regular workers; *assets* = year-end book value of tangible fixed assets; *invest* = annual investment in tangible fixed assets; *estab* = the number of establishments. All variables are from the annual Mining and Manufacturing Survey, which covers establishments with 10 or more workers. All specifications are estimated by ordinary least squares (OLS) at the 5-digit KSIC sector-year level. See Appendix A.9 for detailed variable definitions and data construction.
Source: Statistics Korea (2014–2024).

capacity was already in place.[20] The substantial post-event expansion thus reflects additional investment in existing capacity rather than entry into new product lines, in line with the production responses in Table 2. Third, examined at monthly frequency, Fact 5 reveals that export quantities rise first and remain elevated, while the price decline follows with a lag and deepens over time. This Q-first, P-lag dynamic reflects productivity gains through scale economies. See Appendix A.5, Appendix A.7, and Appendix A.8 for details.

Alternative Mechanisms. These pieces of evidence, together with the production-level estimates, are difficult to reconcile with alternative mechanisms for the export expansion and price decline documented in Fact 5. A first alternative is pure substitution toward cheaper sources, under which Korean firms replace Japanese inputs with less expensive alternatives, thereby lowering production costs. This is inconsistent with the composition of the production response: a pure switch in input sources would leave the production process unchanged, whereas the machinery stock shows the largest expansion among all inputs—capital deepening rather than cheaper sourcing. A second alternative is markup cutting, under which the price decline reflects compressed margins rather than productivity gains. This is inconsistent with the simultaneous rise in all factor inputs and with the Q-first, P-lag export dynamics, both of which are at odds with an immediate price cut driven by margin compression. Moreover, given the rise in domestic demand and market share, lowering export prices to expand exports is implausible and contradicts variable markup theory. A third alternative is pure capacity reallocation, under which existing resources within Korea are redirected toward *Japan-dominant* sectors without genuine expansion. This is inconsistent with the production response: redirecting existing resources would require neither new investment nor new establishments, yet both rise, together with the machinery stock and employment.

Combined with the trade-side findings—the disproportionate decline in Korea's imports of *Japan-dominant* goods (Fact 3), the absence of substitution from alternative origins (Fact 4), and the simultaneous rise in Korean exports of these goods and decline in their prices (Fact 5)—this expansion in domestic production extends beyond mere import substitution. Together with the additional pieces of evidence, the patterns point to scale economies in production as the underlying mechanism. Under scale economies,

[20] This capability is also supported by the similar export structures of the two countries. For Korea, Japan ranks as the most similar export competitor in the global market. For Japan, Korea ranks as the second-most similar export competitor. See Appendix A.7 for details.

characterized by a downward-sloping supply curve, an increase in production raises productivity, which in turn lowers prices and boosts exports.

4 Model

This section presents an Armington model of trade featuring external economies of scale to demonstrate how foreign export controls can lead to an increase in exports accompanied by a price decline in the target country. The model follows Kucheryavyy et al. (2023) and Bartelme et al. (2025).

4.1 Environment

Consider a general equilibrium trade model with multiple countries, indexed by i (origin) or j (destination), and multiple sectors, indexed by k. Each sector features product differentiation by country of origin (the Armington assumption), meaning each country produces a unique variety within each sector. Labor is the only factor of production and is immobile across countries but perfectly mobile across sectors within a country. L_j and w_j denote the inelastic labor supply and wage level in country j, respectively.

Preferences and Demand. Each country has a representative consumer with two-tier preferences across sectors and goods. The upper-tier preferences are represented by a Cobb-Douglas utility function:

$$U_j = \prod_{k=1}^{K} Q_{j,k}^{\beta_{j,k}},$$

where $Q_{j,k}$ is the consumption index for sector k in country j, and $\beta_{j,k}$ is the sector-level expenditure share, such that $\sum_k \beta_{j,k} = 1$. The lower-tier preferences are of the CES form, with the elasticity of substitution $(1 + \theta_k)$ between goods from different origins within sector k:

$$Q_{j,k} = \left(\sum_i q_{ij,k}^{\frac{\theta_k}{1+\theta_k}} \right)^{\frac{1+\theta_k}{\theta_k}},$$

where $q_{ij,k}$ is the quantity of good k produced in origin i and consumed in destination j. Then, the demand in destination j for good k produced in origin i is:

$$q_{ij,k} = p_{ij,k}^{-1-\theta_k} P_{j,k}^{\theta_k} \beta_{j,k} w_j L_j, \tag{1}$$

where $p_{ij,k}$ is the price of good k produced in origin i and sold in destination j, and $P_{j,k}$ is the price index for sector k in destination j given by $P_{j,k} = \left(\sum_i p_{ij,k}^{-\theta_k} \right)^{-\frac{1}{\theta_k}}$.

Production and Pricing. The production technology exhibits external economies of scale, giving rise to increasing returns. Specifically, the labor productivity of sector k in country i is given by $\tilde{A}_{i,k} = A_{i,k} L_{i,k}^{\gamma_k}$, where $A_{i,k}$ is an exogenous productivity parameter and $L_{i,k}$ is the total employment in sector k in country i. The parameter γ_k—the *scale elasticity*—governs the strength of scale economies in sector k. Note that if $\gamma_k = 0$, production exhibits constant returns to scale as in the standard Armington model. This production technology, together with the iceberg costs incurred in bilateral trade, $\tau_{ij,k} \geq 1$, yields the unit cost of good k produced in origin i for sale in destination j as follows:

$$c_{ij,k} = \frac{\tau_{ij,k} w_i}{A_{i,k} L_{i,k}^{\gamma_k}}. \tag{2}$$

The unit cost is a decreasing function of the sector size $L_{i,k}$, with an elasticity of $-\gamma_k$. Assuming that all markets are perfectly competitive ($p_{ij,k} = c_{ij,k}$) and using the zero-profit condition ($w_i L_{i,k} = Y_{i,k}$) to substitute for $L_{i,k}$ in equation (2), the price equation is given by:

$$p_{ij,k} = \frac{\tau_{ij,k} w_i^{1+\gamma_k}}{A_{i,k} Y_{i,k}^{\gamma_k}}, \tag{3}$$

where $Y_{i,k}$ is the total sales of good k across destinations by origin i such that $Y_{i,k} = \sum_j X_{ij,k}$. This equation highlights how scale economies impact the relationship between the total sales $Y_{i,k}$ and the price $p_{ij,k}$. Specifically, when scale economies exist (i.e., $\gamma_k > 0$), an increase in $Y_{i,k}$ enhances productivity, leading to a reduction in $p_{ij,k}$.

4.2 Bilateral Trade

To derive the bilateral trade equation, equation (3) is first substituted into the demand function (1). Applying the Cobb-Douglas expenditure share $\beta_{j,k}$ to income $w_j L_j$ gives the total expenditure on good k by destination j, $\beta_{j,k} w_j L_j = X_{j,k}$, where $X_{j,k} = \sum_i X_{ij,k}$. These substitutions yield the expenditure by destination j on good k produced in origin i as follows:

$$X_{ij,k} = \left(\frac{\tau_{ij,k} w_i^{1+\gamma_k}}{A_{i,k}} \right)^{-\theta_k} Y_{i,k}^{\gamma_k \theta_k} X_{j,k} P_{j,k}^{\theta_k}, \tag{4}$$

where $X_{ij,k} = p_{ij,k} q_{ij,k}$.

The key term in this equation is the exponent on total sales, $\gamma_k \theta_k$, which represents the *output elasticity*: the elasticity of bilateral trade ($X_{ij,k}$) with respect to total sales ($Y_{i,k}$). This output elasticity arises from two channels. First, a 1% increase in total sales ($Y_{i,k}$) lowers price ($p_{ij,k}$) by γ_k percent; this is the scale elasticity channel from equation (3). Second, a 1% decrease in price increases sales to destination j ($X_{ij,k}$) by θ_k percent; this is the trade elasticity channel from equation (1). The product of these two effects, $\gamma_k\theta_k$, gives the total output elasticity. If $\gamma_k = 0$, indicating the absence of scale economies (the standard Armington model), an origin's total sales ($Y_{i,k}$) do not affect its sales in destination j ($X_{ij,k}$). In contrast, if $\gamma_k > 0$, indicating the presence of scale economies, an increase in $Y_{i,k}$ leads to an increase in $X_{ij,k}$.

4.3 General Equilibrium

The model's general equilibrium is defined by two market clearing conditions: the labor market and the goods market. First, the labor market clears when total employment in country i equals its labor supply, $\sum_k L_{i,k} = L_i$; combined with the zero-profit condition ($w_i L_{i,k} = Y_{i,k}$), this implies that country i's total income equals its total wage bill, $Y_i = w_i L_i$. Second, the goods market clears when total output equals total sales across all sectors and destinations, $Y_i = \sum_k \sum_j X_{ij,k}$. For the economy to be in equilibrium, both conditions must hold simultaneously, and combining them yields the market clearing condition:

$$w_i L_i = \sum_k \sum_j X_{ij,k}. \tag{5}$$

This condition must hold for all countries simultaneously. The system can be formalized by defining an excess demand function for each country i:

$$Z_i(\mathbf{w}) \equiv \frac{1}{w_i}\left(\sum_k \sum_j X_{ij,k} - w_i L_i \right). \tag{6}$$

The equilibrium is then the wage vector $\mathbf{w}^*$ that solves $Z_i(\mathbf{w}^*) = 0$ for all countries.

4.4 Application: Japan's 2019 Export Controls on Korea

This theoretical framework can rationalize the empirical findings presented in Section 3. To illustrate, I model Japan's announcement of export controls as a perceived increase in the bilateral trade cost, $\tau_{JP\,KR,k}$, where JP and KR denote Japan and Korea.

The model directly explains the decline in Korea's imports from Japan ($X_{JP\,KR,k}$) and the accompanying increase in Korean producers' output. The bilateral trade equation

(4) shows that an increase in $\tau_{JP\,KR,k}$ mechanically reduces $X_{JP\,KR,k}$, consistent with Fact 1 in Section 3. The increase in $\tau_{JP\,KR,k}$ also raises the price of Japanese goods in Korea ($p_{JP\,KR,k}$), as shown in equation (3). This in turn raises the sectoral price index in Korea ($P_{KR,k}$), which directly increases expenditure on domestically produced goods ($X_{KR\,KR,k}$), as shown in the bilateral trade equation (4). This rise in domestic sales consequently increases total sales ($Y_{KR,k}$), consistent with the production response documented in Section 3.

Beyond the effect on domestic sales, the model also predicts that the increase in total sales lowers the price of Korea's exports and increases their quantity, consistent with Fact 5. This occurs because the larger scale of production enhances productivity, which lowers the price of Korean goods in foreign markets ($p_{KR\,j,k}$), as shown in equation (3). This price decline boosts Korea's exports, as shown in the bilateral trade equation (4).

This entire mechanism is driven by the model's key feature: the presence of scale economies ($\gamma_k > 0$). In a standard model without this feature (i.e., $\gamma_k = 0$), the link between an increase in domestic sales and export performance is severed. The resulting increase in total sales ($Y_{i,k}$) would have no impact on prices ($p_{ij,k}$) or exports ($X_{ij,k}$). Such a framework is therefore inconsistent with the empirical findings in Section 3. Thus, the model accounts for these empirical facts within a single framework, with scale economies as the key mechanism that turns a negative import shock into a positive export expansion.

5 Estimation of Scale Elasticity

Motivated by the empirical findings from Section 3, this section structurally estimates the strength of scale economies based on the model presented in Section 4. First, I estimate the scale elasticity in Korea. Second, I extend the analysis to Korea's major import sources other than Japan. For these countries, an increase in their sales to Korea, replacing Japanese goods, can enhance productivity through scale economies, thereby boosting their exports to all other destinations. The estimation results strongly support the presence of scale economies in both Korea and its major import sources. The quantitative analysis in Section 6 then employs scale elasticities consistent with these estimates.

Data. The estimation uses three data sources. The first source is annual bilateral trade data from the CEPII BACI database (Gaulier and Zignago (2010)), covering 5,199 goods at the 6-digit level of the Harmonized System (HS) under the 2012 classification across 226 countries. The sample period spans nine years, from 2015 to 2023. The second source is producer data from the Korean Mining and Manufacturing Survey by Statistics Korea. The data are annual, covering 476 sectors at the 5-digit level of the Korean Standard Industrial Classification (KSIC), the Korean adaptation of the United Nations' International Standard Industrial Classification (ISIC). The sample period again spans nine years, from 2015 to 2023. The third source is a concordance provided by Statistics Korea, used to match the HS-6 goods to KSIC-5 sectors; a single KSIC-5 sector typically corresponds to multiple HS-6 goods. The analysis focuses on manufacturing sectors, which correspond to KSIC-2 codes 10 to 33.[21]

Baseline Specification. The estimation specification is derived from the bilateral trade equation (4). Taking logarithmic differences yields:

$$\Delta \ln X_{ij,k} = \gamma_k \theta_k \Delta \ln Y_{i,k} - \theta_k (1+\gamma_k) \Delta \ln w_i + \Delta \ln \left(X_{j,k} P_{j,k}^{\theta_k} \right) \\ + \theta_k \Delta \ln A_{i,k} - \theta_k \Delta \ln \tau_{ij,k}. \tag{7}$$

The change in origin i's sales to destination j, $\Delta \ln X_{ij,k}$, depends on changes in i's supply-side factors, which comprise the total sales, $Y_{i,k}$, the wage, w_i, and the exogenous productivity, $A_{i,k}$. $\Delta \ln X_{ij,k}$ also depends on changes in destination j's demand-side factors, $X_{j,k} P_{j,k}^{\theta_k}$, and in trade costs from origin i to destination j, $\tau_{ij,k}$.

To obtain a direct estimate of the scale elasticity, γ_k, the baseline specification adjusts $\Delta \ln X_{ij,k}$ by the trade elasticity (θ_k) and includes a set of fixed effects as follows:

$$\frac{1}{\theta_k} \Delta \ln X_{ij,k} = \gamma_k \Delta \ln Y_{i,k} + \alpha_i + \alpha_{j,k} + \alpha_{i,K(k)} + \varepsilon_{ij,k}. \tag{8}$$

The origin fixed effects, α_i, absorb any origin-specific changes, such as w_i in equation (7). The destination-sector fixed effects, $\alpha_{j,k}$, absorb demand shifts within each j and k, such as $X_{j,k} P_{j,k}^{\theta_k}$ in equation (7). The origin-sector fixed effects, $\alpha_{i,K(k)}$, absorb supply-side changes common to all goods within the same 5-digit sector, such as K-level shifts in productivity $A_{i,k}$ in equation (7); $K(k)$ denotes the 5-digit sector to which the 6-digit good k belongs. This level of aggregation is necessary because 6-digit origin-sector fixed effects ($\alpha_{i,k}$) are perfectly collinear with the independent variable, $\Delta \ln Y_{i,k}$. The remaining supply-side shocks specific to the 6-digit good k, including the corresponding component of $A_{i,k}$, are left in the error term $\varepsilon_{ij,k}$; because $\Delta \ln Y_{i,k}$ may be correlated with them, identifying γ_k requires an instrumental variable, which I develop for each analysis below.

[21] Mining sectors are excluded, accounting for only 14 of the 476 KSIC-5 sectors. They are not subject to Japan's export controls, have so few firms that their values are often suppressed, and are rarely matched to the HS-6 goods through the concordance.

5.1 Analysis 1: Korea Only

The first analysis focuses solely on Korea as the origin country. This provides the most direct setting in which to estimate the strength of scale economies, as Korea was directly targeted by the export controls and subsequently experienced a substantial increase in its total output and exports, as documented in Section 3.

Specification. The estimation specification for Korea is adapted from the baseline equation (8) for the case of a single origin country, Korea. It imposes a common scale elasticity ($\gamma_k = \gamma$), pooling the variation across sectors for identification:[22]

$$\frac{1}{\theta_k}\Delta \ln X_{KOR\,j,k} = \gamma \Delta \ln Y_{KOR,k} + \alpha_{KOR} + \alpha_{KOR\,j} + \alpha_{KOR,K(k)} + \varepsilon_{KOR\,j,k}. \tag{9}$$

The estimation uses a long difference between a pre-event period (2015–2018) and a post-event period (2020–2023), excluding the event year 2019.[23] The dependent variable, $\frac{1}{\theta_k}\Delta \ln X_{KOR\,j,k}$ (defined as $\frac{1}{\theta_k}\ln X^{Post}_{KOR\,j,k} - \frac{1}{\theta_k}\ln X^{Pre}_{KOR\,j,k}$), is the log difference in Korea's exports of good k to destination j between the post-event and pre-event periods, adjusted by the trade elasticity (θ_k). The same estimates for θ_k are used as in Bartelme et al. (2025), who in turn take the values from Giri et al. (2021).[24] The independent variable, $\Delta \ln Y_{KOR,k}$, is the log difference in Korea's total sales in good k over the same periods. Due to the unavailability of domestic sales data at the HS-6 level, total sales are proxied by total exports. This limitation is addressed in a robustness check at a more aggregated 5-digit level; including domestic sales yields results consistent with the baseline findings at the 6-digit level.[25] Moreover, because $\Delta \ln Y_{KOR,k}$ is instrumented, this proxy introduces only measurement error, and the estimator remains consistent provided the instrument is uncorrelated with it.

γ is the coefficient of interest, which measures the strength of scale economies and

[22] A sector-specific elasticity (γ_k) is not separately identified in this specification: the independent variable, $\Delta \ln Y_{KOR,k}$, takes a single value per sector, so the elasticity can be identified only from the variation across sectors, yielding a single common estimate.

[23] In principle, the specification could instead be estimated on the annual panel to exploit the variation over time. However, the instrument, introduced below, takes one fixed value for each sector, based on Japan's 2018 import share; its cross-sector variation is identical in every year, so the annual panel would add no identifying variation beyond the single difference between the pre-event and post-event periods.

[24] Giri et al. (2021) provide estimates for manufacturing sectors at the SIC-2 level; these estimates are assigned to KSIC-2 sectors, and all HS-6 goods within the same KSIC-2 sector are therefore assumed to share a common trade elasticity. The elasticity is taken from the literature rather than estimated, following Bartelme et al. (2025) and Kucheryavyy et al. (2023). The full list of values is provided in Appendix E.3.

[25] This is consistent with Bartelme et al. (2025), whose framework implies that the scale elasticity can be recovered from any subset of an origin's destination sales; estimating it from exports alone, excluding domestic sales, therefore yields a similar estimate.

is referred to as the scale elasticity. The estimate of γ has two equivalent interpretations based on the model: (i) it is the elasticity of productivity with respect to total sales, as defined by the production technology ($\tilde{A}_{i,k} = A_{i,k} L_{i,k}^{\gamma_k}$); and (ii) it is the elasticity of price with respect to total sales, as shown in equation (3). Since the specification imposes a common elasticity across sectors, the estimate of γ is interpreted as a weighted average of the sector-level elasticities (γ_k).

The remaining terms in the specification (9) are a set of fixed effects that control for other sources of variation, with each effect corresponding to a theoretical term in equation (7). The origin fixed effect (α_{KOR}) absorbs any aggregate shocks to Korea that are common across all js and ks, such as changes in the wage ($\Delta \ln w_{KOR}$) in equation (7). The Korea-destination fixed effects ($\alpha_{KOR\,j}$) absorb any shocks specific to destination j, such as changes in overall demand for Korean goods. The origin-sector fixed effects ($\alpha_{KOR,K(k)}$) absorb supply-side confounders common within each 5-digit sector K, such as productivity. They are kept at the 5-digit K-level rather than the 6-digit k-level because $\alpha_{KOR,k}$ would absorb the variation in $\Delta \ln Y_{KOR,k}$ that identifies γ. The destination-sector fixed effects $\alpha_{j,k}$ are not included in this Korea-only specification. With only a single origin country, each destination-sector cell contains a single observation, so these fixed effects would absorb all variation in the data. This limitation is addressed in Section 5.2, where the inclusion of multiple origins allows $\alpha_{j,k}$ to be incorporated.

A key identification challenge is that the error term ($\varepsilon_{KOR\,j,k}$) in specification (9) contains shocks from the theoretical model (equation (7)) that are not fully absorbed by the fixed effects, including shocks to destination-sector specific demand ($\Delta \ln \left(X_{j,k} P_{j,k}^{\theta_k} \right)$), bilateral trade costs ($\Delta \ln \tau_{KOR\,j,k}$), and fundamental productivity ($\Delta \ln A_{KOR,k}$).[26] The independent variable, the change in total sales ($\Delta \ln Y_{KOR,k}$), is likely to respond endogenously to these shocks, implying $E[\Delta \ln Y_{KOR,k} \times \varepsilon_{KOR\,j,k}] \neq 0$. Therefore, an instrumental variable that isolates exogenous demand-side variation is required to obtain an unbiased estimate of the scale elasticity, γ.

Instrumental Variable. To address this endogeneity, the estimation requires an instrumental variable that is correlated with $\Delta \ln Y_{KOR,k}$ but uncorrelated with the error term ($\varepsilon_{KOR\,j,k}$), in particular the unobserved supply-side shocks. I therefore construct an instrument that captures the exogenous demand shock from the 2019 Korea-Japan Trade Dispute. The instrument is the product of the trade elasticity (θ_k) and Japan's pre-event

[26] These productivity shocks vary at the 6-digit good level, below the 5-digit fixed effects. In the model, they also govern the good-level variation in factory-gate prices and outward multilateral resistance.

(2018) import share in Korea for good k, denoted as $IV_{KOR,k}$:

$$IV_{KOR,k} = \theta_k Share^{2018}_{JPN\,KOR,k},$$

where $Share^{2018}_{JPN\,KOR,k} = \frac{X_{JPN\,KOR,k}}{\sum_{i \neq KOR} X_{i\,KOR,k}}$.

This instrumental variable is designed to capture the sector-specific exposure to the 2019 Trade Dispute. The intuition is straightforward: sectors where Korea was more reliant on Japanese imports (i.e., a higher $Share^{2018}_{JPN\,KOR,k}$) were poised to experience a larger increase in demand for domestically produced goods following the shock. This increase in domestic demand should, in turn, lead to greater sales growth for Korean producers of these goods. This mechanism is strongly supported by the empirical evidence in Section 3. Fact 3 shows that the decline in imports from Japan was the largest in these high-share goods, and Korean producers' output also increased the most in the corresponding sectors.

The structural model in Section 4 also provides a theoretical justification for this instrumental variable. Within this framework, the trade dispute is represented as an exogenous increase in the bilateral trade cost ($\tau_{JPN\,KOR,k}$).[27] This shock propagates purely through a demand-side channel: it raises the price of Japanese goods ($p_{JPN\,KOR,k}$), which increases Korea's price index for good k ($P_{KOR,k}$) and shifts expenditure toward domestic goods ($X_{KOR\,KOR,k}$), ultimately increasing Korean producers' total sales ($Y_{KOR,k}$). This mechanism establishes the theoretical relevance of the instrument. Furthermore, because this channel is entirely demand-driven, the instrument is by construction uncorrelated with the unobserved supply-side shocks in the error term, satisfying the exclusion restriction. The formal derivation of this mechanism is detailed in Appendix B.1.

Validity of the Instrument. The demand-driven nature of the instrument implies that it should be uncorrelated with the supply-side shocks in the error term. Nevertheless, one specific scenario could induce such a correlation: the exclusion restriction would be violated if supply-side shocks during the post-event period were correlated with $Share^{2018}_{JPN\,KOR,k}$. For instance, the Korean government might have subsidized sectors with a high Japanese import share to boost domestic production, in response to Japan's 2019 announcement of export controls. This would lead to an overestimate of the strength of scale economies.

[27] Fact 2 in Section 3 rationalizes this choice: the import decline was similarly large for targeted and non-targeted goods. This suggests a broad-based increase in the anticipated cost of sourcing from Japan, which $\tau_{JPN\,KOR,k}$ captures. Importantly, this interpretation does not affect the estimation, which identifies the scale elasticity from the response of prices to an exogenous demand shift, whatever its source.

To address this concern, I examine the Korean government's actual policy responses to the 2019 export controls. Immediately after Japan's announcement, the government provided support packages for domestic importers of Japanese goods. These packages focused on two main strategies: first, expediting imports from Japan before export controls could be fully enforced; and second, helping domestic firms shift their sources from Japan to alternative foreign suppliers. These immediate 2019 measures were therefore not supply-side shocks, but responses to import-side challenges. The government also announced longer-term plans to promote domestic production of key materials and components, but the goods these plans covered did not coincide precisely with the *Japan-dominant* goods underlying the instrument.[28] Moreover, their realization was uncertain, given the implementation lags inherent in such policies and further disruptions from the COVID-19 pandemic and the change in administration.[29] None of these policy responses was thus systematically correlated with the instrument.

As a further safeguard, any such sector-specific subsidies would be absorbed by the origin-sector fixed effects ($\alpha_{KOR,K(k)}$) in specification (9).[30] These fixed effects are defined at the 5-digit level, which is more disaggregated than the 2-digit and 3-digit industry levels at which industrial support policies are typically administered, so that they fully absorb any such sector-specific support. Consistent with this, the estimate of γ remains stable when the fixed effects are instead defined at the more aggregated 2-digit and 3-digit levels, as confirmed by the robustness checks below. Finally, the direction of the estimation bias is itself reassuring: the OLS estimate is smaller than the IV estimate, a downward bias—the opposite of the upward bias that such subsidies, a positive supply-side shock, would produce, as discussed in the Results below.

Results. Table 3 reports the estimation results for equation (9). Column (1) presents the Two-Stage Least Squares (2SLS) estimate, where the log change in total sales ($\Delta \ln Y_{KOR,k}$) is instrumented by the product of the trade elasticity and Japan's 2018 import share ($\theta_k Share^{2018}_{JPN\,KOR,k}$). The 2SLS estimate for the scale elasticity (γ) is 0.164 and is statistically significant. This result provides strong evidence for scale economies in Korea, implying

[28]These plans covered only about 100 goods, far fewer than the 659 goods for which Japan was the leading supplier to Korea in 2018 (see Appendix A.3). They are thus too narrow to account for the broad-based expansion in production and exports across these goods, as documented in Section 3.

[29]Government subsidies must pass through several administrative stages—budget drafting, parliamentary approval, allocation in the following fiscal year, and a further screening by industry associations—before reaching firms. Each stage adds delay and uncertainty that can erode the original policy intent. For example, the U.S. CHIPS program, first proposed in 2020, finalized its major awards to firms only at the end of 2024, with actual disbursements following even later.

[30]k refers to a good at the HS-6 level, while K refers to the more aggregated 5-digit KSIC sector to which good k belongs.

Table 3: Estimates of Scale Elasticity (HS-6)

	$\frac{1}{\theta_k}\Delta \ln X_{KOR\,j,k}$		
	2SLS	OLS	Reduced-form
	(1)	(2)	(3)
$\Delta \ln Y_{KOR,k}$	0.164^{***}	0.104^{***}	
	(0.049)	(0.002)	
$\theta_k Share^{2018}_{JPN\,KOR,k}$			0.006^{***}
			(0.002)
First-stage Coefficient	0.036		
First-stage F-statistic	422.47		
Destination Fixed Effects	Yes	Yes	Yes
Sector (KSIC-5) Fixed Effects	Yes	Yes	Yes
Number of Goods	4,634	4,634	4,634
Observations	220,763	220,763	220,763

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Table 3 reports the estimation results for the scale elasticity (γ) at the HS-6 level using specification (9). Column (1) presents the main 2SLS estimate, with its corresponding first-stage coefficient and F-statistic reported below. For this 2SLS estimation, the log difference in total sales ($\Delta \ln Y_{KOR,k}$) is instrumented by the product of the trade elasticity and Japan's 2018 import share in Korea ($\theta_k Share^{2018}_{JPN\,KOR,k}$). Columns (2) and (3) show the OLS and reduced-form estimates, respectively. The independent variable ($\Delta \ln Y_{KOR,k}$) is proxied by the log difference in total exports. The trade elasticity (θ_k) is taken from Giri et al. (2021) following Bartelme et al. (2025), assuming a common elasticity for all HS-6 goods within the same KSIC-2 sector. The analysis focuses on manufacturing sectors, which correspond to KSIC-2 codes 10 to 33.

Source: CEPII BACI (Gaulier and Zignago (2010)) and Statistics Korea.

that a 1% increase in a sector's total sales increases its productivity by 0.164% on average. The result also supports the model's key mechanism—a demand-driven increase in sales enhancing productivity, which in turn lowers price and boosts exports—as documented in Section 3. The first-stage result confirms the relevance of the instrument, showing a strong positive correlation with the endogenous variable.

The OLS estimate for the scale elasticity is 0.104 (Column (2) of Table 3), substantially smaller than the corresponding 2SLS estimate of 0.164. This downward bias suggests that the change in total sales ($\Delta \ln Y_{KOR,k}$) is negatively correlated with unobserved supply-side shocks ($\varepsilon_{KOR\,j,k}$). Such a bias is expected if sectors experiencing an increase in sales also faced negative supply-side shocks, such as rising production costs. Crucially, this finding alleviates the main threat to the exclusion restriction: if the dominant unobserved shock had been sector-specific government subsidies (a positive supply-side shock), we would expect an upward bias, not the downward bias observed in the data.

Column (3) of Table 3 presents the reduced-form estimate, showing the direct effect of the instrumental variable on exports. The coefficient is positive and significant, indicating that sectors with greater exposure to the shock (i.e., a higher instrument value) experienced greater subsequent export growth. The result is consistent with Fact 5 in Section 3 and provides direct, reduced-form evidence for the central mechanism in this paper: that the export-control-induced demand shock expands Korea's exports through scale economies. This finding also serves as a clear demonstration of the home-market effect.

Comparison with the Literature. The estimate of the scale elasticity ($\gamma = 0.164$) is consistent with the findings of Bartelme et al. (2025). Although this study adopts their theoretical framework, the empirical strategy differs: Bartelme et al. (2025) estimate sector-specific elasticities (γ_k) using long-run demand variation, whereas this study estimates a single, common elasticity (γ) using a short-run trade policy shock. Despite these differences, the results are consistent: their preferred IV estimates of γ_k range from 0.11 to 0.31 across sectors, and the single common elasticity of 0.164 estimated in this paper falls well within this range. More precisely, weighting the sector-level estimates of Bartelme et al. (2025) by Korea's pre-event exports implies a value of 0.226 for the common elasticity, which lies within the 95% confidence interval of the estimate of 0.164 (standard error 0.049). Similarly, for the cross-country analysis in Section 5.2, the corresponding weighted average is 0.214, closely matching the estimate of 0.206 from that analysis.[31]

[31]The weighted averages are computed from the sector-level estimates listed in Appendix E.3, using pre-event (2015–2018) export values as weights over the estimation sample. For the cross-country analysis,

The findings in this paper are also consistent with those of Breinlich et al. (2026), which uses a theoretically similar model to study a different event: the permanent normalization of US trade relations with China. A key difference, however, is that their specification estimates the output elasticity—the product of the trade and scale elasticities—while the baseline in this paper estimates the scale elasticity (γ) directly. To make a direct comparison, I re-estimate the main specification (9) with the dependent variable unadjusted by the trade elasticity. This exercise yields an output elasticity of 0.806, which is consistent with both their structural estimate of 0.74 and the calibrated value of 0.821 used in their quantitative analysis.[32] The detailed results of this estimation are presented in Appendix B.2.

This close alignment with the findings of two recent studies, given the different identification strategies and events analyzed, strengthens the external validity of the estimated scale elasticity.

Robustness. To check the robustness of the main estimate in Table 3, the analysis is re-estimated at a more aggregated KSIC-5 sector level, where domestic sales data are available.[33] This allows the endogenous variable, $\Delta \ln Y_{KOR,k}$, to be measured as actual total sales, rather than being proxied by total exports as in the main analysis. The specification is:

$$\frac{1}{\theta_k} \Delta \ln X_{KOR\,j,k} = \gamma \Delta \ln Y_{KOR,k} + \alpha_{KOR} + \alpha_{KOR\,j} + \delta' \Delta \ln \mathbf{Z}_{KOR,k} + \varepsilon_{KOR\,j,k}. \tag{10}$$

In the main analysis, the origin-sector fixed effects ($\alpha_{KOR,K(k)}$) are defined at the 5-digit (KSIC-5) sector level and absorb supply-side confounders such as productivity. Since the endogenous variable, $\Delta \ln Y_{KOR,k}$, now varies at the same 5-digit level, these fixed effects would be perfectly collinear with it. They are therefore replaced by a vector of sector-level controls, $\Delta \ln \mathbf{Z}_{KOR,k}$, that proxy for the changes in the unobserved productivity ($\Delta \ln A_{KOR,k}$): the log changes in the average wage per worker, machinery and equipment, tangible fixed assets, and investment.[34] All other terms are defined as in the main analysis but applied at the 5-digit level.

exports are pooled across all origin countries.

[32]The structural estimate is from Column (c) of Table 5 in Breinlich et al. (2026).

[33]Total sales ($Y_{KOR,k}$) are taken from the Korean Mining and Manufacturing Survey by Statistics Korea, and exports ($X_{KOR\,j,k}$) are aggregated from the HS-6 level to the corresponding KSIC-5 level using a concordance provided by Statistics Korea.

[34]These input measures are recorded in the Korean Mining and Manufacturing Survey, at the KSIC-5 level, but not in the HS-6 trade data; the main analysis therefore relies on the origin-sector fixed effects instead.

Table 4: Estimates of Scale Elasticity (KSIC-5)

	$\frac{1}{\theta_k}\Delta \ln X_{KOR\,j,k}$		
	2SLS (1)	OLS (2)	Reduced-form (3)
$\Delta \ln Y_{KOR,k}$	0.166^{***} (0.061)	0.078^{***} (0.009)	
$\theta_k Share^{2018}_{JPN\,KOR,k}$			0.011^{***} (0.004)
First-stage Coefficient	0.064		
First-stage F-statistic	1130.58		
Destination Fixed Effects	Yes	Yes	Yes
Sector-level Controls	Yes	Yes	Yes
Number of Sectors	393	393	393
Observations	46,538	46,538	46,538

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Table 4 reports the estimation results for the scale elasticity (γ) at the KSIC-5 level using specification (10). Column (1) presents the 2SLS estimate, with its corresponding first-stage coefficient and F-statistic reported below. For this 2SLS estimation, the log difference in total sales ($\Delta \ln Y_{KOR,k}$) is instrumented by the product of the trade elasticity and Japan's 2018 import share in Korea ($\theta_k Share^{2018}_{JPN\,KOR,k}$). Columns (2) and (3) show the OLS and reduced-form estimates, respectively. Total sales ($Y_{KOR,k}$) are taken from the Korean Mining and Manufacturing Survey by Statistics Korea, and the corresponding exports ($X_{KOR\,j,k}$) are aggregated from the HS-6 level to the KSIC-5 level using a concordance provided by Statistics Korea. The sector-level controls are the log changes in the average wage per worker, machinery and equipment, tangible fixed assets, and investment. The trade elasticity (θ_k) is taken from Giri et al. (2021) following Bartelme et al. (2025), assuming a common elasticity for all KSIC-5 sectors within the same KSIC-2 sector. The analysis focuses on manufacturing sectors, which correspond to KSIC-2 codes 10 to 33. Source: CEPII BACI (Gaulier and Zignago (2010)) and Statistics Korea.

The estimation results for this robustness check, presented in Table 4, are highly consistent with the baseline findings. The 2SLS estimate for the scale elasticity (γ) is 0.166 (Column (1)), nearly identical to the baseline estimate of 0.164. This confirms that the main finding is robust to both the change in aggregation level and the inclusion of domestic sales in the measure of total sales.[35] Furthermore, other key patterns from the main analysis also hold. The OLS estimate in Column (2) is 0.078, again substantially smaller than the 2SLS estimate of 0.166. This downward bias alleviates the main threat to the exclusion restriction, as in the main analysis. The reduced-form estimate in Column (3) is also positive and significant, aligning with the main analysis.

A further robustness check tests whether the main estimate is sensitive to the aggregation level of the origin-sector fixed effects ($\alpha_{KOR,K(k)}$). While the baseline specification uses the 5-digit level for K, I re-estimate equation (9) using more aggregated 2-digit and 3-digit levels. The results, presented in Table A12 and Table A13, confirm that the estimate of the scale elasticity (γ) remains positive, significant, and of a similar magnitude across all specifications. This demonstrates that the main finding is not driven by the specific choice of the fixed effect level, further strengthening the robustness of the results.

5.2 Analysis 2: Korea's Major Import Sources

The second analysis extends the estimation to Korea's major import sources, namely the 20 largest sources of its imports other than Japan.[36] In response to the decline in imports from Japan, these countries could have expanded their sales to the Korean market. Such a demand-driven increase in their total sales, if scale economies are present, would in turn enhance their productivity and boost their exports to all other destinations. Estimating the scale elasticity across these countries thus provides a cross-country test for the presence of scale economies.

Specification. The estimation specification for the cross-country analysis is adapted from the baseline equation (8) for the case of multiple origins. It again imposes a com-

[35] This near-equality is what the framework of Bartelme et al. (2025) implies: the scale elasticity can be recovered from any subset of an origin's destination sales, so estimating it from exports alone, excluding domestic sales, yields a similar estimate.

[36] These countries account for approximately 83% of Korea's imports from all sources other than Japan in 2018. The remaining origins each have a small import share and correspondingly limited exposure to the Korean market. The results are robust to alternative selection criteria, as shown in the robustness checks below.

mon scale elasticity ($\gamma_k = \gamma$), pooling the variation across sectors for identification:

$$\frac{1}{\theta_k}\Delta \ln X_{ij,k} = \gamma \Delta \ln Y_{i,k} + \alpha_i + \alpha_{ij} + \alpha_{j,k} + \alpha_{i,K(k)} + \varepsilon_{ij,k}. \tag{11}$$

Its key difference from the Korea-only specification (9) is the inclusion of the destination-sector fixed effects ($\alpha_{j,k}$). With multiple origins in the sample, each destination-sector cell contains multiple observations, so these fixed effects can now be included, providing more complete control of demand-side factors.

The main variables are defined consistently with the Korea-only analysis. The estimation again uses a long difference between the pre-event period (2015–2018) and the post-event period (2020–2023), excluding the event year 2019. The dependent variable, $\frac{1}{\theta_k}\Delta \ln X_{ij,k}$, is the log difference in origin i's exports of good k to destination j between these two periods, adjusted by the trade elasticity (θ_k). The trade elasticity is taken from Bartelme et al. (2025), who in turn take the values from Giri et al. (2021). The independent variable, $\Delta \ln Y_{i,k}$, is the log difference in origin i's total sales in good k over the same periods, proxied by total exports owing to the unavailability of domestic sales data at the HS-6 level, as in the Korea-only analysis.

The coefficient of interest is again the scale elasticity (γ). The remaining terms in specification (11) are a set of fixed effects that control for other sources of variation, with each effect corresponding to a theoretical term in equation (7). The origin fixed effects (α_i) absorb aggregate shocks to each origin that are common across all destinations and goods. The origin-destination fixed effects (α_{ij}) absorb shocks specific to a given trade partnership. The destination-sector fixed effects ($\alpha_{j,k}$) absorb demand shifts within each destination and sector. The origin-sector fixed effects ($\alpha_{i,K(k)}$) absorb supply-side confounders common within each 5-digit sector K.

A key identification challenge is that the error term ($\varepsilon_{ij,k}$) in specification (11) contains shocks from the theoretical model (equation (7)) that are not fully absorbed by the fixed effects, including shocks to fundamental productivity ($\Delta \ln A_{i,k}$) and bilateral trade costs ($\Delta \ln \tau_{ij,k}$). The independent variable, the change in total sales ($\Delta \ln Y_{i,k}$), is likely to respond endogenously to these shocks, implying $E[\Delta \ln Y_{i,k} \times \varepsilon_{ij,k}] \neq 0$. Therefore, an instrumental variable that isolates exogenous demand-side variation is required to obtain an unbiased estimate of the scale elasticity, γ.

Instrumental Variable. To address this endogeneity, the estimation requires an instrumental variable that is correlated with $\Delta \ln Y_{i,k}$ but uncorrelated with the error term ($\varepsilon_{ij,k}$), in particular the unobserved supply-side shocks. I therefore construct an instrument that captures the exogenous demand shock from the 2019 Korea-Japan Trade Dispute, specific to each origin i and good k. The instrument is the product of the trade elasticity (θ_k) and two pre-event (2018) shares: Japan's import share in Korea and origin i's export share to Korea. Formally, it is defined as:

$$IV_{i,k} = \theta_k \, Share^{2018}_{JPN\,KOR,k} \, EXShare^{2018}_{i\,KOR,k},$$

where $Share^{2018}_{JPN\,KOR,k} = \frac{X_{JPN\,KOR,k}}{\sum_{i \neq KOR} X_{i\,KOR,k}}$ and $EXShare^{2018}_{i\,KOR,k} = \frac{X_{i\,KOR,k}}{\sum_{j \neq i} X_{ij,k}}$.

This instrumental variable is designed to capture the (i,k)-specific exposure to the demand shock in Korea through its two share components. First, Japan's import share in Korea ($Share^{2018}_{JPN\,KOR,k}$) proxies for the magnitude of the shock in the Korean market. Second, origin i's export share to Korea ($EXShare^{2018}_{i\,KOR,k}$) proxies for the origin's exposure to that shock. The intuition is that the positive impact on an origin's sales is greater when it both supplies a large share of its exports to Korea (a high $EXShare^{2018}_{i\,KOR,k}$) and faces a larger demand shock there (a high $Share^{2018}_{JPN\,KOR,k}$).

The structural model in Section 4 also provides a theoretical justification for this instrumental variable. Within this framework, the trade dispute is represented as an exogenous increase in the bilateral trade cost ($\tau_{JPN\,KOR,k}$). This shock propagates purely through a demand-side channel: it raises the price of Japanese goods ($p_{JPN\,KOR,k}$), which increases Korea's price index for good k ($P_{KOR,k}$) and shifts expenditure toward goods from other countries ($X_{i\,KOR,k}$), ultimately increasing those countries' total sales ($Y_{i,k}$). This mechanism establishes the theoretical relevance of the instrument. Furthermore, because this channel is driven by a demand shock external to origin i, the instrument is by construction uncorrelated with origin i's own supply-side shocks, satisfying the exclusion restriction. The formal derivation of this mechanism is detailed in Appendix C.1.

Validity of the Instrument. The demand-driven nature of the instrument implies that it should be uncorrelated with origin i's unobserved supply-side shocks. A threat would arise if foreign governments subsidized producers in sectors exposed to the increased Korean demand, biasing the scale elasticity upward. The structure of the instrument, however, makes such a threat narrow: because the shock originates in Korea and is external to origin i, the instrument can correlate with origin i's supply-side shocks only through its export share to Korea. This would require numerous governments to target subsidies at precisely the sectors most exposed to the Korean market—an implausible basis for industrial policy, which is designed around domestic priorities. As a further safeguard, any such subsidies would be absorbed by the origin-sector fixed effects ($\alpha_{i,K(k)}$), defined at a more disaggregated level (5-digit) than the 2-digit and 3-digit levels at which industrial policy is typically administered. Consistent with this, γ remains stable across the more aggregated fixed-effect definitions, as shown in the robustness checks below. Finally, the estimation bias is reassuring: the OLS estimate lies below the IV estimate, the opposite of the upward bias such subsidies would produce.

Results. Table 5 reports the estimation results for equation (11). Column (1) presents the Two-Stage Least Squares (2SLS) estimate, where the log change in total sales ($\Delta \ln Y_{i,k}$) is instrumented by the product of the trade elasticity, Japan's 2018 import share, and the origin's export share to Korea ($\theta_k Share^{2018}_{JPN\,KOR,k} EXShare^{2018}_{i\,KOR,k}$). The 2SLS estimate for the scale elasticity (γ) is 0.206 and is statistically significant, providing evidence that scale economies are also present in the cross-country sample and implying that a 1% increase in a sector's total sales increases its productivity by 0.206% on average. This estimate is consistent with the baseline of 0.164 from the Korea-only analysis, and this agreement across two analyses with different samples and specifications reinforces the robustness of the finding.[37] It also supports the model's key mechanism—a demand-driven increase in sales enhancing productivity, which in turn lowers price and boosts exports. The first-stage result confirms the relevance of the instrument, showing a positive correlation with the endogenous variable.

The OLS estimate in Column (2) is smaller than the 2SLS estimate. This downward bias suggests that the change in total sales ($\Delta \ln Y_{i,k}$) is negatively correlated with unobserved supply-side shocks ($\varepsilon_{ij,k}$). Such a bias is expected if high-growth sectors also faced negative supply-side shocks, such as rising production costs. In line with the Korea-only analysis, this bias alleviates the main threat to the exclusion restriction: if the dominant unobserved shock had been sector-specific subsidies (a positive supply-side shock), we would expect an upward bias, not the downward bias observed in the data. Column (3) reports the reduced-form estimate, showing the direct effect of the instrument on exports ($\Delta \ln X_{ij,k}$). The coefficient is positive and significant, providing direct evidence for the main mechanism: export growth was greater for origin-good pairs with a higher pre-event export share to Korea (a high $EXShare^{2018}_{i\,KOR,k}$) and a larger demand shock in the Korean market (a high $Share^{2018}_{JPN\,KOR,k}$).

[37] The cross-country estimate of 0.206 lies within the 95% confidence interval of the Korea-only estimate of 0.164 (standard error 0.049).

Table 5: Cross-Country Estimates of Scale Elasticity (HS-6)

	$\frac{1}{\theta_k}\Delta \ln X_{ij,k}$		
	2SLS	OLS	Reduced-form
	(1)	(2)	(3)
$\Delta \ln Y_{i,k}$	0.206^{***}	0.101^{***}	
	(0.049)	(0.001)	
$\theta_k Share^{2018}_{JPN\,KOR,k} EXShare^{2018}_{i\,KOR,k}$			0.037^{***}
			(0.008)
First-stage Coefficient	0.180		
First-stage F-statistic	12.03		
Origin Fixed Effects	Yes	Yes	Yes
Origin-Destination Fixed Effects	Yes	Yes	Yes
Destination-Good Fixed Effects	Yes	Yes	Yes
Origin-Sector (KSIC-5) Fixed Effects	Yes	Yes	Yes
Number of Goods	4,667	4,667	4,667
Number of Origins	20	20	20
Observations	4,479,035	4,479,035	4,479,035

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Table 5 reports the cross-country estimation results for the scale elasticity (γ) at the HS-6 level using specification (11). Column (1) presents the main 2SLS estimate, with its corresponding first-stage coefficient and F-statistic reported below. For this 2SLS estimation, the log difference in total sales ($\Delta \ln Y_{i,k}$) is instrumented by the product of the trade elasticity, Japan's 2018 import share in Korea, and origin i's 2018 export share to Korea ($\theta_k Share^{2018}_{JPN\,KOR,k} EXShare^{2018}_{i\,KOR,k}$). Columns (2) and (3) show the OLS and reduced-form estimates, respectively. The independent variable ($\Delta \ln Y_{i,k}$) is proxied by the log difference in total exports. The trade elasticity (θ_k) is taken from Giri et al. (2021) following Bartelme et al. (2025), assuming a common elasticity for all HS-6 goods within the same KSIC-2 sector. The analysis focuses on manufacturing sectors, which correspond to KSIC-2 codes 10 to 33, and on Korea's 20 largest import sources other than Japan. Standard errors are clustered at the origin-good level.
Source: CEPII BACI (Gaulier and Zignago (2010)) and Statistics Korea.

Robustness. To test the robustness of the main estimate in Table 5, I re-examine specification (11) by varying the level of aggregation for K in the origin-sector fixed effects ($\alpha_{i,K(k)}$), following the same approach as in Section 5.1. While the baseline uses the 5-digit level for K, this exercise re-estimates the specification using more aggregated 2-digit and 3-digit levels. The results, presented in Table A14 (2-digit) and Table A15 (3-digit), are highly consistent with the baseline 5-digit estimates in Table 5: the 2SLS, OLS, and reduced-form estimates all remain stable in sign, significance, and magnitude. This confirms that the main finding is not sensitive to the aggregation level of the origin-sector fixed effects.

A further robustness check examines the sensitivity of the estimate to the composition of origin countries. Restricting the sample to Korea's 10 largest import sources yields a scale elasticity of 0.181, as presented in Table A16. Restricting it to the four major suppliers to Korea that compete with Japan—China, the United States, Germany, and Taiwan—yields 0.172, as presented in Table A17.[38] Both estimates remain close to the cross-country baseline of 0.206, showing that the result does not depend on which countries are included. Moreover, they are close to the Korea-only estimate of 0.164, despite the different sample and specification. Overall, the estimate of the scale elasticity is robust to the aggregation level of the fixed effects and to the composition of the country sample; in every case it falls within the range of sector-level elasticities reported by Bartelme et al. (2025).

6 Quantitative Analysis

This section quantifies the welfare effects of the 2019 Korea-Japan Trade Dispute, using the structural model from Section 4 and scale elasticities consistent with the estimates from Section 5. To understand the extent to which scale economies shape these welfare effects, the exercise compares the outcomes across two scenarios: (i) one without scale economies ($\gamma_k = 0$) and (ii) one with scale economies ($\gamma_k > 0$). The results show that the presence of scale economies mitigates the negative welfare impact of the export controls in the target country (Korea), while exacerbating it in the imposing country (Japan).

[38] These four are drawn from Korea's major import sources and selected for their export similarity with Japan, based on pre-event (2018) trade.

6.1 Solving for Equilibrium Changes

To quantify the general equilibrium effects of the trade policy shock, the model is solved in proportional changes using exact hat algebra, as in Dekle et al. (2008). This approach defines the proportional change in any variable x as the ratio of its counterfactual value (x') to its initial value (x), denoted as $\hat{x} = x'/x$.

Applying the hat algebra approach to the market clearing condition (5) yields the central equation for the change in each country's wage ($\hat{w}_i$). This equation depends on the changes in expenditure shares ($\hat{\lambda}_{ij,k}$), which in turn depend on the changes in sectoral labor allocation ($\hat{L}_{i,k}$); the labor allocation itself depends on wages and expenditure shares. The equilibrium is therefore determined by simultaneously solving the following system of three equations:

$$\hat{w}_i = \frac{1}{Y_i} \sum_k \sum_j \lambda_{ij,k} X_{j,k} \hat{\lambda}_{ij,k} \hat{w}_j, \tag{12}$$

$$\hat{\lambda}_{ij,k} = \frac{\hat{\tau}_{ij,k}^{-\theta_k} \hat{L}_{i,k}^{\gamma_k \theta_k} \hat{w}_i^{-\theta_k}}{\sum_\ell \lambda_{\ell j,k} \hat{\tau}_{\ell j,k}^{-\theta_k} \hat{L}_{\ell,k}^{\gamma_k \theta_k} \hat{w}_\ell^{-\theta_k}}, \tag{13}$$

$$\hat{L}_{i,k} = \frac{1}{Y_{i,k} \hat{w}_i} \sum_j X_{ij,k} \hat{\lambda}_{ij,k} \hat{w}_j, \tag{14}$$

where $\lambda_{ij,k}$ is the initial share of origin i in destination j's expenditure on goods from sector k, $\lambda_{ij,k} = X_{ij,k}/X_{j,k}$. The detailed derivation of this system is provided in Appendix D.

This system of equations can be solved numerically for the changes in the endogenous variables ($\hat{w}_i$, $\hat{\lambda}_{ij,k}$, $\hat{L}_{i,k}$) given three inputs: data from the initial equilibrium (Y_i, $\lambda_{ij,k}$, $X_{j,k}$, $Y_{i,k}$, $X_{ij,k}$), the model's parameters (γ_k, θ_k), and an exogenous shock to trade costs ($\hat{\tau}_{ij,k}$).

6.2 Model Calibration

Data and Initial Equilibrium. The initial equilibrium for the quantitative analysis is constructed to be consistent with the preceding estimation in Section 5. The analysis uses bilateral trade values ($X_{ij,k}$) at the HS-6 product level from the CEPII BACI (Gaulier and Zignago (2010)) for the pre-event period (2015–2018). All other variables required for the simulation are constructed from these bilateral trade values, summed over the pre-event period. These include total output (Y_i), total sectoral output ($Y_{i,k}$), total sectoral expenditure ($X_{j,k}$), and expenditure shares ($\lambda_{ij,k}$).

A key challenge is that the CEPII BACI dataset lacks information on domestic sales ($X_{ii,k}$). This variable is therefore imputed using a proportionality assumption, which posits that the domestic sales share of a 6-digit sector is identical to that of its corresponding aggregate 2-digit sector. The aggregate sector-level data required to calculate these shares are sourced from the International Trade and Production Database for Simulation (Borchert et al. (2024)). The detailed procedure for this imputation is provided in Appendix E.1.

The quantitative analysis focuses on manufacturing sectors, consistent with the empirical analysis in Section 3 and the estimation of scale elasticity in Section 5.[39] The country sample consists of Korea, Japan, and the 20 major import sources from Section 5.2. For computational tractability, all remaining countries are consolidated into a single "Rest of the World" (RoW), yielding 23 country units. The full list of countries included in the analysis is provided in Appendix E.2.

Key Parameters. The scale elasticity, γ_k, is calibrated in two ways: the baseline uses the sector-level estimates of Bartelme et al. (2025), and the alternative calibrations use the estimates from Section 5. Because the pooled estimates from Section 5 measure an average of the sector-level elasticities, imposing them in a sector with high trade elasticity (θ_k) would violate the condition $\gamma_k \theta_k < 1$ that ensures a unique counterfactual equilibrium (Kucheryavyy et al. (2023)). Therefore, the baseline adopts the sector-level estimates of Bartelme et al. (2025), which satisfy the condition in every sector, while the alternative calibrations retain the pooled estimates from Section 5, conservatively setting the scale elasticity to zero in the sectors that violate it. The two sets of estimates thus capture the same scale economies at different levels of aggregation, and their mutual consistency is shown in Section 5. The full list of the sector-level scale elasticities is provided in Appendix E.3.

Another key parameter is the trade elasticity, θ_k. To maintain consistency with the preceding estimation section, the values are taken from Bartelme et al. (2025), who in turn take them from Giri et al. (2021). These estimates are available at the SIC-2 level; they are assigned to KSIC-2 sectors, and all HS-6 goods within the same KSIC-2 sector are therefore assumed to share the same trade elasticity. The full list of the sector-level trade elasticities is provided in Appendix E.3.

[39] Both the scale elasticity estimated in Section 5 and the sector-level elasticities used in the calibration are identified on manufacturing sectors; the quantitative analysis thus applies these parameters within the domain where they are identified. The dispute itself was also concentrated in manufacturing, as documented in Section 3.

Shock Calibration. The final key input for the quantitative analysis is the shock from the 2019 Korea-Japan Trade Dispute. This analysis interprets the shock as an increase in the bilateral trade cost on Korea's imports from Japan ($\hat{\tau}_{JPN\,KOR,k}$). This interpretation is grounded in the empirical findings of Section 3. Although the controls were never enforced in practice, the announcement itself led to a substantial decline in Korea's imports from Japan (Fact 1). Moreover, the size of the decline was governed not by whether a good was targeted, but by its dependence on Japan (Facts 2 and 3), indicating that firms perceived a broad-based increase in the cost of sourcing from Japan that was larger the greater their dependence.[40]

Accordingly, the shock is applied to all sectors, not only to the goods directly targeted by the announcement.[41] Since this shock is unobserved, it is calibrated by inverting the model in Section 4. Specifically, $\hat{\tau}_{JPN\,KOR,k}$ is calibrated to a value that makes the model's predicted change in imports exactly match the change observed in the data ($\hat{X}^{obs}_{JPN\,KOR,k}$); the calibrated shock thus measures the cost-equivalent of the threat, revealed by firms' observed import responses. This calibration is performed for each sector k using the following equation:

$$\log \hat{\tau}_{JPN\,KOR,k} = \frac{1}{-\theta_k(1-\lambda_{JPN\,KOR,k})} \log \hat{X}^{obs}_{JPN\,KOR,k}. \tag{15}$$

This equation is derived from the model and captures the total effect of the shock on Korea's imports from Japan. This total effect is composed of a partial equilibrium effect and a first-order general equilibrium effect that incorporates the shock's impact on Korea's sectoral price index ($P_{KOR,k}$). The details are provided in Appendix E.4.

The inputs for this equation are sourced as follows. The observed change in imports ($\hat{X}^{obs}_{JPN\,KOR,k}$) is calculated directly from pre-shock (2015–2018) and post-shock (2020–2023) data; this is the sole instance where post-shock data enters the calibration.[42] The trade elasticity (θ_k) is as previously described, and the initial shares ($\lambda_{JPN\,KOR,k}$) are constructed from the pre-shock data.

The role of scale economies in the welfare results—mitigating the impact on Korea while exacerbating it for Japan—is robust to an alternative interpretation of the shock as

[40]Firms more reliant on Japan faced larger potential costs should the controls be enforced, prompting larger precautionary adjustments, such as diversifying suppliers and accumulating inventories.

[41]To isolate the effect of the 2019 Korea-Japan Trade Dispute, the counterfactual exercise assumes that all other trade costs remain unchanged ($\hat{\tau}_{ij,k} = 1$ for all $(i,j) \neq (JPN, KOR)$).

[42]Two adjustments are made to ensure the calibrated shocks are economically plausible. First, for any sectors where imports increased, which would imply a trade cost reduction, the calibrated shock is floored at one. Second, to prevent a large decrease in imports from generating an implausibly large shock, its distribution is winsorized at the 95th percentile.

Table 6: Model Predictions for Untargeted Moments

	(1) $\hat{Y}_{KOR,k}$	(2) $\hat{p}_{KOR\,j,k}$	(3) $\hat{q}_{KOR\,j,k}$
A. Scale Economies ($\gamma_k > 0$)			
Dominant Sectors	1.226	0.978	1.240
Non-Dominant Sectors	0.994	1.004	0.982
B. No Scale Economies ($\gamma_k = 0$)			
Dominant Sectors	1.163	1.002	0.996
Non-Dominant Sectors	0.999	1.002	0.981

Notes: Table 6 reports the quantitative model's predicted outcomes for the untargeted moments. Each cell reports the mean of the proportional changes ($\hat{x} = x'/x$) within each group; values above one indicate an increase. Column (1) shows the mean predicted change in total sales ($\hat{Y}_{KOR,k}$), which corresponds to the increase in Korean producers' output documented in Section 3. Columns (2) and (3) show the mean predicted changes in export prices ($\hat{p}_{KOR\,j,k}$) and quantities ($\hat{q}_{KOR\,j,k}$), which correspond to Fact 5 in Section 3. Panel A presents the results from the scenario with scale economies ($\gamma_k > 0$), using the same calibration as the baseline in Table 7. Panel B presents the results from the scenario without scale economies ($\gamma_k = 0$). *Japan-dominant* (Dominant) sectors are defined as those for which Japan was the leading supplier to Korea during the pre-event period (2015–2018). Non-Dominant sectors are all remaining sectors.

a shift in preferences away from Japanese goods, as formalized in Appendix E.5.[43]

6.3 Model Validation

Prior to quantifying the welfare changes, this subsection tests whether the model can replicate key empirical findings that are not targeted in calibration. The preceding shock calibration is designed to match a targeted moment: the observed decline in Korea's imports from Japan (Fact 3 in Section 3). Therefore, a validation test of the model lies in its ability to endogenously generate the untargeted moments documented in Section 3: the increase in Korean producers' output and the simultaneous decrease in export prices and increase in export quantities (Fact 5). Furthermore, to isolate the role of scale economies as the key mechanism, the exercise compares the predictions from a scenario with scale economies ($\gamma_k > 0$) to those from one where this channel is shut down ($\gamma_k = 0$).

The validation exercise confirms that the model replicates the untargeted moments, as shown in Panel A of Table 6. First, Column (1) shows that total sales ($\hat{Y}_{KOR,k}$) in

[43] In the Armington framework, an increase in the bilateral trade cost and a decline in the taste for Japanese goods are observationally equivalent in trade data. All equilibrium responses are therefore identical across the two interpretations; only the level of Korea's welfare change differs.

Japan-dominant sectors increase by 22.6%, while sales in *Non-dominant* sectors remain almost unchanged, consistent with the increase in Korean producers' output documented in Section 3. More importantly, the model also generates Fact 5. Column (2) shows that export prices ($\hat{p}_{KOR\,j,k}$) in *Japan-dominant* sectors decrease by 2.2%, while prices in *Non-dominant* sectors remain almost unchanged. In terms of export quantities ($\hat{q}_{KOR\,j,k}$), Column (3) shows that *Japan-dominant* sectors experience a 24.0% increase, while *Non-dominant* sectors experience a 1.8% decrease. The differences between the two groups in these three untargeted moments are also statistically significant, as confirmed by t-tests (see Appendix F).

By contrast, Panel B of Table 6 shows that the scenario without scale economies ($\gamma_k = 0$) fails this validation test. While this scenario still generates a relative increase in total sales, it cannot generate the export response observed in Fact 5. Specifically, in the absence of scale economies, the model predicts that, in *Japan-dominant* sectors, export prices do not fall (Column (2)) and export quantities do not increase (Column (3)). This demonstrates that scale economies are indeed the key mechanism behind the full set of empirical findings in Section 3. The result also holds in an alternative test that correlates the model's predictions for the three untargeted moments ($\hat{Y}_{KOR,k}$, $\hat{p}_{KOR\,j,k}$, and $\hat{q}_{KOR\,j,k}$) with Japan's pre-event share in Korea; the full pattern is again found only in the scenario with scale economies ($\gamma_k > 0$). Appendix F provides the detailed analysis.

6.4 Welfare Changes

Building on the preceding validation, this subsection quantifies the welfare changes from the dispute. Table 7 reports the percentage change in welfare relative to the initial equilibrium, measured as the change in real income, $\hat{U}_j = \hat{w}_j / \hat{P}_j$.[44] This value is computed from the solved equilibrium changes for the endogenous variables; the full derivation is provided in Appendix G.

The first two columns of Table 7 enable a direct comparison that reveals the crucial role of scale economies in shaping the welfare outcomes. Column (1) presents the scenario without scale economies ($\gamma_k = 0$), and Column (2) presents the baseline, which uses the sector-level estimates of Bartelme et al. (2025). The presence of scale economies mitigates the welfare loss in Korea, while exacerbating it in Japan. Specifically, Korea's welfare loss is reduced from -0.282% to -0.226%, offsetting 20% of the decline in the absence of scale economies. Conversely, Japan's welfare loss increases from -0.051%

[44]This definition assumes constant total labor ($\hat{L}_j = 1$), so the change in nominal income is equal to the change in the wage ($\hat{w}_j$).

Table 7: Welfare Changes: With vs. Without Scale Economies

	No Scale	Scale Economies		
		Baseline		
	(1)	(2)	(3)	(4)
A. Welfare Changes (%)				
Korea	−0.282	−0.226	−0.252	−0.249
Japan	−0.051	−0.057	−0.053	−0.049
B. Scale Elasticity (γ_k)				
Value	0	0.110–0.310	0.164	0.206
Source	–	Bartelme et al. (2025)	Section 5.1	Section 5.2

Notes: Table 7 reports the welfare effects of the 2019 Korea-Japan Trade Dispute under the four scenarios. Panel A shows the percentage change in welfare for Korea and Japan, calculated as $(\hat{U}_j - 1) \times 100$, where $\hat{U}_j$ is the change in real income. Panel B specifies the scale elasticity (γ_k) used in each scenario, which is applied identically to all countries. Column (1) presents the scenario without scale economies ($\gamma_k = 0$). Column (2) is the baseline, which sets γ_k to the sector-level estimates of Bartelme et al. (2025), listed in Appendix E.3. Columns (3) and (4) apply the pooled estimates uniformly across sectors: 0.164 from Section 5.1 and 0.206 from Section 5.2. The scale elasticity is conservatively set to zero in the sectors where imposing these estimates would violate the condition $\gamma_k \theta_k < 1$, as described under Key Parameters in Section 6.2.

to −0.057%, representing an 11% amplification of the decline in the absence of scale economies.

Scale economies shape these welfare effects through prices and productivity rather than through wages. In Korea, the shock shifts expenditure toward domestic goods; the resulting expansion of domestic sectors raises productivity, and the induced price declines absorb part of the increase in Korea's price index. This price channel raises Korea's welfare nearly twice as much as the smaller wage gain lowers it, yielding the net mitigation in Column (2). In Japan, the contraction of its exposed sectors (i.e., those in which Japan had been the leading supplier to Korea) lowers productivity through the loss of scale, so its price index falls by less than it would without scale economies. Notably, Japan's wage falls by less under scale economies, yet its welfare loss deepens: the smaller price decline lowers Japan's welfare more than the smaller wage decline raises it, indicating that the scale channel operates through the price index rather than the labor market. The scale effects are also highly asymmetric: the effect on Korea is roughly ten times as large as that on Japan, as the adjustment to the shock is concentrated in Korea.

Columns (3) and (4) present the alternative calibrations, which apply the pooled estimates from Sections 5.1 and 5.2 uniformly across sectors and countries. Under the Korea-only estimate of 0.164 (Column (3)), the central finding—that the presence of scale economies mitigates the welfare loss in Korea while exacerbating it in Japan—remains intact. Under the cross-country estimate of 0.206 (Column (4)), the mitigating effect for Korea still holds, while Japan's welfare loss is slightly lessened compared to the No Scale scenario. This reversal is a consequence of imposing a high common elasticity on all sectors. With a common elasticity, labor reallocated from Japan's exposed sectors generates equally strong productivity gains in the non-exposed sectors, which comprise the large majority. These gains in turn offset the losses from the contracting sectors. The offset strengthens with the value of the common elasticity: 0.164 only dampens the exacerbation in Japan, while 0.206 overturns it. Under the sector-level baseline, the non-exposed sectors do not necessarily carry strong scale economies, and the reversal does not arise.

Finally, the role of scale economies documented in Table 7 is unchanged when the shock is interpreted as a shift in preferences away from Japanese goods; only the level of Korea's welfare change differs across the two interpretations. See Appendix E.5 for details.

7 Conclusion

Can export controls backfire by increasing productivity and exports in the target country? Does this response in turn exacerbate the welfare loss in the imposing country? I answer these questions in the context of the 2019 Korea-Japan Trade Dispute. Amid heightened political tensions, Japan removed Korea from its "White List" of preferential trade partners, placing all national strategic items exported to Korea under potential export controls. This confronted Korean firms with immediate uncertainty, as their imports of these items required individual approval that could be denied at the discretion of Japanese officials. Although no export restrictions were imposed in practice, the threat alone triggered substantial shifts in Korea's imports, exports, and domestic production.

The empirical analysis first documents a substantial decline in Korea's imports from Japan. The extent of this decline was similar between targeted and non-targeted goods. However, the decline was disproportionately larger in sectors where Japan had been the leading supplier to Korea. More surprisingly, Korea's export quantities in these same sectors expanded while export prices simultaneously fell. Domestic production in these sectors also expanded broadly with capital deepening. Combined with additional pieces of evidence, these patterns point to scale economies in production as the underlying mechanism.

Motivated by these empirical findings, I structurally estimate the strength of scale economies. To address the potential endogeneity of total sales, the estimation employs an instrumental variable strategy that isolates exogenous demand-side variation arising from the dispute. The results provide strong evidence for scale economies in Korea, indicating that a 1% increase in a sector's total sales raises its productivity by 0.164% on average. The quantitative analysis, using scale elasticities consistent with these estimates, reveals the crucial role of scale economies in shaping the welfare outcomes: their presence mitigates the welfare loss in Korea while exacerbating it in Japan. Quantitatively, this amounts to offsetting 20% of Korea's welfare loss and amplifying Japan's loss by 11%, relative to a scenario without scale economies.

This paper demonstrates how export controls can backfire: by increasing productivity in the target country through scale economies, such policies can lead to unintended and self-harming consequences for the imposing country. As export controls have emerged as a pivotal tool in geopolitics, the paper serves as a cautionary tale about the arbitrary use of trade policy for political ends.

References

Alessandria, George, Shafaat Khan, and Armen Khederlarian, "Taking Stock of Trade Policy Uncertainty: Evidence from China's Pre-WTO Accession," *Journal of International Economics*, 2024, *150.*

Amiti, Mary, Stephen J. Redding, and David E. Weinstein, "The Impact of the 2018 Tariffs on Prices and Welfare," *Journal of Economic Perspectives*, November 2019, *33* (4), 187–210.

Antweiler, Werner and Daniel Trefler, "Increasing Returns and All That: A View from Trade," *American Economic Review*, 2002, *92* (1).

Bartelme, Dominick, Arnaud Costinot, Dave Donaldson, and Andrés Rodríguez-Clare, "The Textbook Case for Industrial Policy: Theory Meets Data," *Journal of Political Economy*, 2025, *133* (5).

Borchert, Ingo, Mario Larch, Serge Shikher, and Yoto Yotov, "Globalization, Trade, and Inequality: Evidence from a New Database," *School of Economics Working Paper, Drexel University*, 2024, *2024-6.*

Breinlich, Holger, Elsa Leromain, Dennis Novy, and Thomas Sampson, "Import Liberalization as Export Destruction? Evidence from the United States," *American Economic Journal: Macroeconomics*, July 2026, *18* (3), 77–111.

Clayton, Christopher, Matteo Maggiori, and Jesse Schreger, "A Framework for Geoeconomics," *Econometrica*, 2026, *94* (1), 105–136.

Costinot, Arnaud, Dave Donaldson, Margaret Kyle, and Heidi Williams, "The More We Die, The More We Sell? A Simple Test of the Home-Market Effect," *Quarterly Journal of Economics*, 2019, *134* (2).

Crowley, Meredith, Ning Meng, and Huasheng Song, "Tariff Scares: Trade Policy Uncertainty and Foreign Market Entry by Chinese Firms," *Journal of International Economics*, 2018, *114*, 96–115.

Davis, Donald and David Weinstein, "Market Access, Economic Geography and Comparative Advantage: An Empirical Test," *Journal of International Economics*, 2003, *59.*

Dekle, Robert, Jonathan Eaton, and Samuel Kortum, "Global Rebalancing with Gravity: Measuring the Burden of Adjustment," *IMF Staff Papers*, 2008, *55* (3).

Fajgelbaum, Pablo D. and Amit K. Khandelwal, "The Economic Impacts of the US–China Trade War," *Annual Review of Economics*, 2022, *14* (1), 205–228.

Fajgelbaum, Pablo D, Pinelopi K Goldberg, Patrick J Kennedy, and Amit K Khandelwal, "The Return to Protectionism," *Quarterly Journal of Economics*, 2020, *135* (1), 1–55.

Fajgelbaum, Pablo, Pinelopi Goldberg, Patrick Kennedy, Amit Khandelwal, and Daria Taglioni, "The US-China Trade War and Global Reallocations," *American Economic Review: Insights*, 2024, *6* (2), 295–312.

Fan, Jingting, Kyle Handley, and Nuno Limão, "Economic Consequences of US National (In)Security Export Controls," 2026. Unpublished.

Feenstra, Robert C., "New Product Varieties and the Measurement of International Prices," *American Economic Review*, 1994, *84* (1).

Finger, J. M. and M. E. Kreinin, "A Measure of 'Export Similarity' and Its Possible Uses," *The Economic Journal*, 1979, *89.*

Fisher, Franklin M. and Karl Shell, *The Economic Theory of Price Indices: Two Essays on the Effects of Taste, Quality, and Technological Change*, New York: Academic Press, 1972.

Gaulier, Guillaume and Soledad Zignago, "BACI: International Trade Database at the Product-Level. The 1994-2007 Version," *CEPII Working Paper*, 2010.

Giri, Rahul, Kei-Mu Yi, and Hakan Yilmazkuday, "Gains from Trade: Does Sectoral Heterogeneity Matter?," *Journal of International Economics*, 2021, *129.*

Handley, Kyle and Nuno Limão, "Trade and Investment under Policy Uncertainty: Theory and Firm Evidence," *American Economic Journal: Economic Policy*, 2015, *7* (4), 189–222.

_ **and** _ , "Policy Uncertainty, Trade, and Welfare: Theory and Evidence for China and the United States," *American Economic Review*, 2017, *107* (9), 2731–83.

Hayakawa, Kazunobu, Keiko Ito, Kyoji Fukao, and Ivan Deseatnicov, "The Impact of the Strengthening of Export Controls on Japanese Exports of Dual-Use Goods," *International Economics*, 2023, *174*, 160–179.

Head, Keith and John Ries, "Increasing Returns versus National Product Differentiation as an Explanation for the Pattern of U.S.-Canada Trade," *American Economic Review*, 2001, *91* (4).

Itskhoki, Oleg and Dmitry Mukhin, "Sanctions and the Exchange Rate," *Review of Economic Studies*, 2026, *93* (4), 2680–2714.

_ **and Elina Ribakova**, "The Economics of Sanctions: From Theory into Practice," *Brookings Papers on Economic Activity*, Fall 2024, pp. 425–470.

Juhàsz, Réka, "Temporary Protection and Technology Adoption: Evidence from the Napoleonic Blockade," *American Economic Review*, 2018, *108* (11).

Kim, In Kyung and Kyoo il Kim, "No Beer No Friends: Quantifying the Effect of the Beer Boycott," *Journal of Industrial Economics*, 2022, *70* (3), 711–751.

Kim, Younghoon, "Responding to Semiconductor Supply Chain Disruptions: Evidence from South Korea," 2024. Working Paper, University of California, Los Angeles.

Krugman, Paul, "Scale Economies, Product Differentiation, and the Pattern of Trade," *American Economic Review*, 1980, *70* (5).

Kucheryavyy, Konstantin, Gary Lyn, and Andrés Rodríguez-Clare, "Grounded by Gravity: A Well-Behaved Trade Model with Industry-Level Economies of Scale," *American Economic Journal: Macroeconomics*, 2023, *15* (2).

Lashkaripour, Ahmad and Volodymyr Lugovskyy, "Profits, Scale Economies, and the Gains from Trade and Industrial Policy," *American Economic Review*, 2023, *113* (10).

Makioka, Ryo and Hongyong Zhang, "The Impact of Export Controls on International Trade: Evidence from the Japan–Korea Trade Dispute in the Semiconductor Industry," *Journal of the Japanese and International Economies*, 2024, *74.*

Mayer, Thierry, Isabelle Méjean, and Mathias Thoenig, "Can Sanctions Deter Wars? The Russia-Ukraine Case," *AEA Papers and Proceedings*, 2026, *116*, 114–118.

Moll, Benjamin, Moritz Schularick, and Georg Zachmann, "The Power of Substitution: The Great German Gas Debate in Retrospect," *Brookings Papers on Economic Activity*, Fall 2023, pp. 395–455.

Morgan, T. Clifton, Constantinos Syropoulos, and Yoto V. Yotov, "Economic Sanctions: Evolution, Consequences, and Challenges," *Journal of Economic Perspectives*, 2023, *37* (1), 3–30.

Pierce, Justin R. and Peter K. Schott, "The Surprisingly Swift Decline of US Manufacturing Employment," *American Economic Review*, 2016, *106* (7), 1632–1662.

Redding, Stephen J. and David E. Weinstein, "Measuring Aggregate Price Indices with Taste Shocks: Theory and Evidence for CES Preferences," *Quarterly Journal of Economics*, 2020, *135* (1).

A Appendix to Section 3

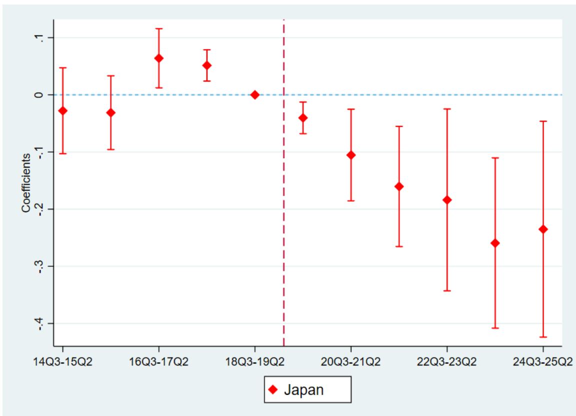
A.1 Robustness Checks

Table A1: Empirical Facts (PPML)

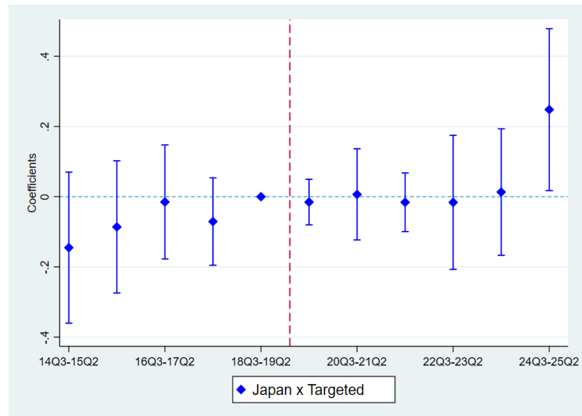
	Fact 1	Fact 2	Fact 3	Fact 4
	(1)	(2)	(3)	(4)
	m	m	m	m
JP × Post	-0.178^{***}			
	(0.059)			
JP × Post × Targeted		0.096		
		(0.080)		
JP × Post × Dominant			-0.163^{**}	
			(0.068)	
Dominant × Post				-0.108
				(0.082)
Origin × Good FE	Yes	Yes	Yes	Yes
Good × Time FE	Yes	Yes	Yes	
Origin × Time FE		Yes	Yes	Yes
No. of Goods	4,211	4,211	4,211	4,266
No. of Targeted Goods		946		
No. of Dominant Goods			594	591
Observations	6,339,754	6,304,945	6,304,945	6,468,474
R-squared	0.97	0.97	0.97	0.96

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

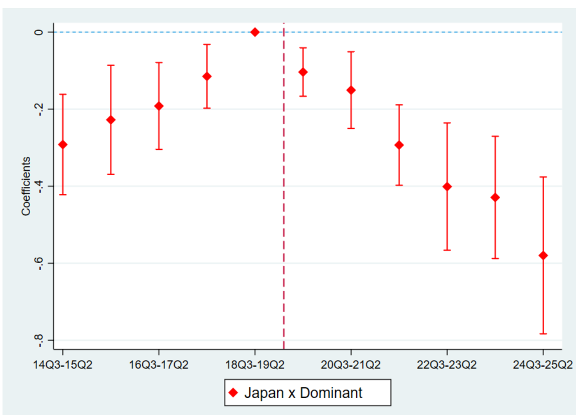
Notes: Table A1 presents the estimation results for the difference-in-differences specifications on Korea's imports from Section 3, estimated by Poisson pseudo-maximum likelihood (PPML). Columns (1) through (4) correspond to Facts 1 through 4, respectively. The dependent variable is Korea's import value. The sample is restricted to intermediate and capital goods. All specifications are conducted at the HS-6 product level. Standard errors are clustered at the origin and good levels.
Source: Korea Customs Service (2014–2025).



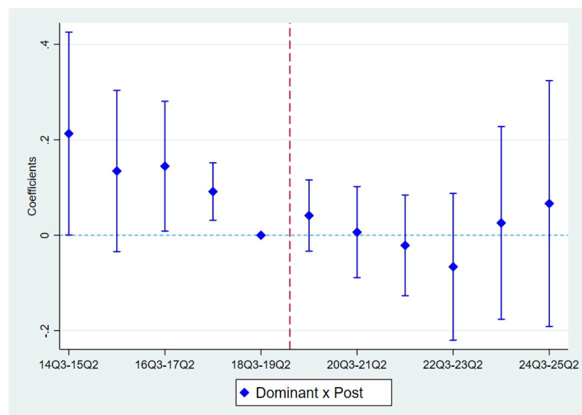
(A) Fact 1: Imports from Japan



(B) Fact 2: Targeted vs. Non-Targeted



(C) Fact 3: Dominant vs. Non-Dominant



(D) Fact 4: Imports from RoW

Figure A1: Empirical Facts (Event Study, PPML)

Notes: Figure A1 illustrates the Poisson pseudo-maximum likelihood (PPML) estimation results for the event-study specifications on Korea's imports from Section 3. Panels (A) through (D) correspond to Facts 1 through 4, respectively. The sample is restricted to intermediate and capital goods. All specifications are conducted at the HS-6 product level, using quarterly data for the period from Q3 2014 to Q2 2025.
Source: Korea Customs Service (2014–2025).

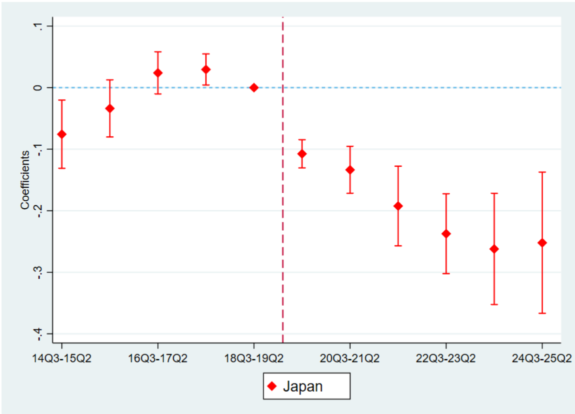
Table A2: Empirical Facts (Including Consumer Goods, Facts 1–3)

	Fact 1	Fact 1	Fact 2	Fact 2	Fact 3	Fact 3
	(1)	(2)	(3)	(4)	(5)	(6)
	$\ln m$	m	$\ln m$	m	$\ln m$	m
JP × Post	-0.186^{***}	-0.197^{***}				
	(0.041)	(0.055)				
JP × Post × Targeted			-0.073^{***}	0.157^{**}		
			(0.022)	(0.078)		
JP × Post × Dominant					-0.176^{***}	-0.168^{**}
					(0.028)	(0.068)
Origin × Good FE	Yes	Yes	Yes	Yes	Yes	Yes
Good × Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin × Time FE			Yes	Yes	Yes	Yes
No. of Goods	5,420	5,701	5,420	5,701	5,420	5,701
No. of Targeted Goods			1,006	1,006		
No. of Dominant Goods					669	669
Observations	3,157,735	9,390,607	3,156,742	9,357,592	3,156,742	9,357,592
R-squared	0.82	0.97	0.82	0.97	0.82	0.97
Estimator	OLS	PPML	OLS	PPML	OLS	PPML

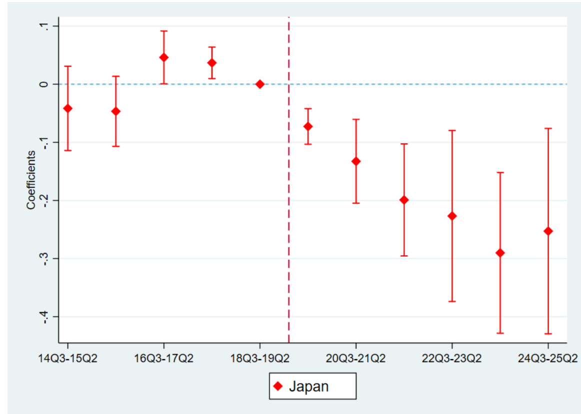
Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Table A2 presents the estimation results for the difference-in-differences specifications on Korea's imports from Section 3, including consumer goods in the sample. Columns (1)–(2), (3)–(4), and (5)–(6) correspond to Facts 1, 2, and 3, respectively, with the first column of each pair reporting ordinary least squares (OLS) estimates and the second reporting Poisson pseudo-maximum likelihood (PPML) estimates. The dependent variable is the log of Korea's import value in the OLS columns and Korea's import value in the PPML columns. All specifications are conducted at the HS-6 product level. Standard errors are clustered at the origin and good levels.
Source: Korea Customs Service (2014–2025).

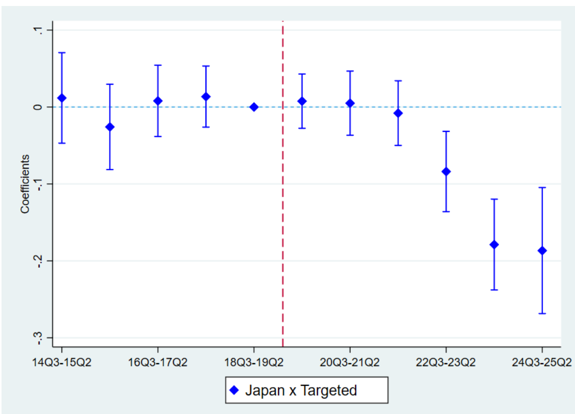
Discussion of Fact 2. In the intermediate and capital goods sample, the difference between *Targeted* and *Non-targeted* goods is not statistically significant under both OLS and PPML estimation, as shown in Column (2) of Table 1 and Table A1. By contrast, in the sample including consumer goods (Columns (3) and (4) of Table A2), the two estimators yield statistically significant coefficients with opposing signs: OLS reports a larger decline in *Targeted* goods, while PPML reports a larger decline in *Non-targeted* goods. Since the only difference between the two samples is the inclusion of consumer goods, this divergence reflects the high incidence of zero import values among *Non-targeted* consumer goods, which OLS drops from the log-linear regression but PPML retains. While the Korean boycott of Japanese products may have contributed, the precise mechanism behind these zeros remains beyond the scope of this analysis. Restricting the sample to intermediate and capital goods removes this divergence and ensures that the main estimate for Fact 2 is robust to the zero-value distortion observed in consumer goods.



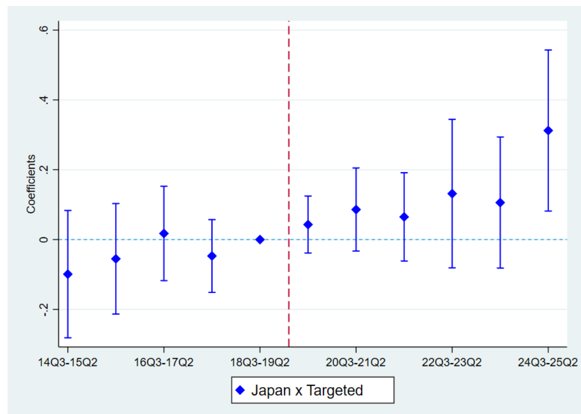
(A) Fact 1: Imports from Japan (OLS)



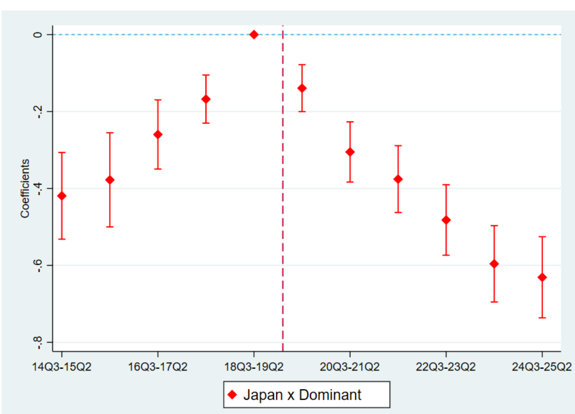
(B) Fact 1: Imports from Japan (PPML)



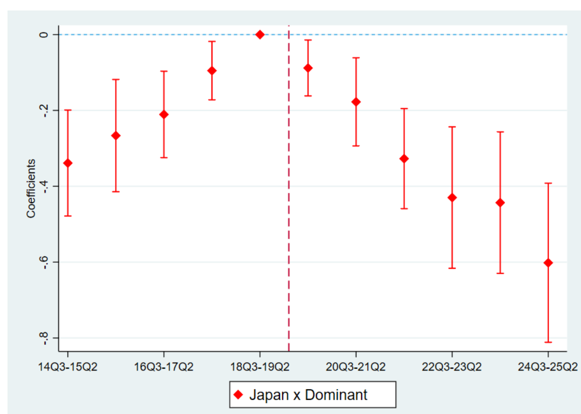
(C) Fact 2: Targeted Goods (OLS)



(D) Fact 2: Targeted Goods (PPML)



(E) Fact 3: JP-Dominant Goods (OLS)



(F) Fact 3: JP-Dominant Goods (PPML)

Figure A2: Empirical Facts (Event Study, Including Consumer Goods, Facts 1–3)

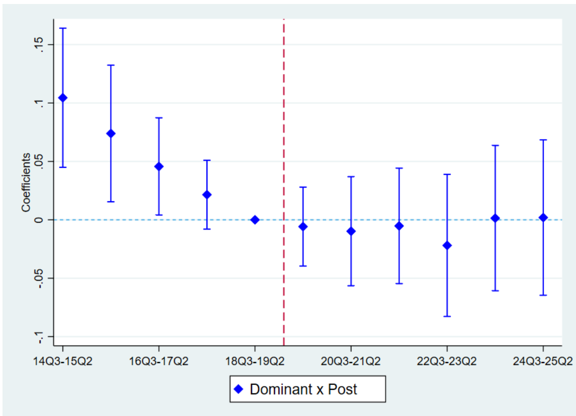
Notes: Figure A2 illustrates the estimation results for the event-study specifications on Korea's imports from Section 3, including consumer goods in the sample. Panels (A)–(B), (C)–(D), and (E)–(F) correspond to Facts 1, 2, and 3, respectively, with the first panel of each pair showing ordinary least squares (OLS) estimates and the second showing Poisson pseudo-maximum likelihood (PPML) estimates. All specifications are conducted at the HS-6 product level, using quarterly data for the period from Q3 2014 to Q2 2025.
Source: Korea Customs Service (2014–2025).

Table A3: Empirical Facts (Including Consumer Goods, Facts 4 and 5)

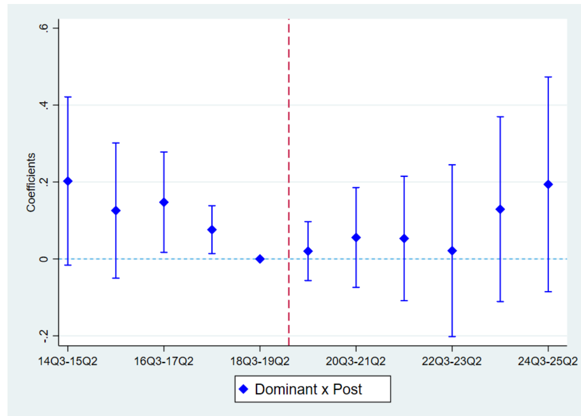
	Fact 4 (1) $\ln m$	Fact 4 (2) m	Fact 5-1 (3) $\ln x^q$	Fact 5-2 (4) $\ln x^p$
Dominant × Post	-0.054^{**} (0.026)	-0.027 (0.114)	0.118^{**} (0.046)	-0.049^{**} (0.022)
Origin × Good FE	Yes	Yes		
Origin × Time FE	Yes	Yes		
Good FE			Yes	Yes
Sector × Time FE			Yes	Yes
No. of Goods	5,690	5,775	4,781	4,781
No. of Dominant Goods	664	664	630	630
Observations	3,007,136	9,644,973	184,508	184,508
R-squared	0.79	0.95	0.89	0.85
Estimator	OLS	PPML	OLS	OLS

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

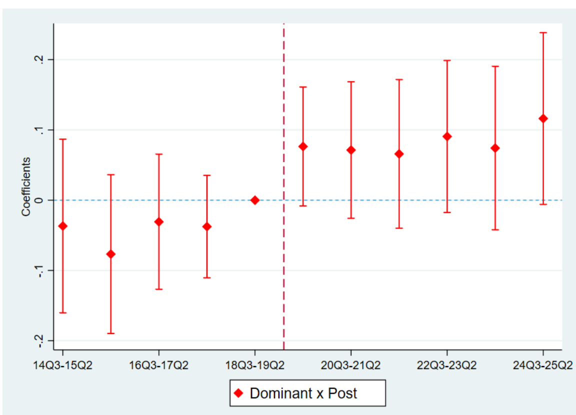
Notes: Table A3 presents the estimation results for the difference-in-differences specifications on Korea's imports and exports from Section 3, including consumer goods in the sample. Columns (1)–(2) correspond to Fact 4, with Column (1) reporting the ordinary least squares (OLS) estimate and Column (2) reporting the Poisson pseudo-maximum likelihood (PPML) estimate. Columns (3) and (4) correspond to Fact 5, reporting the export quantity and price, respectively. Both are estimated by OLS, as PPML is not applicable to Fact 5 because export prices cannot be coded as zero. The dependent variable is the log of Korea's import value in Column (1), Korea's import value in Column (2), the log of Korea's export quantity in Column (3), and the log of Korea's export price in Column (4). All specifications are conducted at the HS-6 product level. Standard errors are clustered at the origin and good levels in Columns (1)–(2), and at the good level in Columns (3)–(4).
Source: Korea Customs Service (2014–2025).



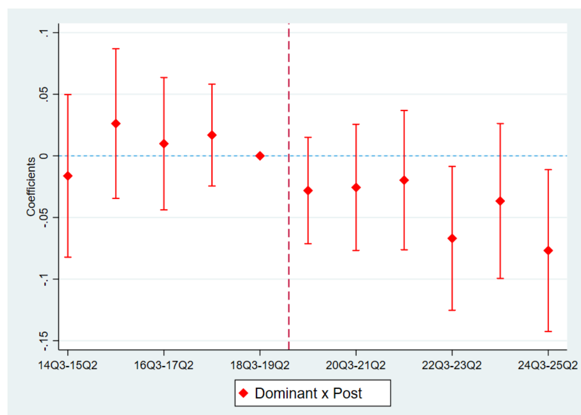
(A) Fact 4: Imports from RoW (OLS)



(B) Fact 4: Imports from RoW (PPML)



(C) Fact 5-1: Korea's Export Quantity (OLS)



(D) Fact 5-2: Korea's Export Price (OLS)

Figure A3: Empirical Facts (Event Study, Including Consumer Goods, Facts 4 and 5)

Notes: Figure A3 illustrates the estimation results for the event-study specifications on Korea's imports and exports from Section 3, including consumer goods in the sample. Panels (A)–(B) correspond to Fact 4, with Panel (A) showing ordinary least squares (OLS) estimates and Panel (B) showing Poisson pseudo-maximum likelihood (PPML) estimates. Panels (C)–(D) correspond to Fact 5, showing the export quantity and price, respectively. Both are estimated by OLS, as PPML is not applicable to Fact 5 because export prices cannot be coded as zero. All specifications are conducted at the HS-6 product level, using quarterly data for the period from Q3 2014 to Q2 2025.
Source: Korea Customs Service (2014–2025).

A.2 Empirical Facts on Japan's Exports to Korea

Data. The empirical analysis in this section uses Japan's export data at the 6-digit level of the Harmonized System (HS) from the CEPII BACI database (Gaulier and Zignago (2010)). The data are annual, covering Japan's exports to 215 destination countries and 5,175 goods at the HS-6 level over 11 years, from 2014 to 2024. The analysis restricts the sample to intermediate and capital goods, consistent with the main analysis in Section 3.

Empirical Strategy. Following the empirical strategy of Section 3, I document the empirical facts using two approaches: a difference-in-differences specification and an event study that traces the time path of the same comparison in annual windows. The exact specification is reported in each fact below. Each specification is estimated under both ordinary least squares (OLS) and Poisson pseudo-maximum likelihood (PPML).

Fact 1: Japan's exports to Korea substantially decreased, both relative to the pre-event period and to exports to other countries.

To investigate the effect of Japan's announcement of export controls on Japan's exports to Korea, I use a difference-in-differences specification:

$$\log x_{j,k,t}^{jp} = \alpha_{j,k} + \alpha_{k,t} + \beta\Big(Korea_j\Big) \times \Big(Post_t\Big) + \varepsilon_{j,k,t},$$

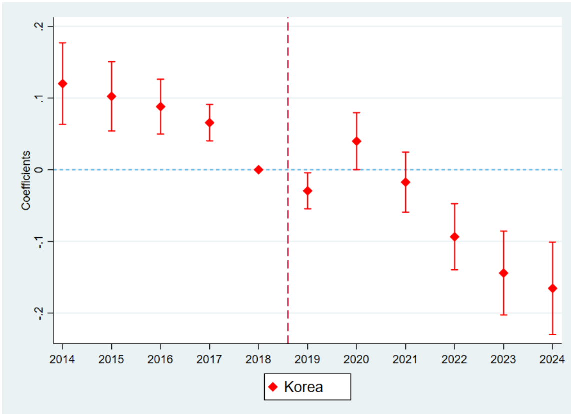
where $x_{j,k,t}^{jp}$ is the value of Japan's exports of good k to destination j at time t. $\alpha_{j,k}$ and $\alpha_{k,t}$ denote destination-good and good-time fixed effects, respectively. $Korea_j$ is a destination indicator variable for Korea and $Post_t$ is a time indicator for the post-event period, which spans from 2019 to 2024. Standard errors are clustered at destination j and good k levels. The coefficient β is estimated to be -0.143, as shown in Column (1) of Table A4. This indicates that Japan's exports to Korea decreased by 13.3 percent, both relative to the pre-event period and to Japan's exports to other countries. Time-varying coefficients β_t, estimated using an event study, are also plotted in Panel (A) of Figure A4, along with 95 percent confidence intervals for the estimates. This event-study analysis confirms that the decline persisted throughout the post-event period. This decline is also robust to PPML estimation, as shown in Column (2) of Table A4 and Panel (B) of Figure A4. This finding aligns with Korea's imports from Japan documented in Fact 1 of Section 3.

Table A4: Empirical Facts (Japan's Exports, Facts 1–3)

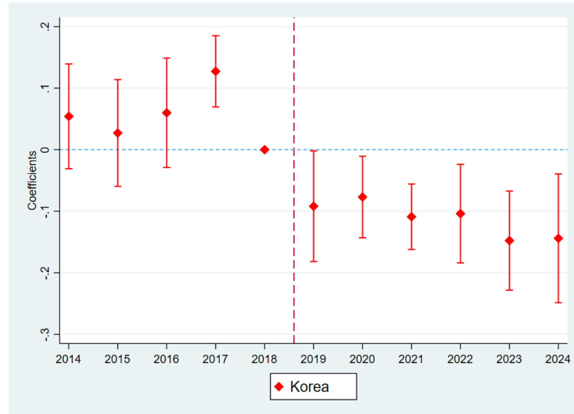
	Fact 1 (1) $\ln x$	Fact 1 (2) x	Fact 2 (3) $\ln x$	Fact 2 (4) x	Fact 3 (5) $\ln x$	Fact 3 (6) x
KOR × Post	-0.143^{***}	-0.166^{***}				
	(0.023)	(0.031)				
KOR × Post × Targeted			0.014	-0.024		
			(0.017)	(0.059)		
KOR × Post × Dominant					-0.150^{***}	-0.189^{***}
					(0.020)	(0.065)
Destination × Good FE	Yes	Yes	Yes	Yes	Yes	Yes
Good × Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Destination × Time FE			Yes	Yes	Yes	Yes
No. of Goods	3,656	3,776	3,656	3,776	3,656	3,776
No. of Targeted Goods			901	901		
No. of Dominant Goods					579	579
Observations	1,742,033	3,223,561	1,742,017	3,222,559	1,742,017	3,222,559
R-squared	0.86	0.97	0.86	0.98	0.86	0.98
Estimator	OLS	PPML	OLS	PPML	OLS	PPML

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

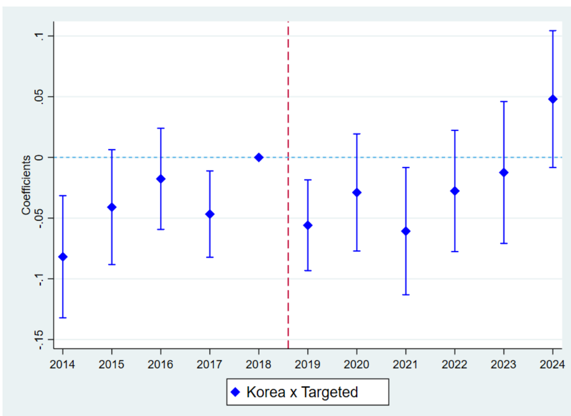
Notes: Table A4 presents the estimation results for the difference-in-differences specifications on Japan's exports to Korea, providing a mirror perspective on the findings in Section 3. Columns (1)–(2), (3)–(4), and (5)–(6) correspond to Facts 1, 2, and 3, respectively, with the first column of each pair reporting ordinary least squares (OLS) estimates and the second reporting Poisson pseudo-maximum likelihood (PPML) estimates. The dependent variable is the log of Japan's export value in the OLS columns and Japan's export value in the PPML columns. The sample is restricted to intermediate and capital goods. All specifications are conducted at the HS-6 product level. Standard errors are clustered at the destination and good levels.
Source: CEPII BACI (Gaulier and Zignago (2010)).



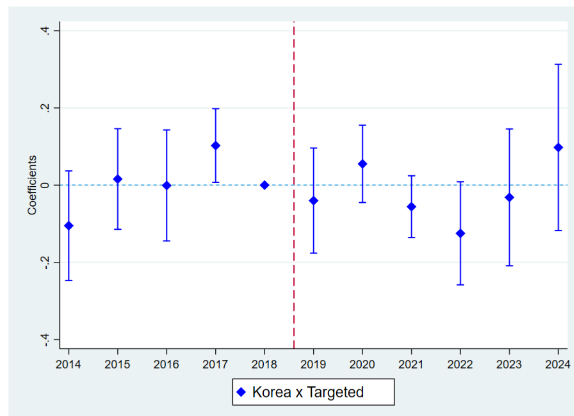
(A) Fact 1: Exports to Korea (OLS)



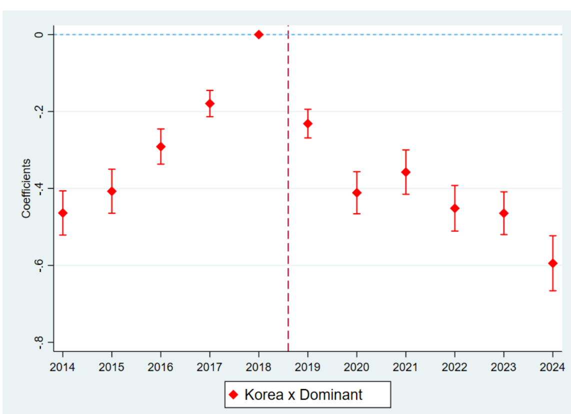
(B) Fact 1: Exports to Korea (PPML)



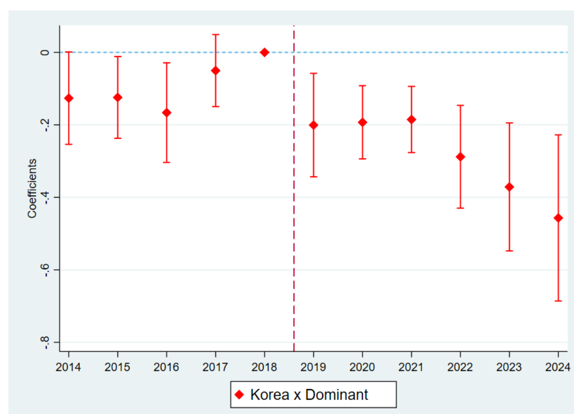
(C) Fact 2: Targeted Goods (OLS)



(D) Fact 2: Targeted Goods (PPML)



(E) Fact 3: JP-Dominant Goods (OLS)



(F) Fact 3: JP-Dominant Goods (PPML)

Figure A4: Empirical Facts (Event Study, Japan's Exports, Facts 1–3)

Notes: Figure A4 illustrates the estimation results for the event-study specifications on Japan's exports to Korea. Panels (A)–(B), (C)–(D), and (E)–(F) correspond to Facts 1, 2, and 3, respectively, with the first panel of each pair showing ordinary least squares (OLS) estimates and the second showing Poisson pseudo-maximum likelihood (PPML) estimates. The sample is restricted to intermediate and capital goods. All specifications are conducted at the HS-6 product level, using annual data from 2014 to 2024.
Source: CEPII BACI (Gaulier and Zignago (2010)).

Fact 2: The extent of the decline in Japan's exports to Korea was similar between targeted and non-targeted goods.

To examine whether the substantial decrease in Japan's exports to Korea is limited to goods directly targeted by Japan's export controls, I use a triple difference-in-differences specification:

$$\log x^{jp}_{j,k,t} = \alpha_{j,k} + \alpha_{k,t} + \alpha_{j,t} + \beta\Big(Korea_j\Big) \times \Big(Post_t\Big) \times \Big(Targeted_k\Big) + \varepsilon_{j,k,t},$$

where $\alpha_{j,t}$ denotes destination-time fixed effects and $Targeted_k$ is an indicator variable for goods that were directly targeted by the export controls. The remaining terms and clustering level are as previously described. The coefficient β is estimated to be 0.014, essentially indistinguishable from zero, as shown in Column (3) of Table A4. This indicates that the extent of the decline in *Non-targeted* goods exported to Korea is not statistically different from that of *Targeted* goods. Moreover, time-varying coefficients β_t, estimated from an event study, further support this finding, as shown in Panel (C) of Figure A4. This finding is also robust to PPML estimation, as shown in Column (4) of Table A4 and Panel (D) of Figure A4. This pattern aligns with Korea's imports from Japan documented in Fact 2 of Section 3.

Fact 3: Japan's exports to Korea decreased disproportionately in goods for which Japan had been the leading supplier to Korea.

To explore whether the decrease was disproportionately larger in sectors where Korea had been most reliant on Japan, I use a triple difference-in-differences specification:

$$\log x^{jp}_{j,k,t} = \alpha_{j,k} + \alpha_{k,t} + \alpha_{j,t} + \beta\Big(Korea_j\Big) \times \Big(Post_t\Big) \times \Big(Dominant_k\Big) + \varepsilon_{j,k,t},$$

where $Dominant_k$ is an indicator variable for *Japan-dominant* goods, defined as goods for which Japan was the leading supplier to Korea during the pre-event period (2018). The remaining terms and clustering level are as previously described. The coefficient β is estimated to be -0.150, as shown in Column (5) of Table A4. This indicates that Japan's exports of *Japan-dominant* goods to Korea declined approximately 13.9 percent more than *Non-dominant* goods. The time-varying coefficients β_t, plotted in Panel (E) of Figure A4, further support this finding. This finding is also robust to PPML estimation, as shown in Column (6) of Table A4 and Panel (F) of Figure A4. This finding aligns with Korea's imports from Japan documented in Fact 3 of Section 3.

A.3 End-Use Composition of Korea's Imports from Japan

Table A5: Korea's Imports from Japan by End-Use Category (2018)

	All Goods		Targeted		Japan-dominant	
(USD bn, No.)	Value	HS-6	Value	HS-6	Value	HS-6
Consumer	4.56	1,013	1.39	56	1.64	74
	(8.4)	(24.0)	(4.4)	(6.3)	(6.3)	(11.2)
Intermediate	35.68	2,604	19.07	615	19.00	439
	(65.4)	(61.7)	(59.8)	(68.9)	(72.4)	(66.6)
Capital	14.36	605	11.42	221	5.59	146
	(26.3)	(14.3)	(35.8)	(24.8)	(21.3)	(22.2)
Total	54.60	4,222	31.88	892	26.23	659
	(100.0)	(100.0)	(100.0)	(100.0)	(100.0)	(100.0)

Notes: Table A5 reports the composition of Korea's imports from Japan in 2018, the pre-event year, classified by the Broad Economic Categories (BEC) end-use: consumer goods (Final Consumption), intermediate goods (Intermediate Consumption), and capital goods (Gross Fixed Capital Formation). For each goods group, the value column reports import value in billions of USD and the HS-6 column reports the number of HS-6 products; shares within each group (in percent) are in parentheses. "All Goods" covers all goods imported from Japan; "Targeted" denotes goods subject to Japan's export controls (see Appendix A.4); "Japan-dominant" denotes goods for which Japan was the leading supplier to Korea in 2018.
Source: Korea Customs Service (2018) and the United Nations Statistics Division (BEC Rev.5).

A.4 Identification of Goods Targeted by Japan's Export Controls

This appendix details the identification of the Japanese national strategic items that became subject to potential export controls following Japan's removal of Korea from its "White List" in September 2019. The practical effect of this policy, which revoked Korea's status as a preferential trade partner, was a significant shift in export approval procedures.

Previously, when Korea had been on the "White List," Japan's strategic items exported to Korea had been eligible for a "General Comprehensive Permit," a simplified export process. Following the exclusion of Korea from the list, these strategic items were then subject to individual export approval—a strict, case-by-case review required for Non-White List countries. Therefore, exports of these items to Korea became contingent on an uncertain approval decision from the Japanese government, which was legally authorized to ban or restrict shipments.

In response to Japan's removal of Korea from its "White List," the Korea Security Agency of Trade and Industry (KOSTI), under Korea's Ministry of Trade, Industry and Energy, identified and published a list of affected items at the HSK-10 level to enable Korean firms to anticipate and mitigate the potential impact.[45] This identification process was based on the strategic (controlled) items list compiled and published annually by Japan's Ministry of Economy, Trade and Industry, which consists of 1,120 items under 15 categories.[46] KOSTI matched these 1,120 strategic items to HSK-10 codes, resulting in a list of 2,697 HSK-10 items.

This study identifies 1,017 unique items at the HS-6 level as *Targeted* goods by mapping the 2,697 HSK-10 items to the HS-6 level. Of these 1,017 *Targeted* goods, 886 were imported from Japan during the pre-event period (July 2018 to June 2019). This mapping to the HS-6 level is necessary because the HSK-10 codes are specific to Korea, whereas the analyses throughout the paper rely on multi-country trade data reported at the internationally standardized HS-6 level (e.g., the BACI database). These *Targeted* goods span a wide range of product categories, including semiconductor integrated circuits, lithium-ion accumulators, optical fibers, aramid fibers, industrial robots, machining centers, spacecraft and satellites, and radar apparatus.

[45] The first six digits of the Harmonized System (HS) code are internationally standardized. Countries often add subsequent digits for more detailed national classification. The HSK-10 (Harmonized System of Korea) adds four digits to the global HS-6 standard.

[46] The 15 categories are: (1) Weapons, (2) Nuclear, (3) Chemical & Biological, (4) Missiles, (5) Advanced Materials, (6) Materials Processing, (7) Electronics, (8) Computers, (9) Telecommunications, (10) Sensors, (11) Navigation, (12) Marine, (13) Propulsion, (14) Miscellaneous, and (15) Sensitive Items.

A.5 Composition of Japan-dominant Sectors

Table A6: Representative Japan-dominant Sectors

KSIC-5 Sector	Japan's Share (%)
Other machinery and equipment (KSIC-2: 29)	
Metal cutting machinery (29223)	49.4
Industrial robots (29280)	39.6
Ball and roller bearings (29141)	36.4
Display component manufacturing machinery (29272)	35.1
Semi-conductor manufacturing machinery (29271)	34.0
Metal shaping machinery (29224)	32.6
Forming machinery for digital additive manufacturing (29222)	22.3
Chemicals and chemical products (KSIC-2: 20)	
Sensitized materials and related chemical preparations (20491)	66.9
Basic organic petrochemicals (20111)	38.0
Adhesives and gelatin (20493)	35.0
Surface-active agents (20421)	33.3
Synthetic rubber (20201)	28.9
Non-metallic mineral products (KSIC-2: 23)	
Flat glass (23111)	76.9
Carbon fibers (23995)	35.4
Abrasive articles (23992)	33.9
Rubber and plastic products (KSIC-2: 22)	
Plastic sheets and plates (22213)	73.1
Plastic films (22212)	45.5
Medical, precision and optical instruments (KSIC-2: 27)	
Electrical measuring and testing instruments (27212)	42.5
Optical lens and elements (27301)	36.8
Fabricated metal products (KSIC-2: 25)	
Nuclear reactors and steam generators (25130)	51.8
Basic metals (KSIC-2: 24)	
Basic steel (24112)	47.7

Notes: Table A6 lists representative *Japan-dominant* sectors at the KSIC-5 level, grouped by KSIC-2 industry and ordered by Japan's share within each group. The KSIC is the Korean adaptation of the United Nations' International Standard Industrial Classification (ISIC). *Japan-dominant* sectors are those in which Japan was the leading supplier to Korea during the pre-event period (2018). "Japan's Share" is Japan's share of each sector's total imports, obtained by aggregating Korea's 2018 imports at the HS-6 level to the KSIC-5 level via a concordance from Statistics Korea.
Source: Korea Customs Service (2018) and Statistics Korea.

Table A7: Distribution of Japan-dominant Sectors by KSIC-2 Industry

KSIC-2 Industry		Japan-dominant	Total
10	Food products	1	43
11	Beverages	2	9
12	Tobacco products	0	1
13	Textiles, except apparel	0	28
14	Wearing apparel and clothing accessories	0	15
15	Leather, luggage and footwear	0	7
16	Wood and products of wood and cork	0	13
17	Pulp, paper and paper products	1	17
18	Printing and reproduction of recorded media	0	8
19	Coke and refined petroleum products	0	4
20	Chemicals and chemical products	11	30
21	Pharmaceuticals and medicinal chemicals	0	5
22	Rubber and plastic products	5	22
23	Non-metallic mineral products	6	32
24	Basic metals	3	24
25	Fabricated metal products	2	33
26	Electronics and communication equipment	1	30
27	Medical, precision and optical instruments	3	17
28	Electrical equipment	3	25
29	Other machinery and equipment	19	43
30	Motor vehicles, trailers and semitrailers	2	15
31	Other transport equipment	0	15
32	Furniture	0	6
33	Other manufacturing	4	20
Total		63	476

Notes: Table A7 reports the distribution of *Japan-dominant* sectors at the KSIC-5 level by KSIC-2 industry. The KSIC is the Korean adaptation of the United Nations' International Standard Industrial Classification (ISIC). *Japan-dominant* sectors are those in which Japan was the leading supplier to Korea during the pre-event period (2018). "Japan-dominant" is the number of *Japan-dominant* KSIC-5 sectors within each KSIC-2 industry. "Total" is the total number of KSIC-5 sectors in that industry. The final row reports the totals across all 476 KSIC-5 sectors, including 14 mining sectors that are not listed separately; none of them is *Japan-dominant*.
Source: Korea Customs Service (2018) and Statistics Korea.

A.6 Overlap between Targeted and Japan-dominant Goods

Table A8: Targeted and Japan-dominant Goods (HS-6 Level)

(No., % of total)	Targeted	Non-targeted	Total
Japan-dominant	181	413	594
	(4.8)	(10.8)	(15.6)
Non-dominant	733	2,483	3,216
	(19.2)	(65.2)	(84.4)
Total	914	2,896	3,810
	(24.0)	(76.0)	(100.0)

Notes: Table A8 cross-classifies the HS-6 goods Korea imported during the pre-event period (July 2018–June 2019) by whether Japan was the leading supplier to Korea (rows) and whether they were targeted by Japan's export controls (columns). The table covers intermediate and capital goods, consistent with the main empirical analysis in Section 3. The first row of each cell reports the number of HS-6 goods and the second its share of the grand total (the bottom-right cell, 3,810); the four interior cells thus sum to 100 percent. "Japan-dominant" denotes goods for which Japan was the leading supplier during the pre-event period; "Targeted" denotes goods subject to Japan's export controls (see Appendix A.4).
Source: Korea Customs Service (2018–2019) and the Korea Security Agency of Trade and Industry.

A.7 Korea's Export Capability and Similarity with Japan

This appendix provides the rationale for Fact 5 in Section 3—the rapid expansion of Korea's exports of *Japan-dominant* goods following Japan's 2019 export control announcement. The analysis shows that this expansion is plausible because Korea already had the necessary export capability, which is documented at two levels: at the product level, Korea was already exporting nearly all of the *Japan-dominant* goods themselves; at the structural level, Korea and Japan share similar export structures, ranking among each other's closest export competitors in the global market. This suggests that the expansion was a scale-up of existing operations rather than entry into new industries.

At the product level, the analysis examines whether Korea already exported the *Japan-dominant* goods themselves, using Korea's trade data from the Korea Customs Service. Following Section 3, *Japan-dominant* goods are defined as products at the 6-digit level of the Harmonized System (HS) for which Japan was the leading supplier to Korea during the pre-event period (July 2018–June 2019). Consistent with the main analysis, the sample covers intermediate and capital goods. Panel A of Table A9 first identifies *Japan-dominant* goods based on Korea's import data over the pre-event period. Of Korea's 3,810 imported goods, 594 are classified as *Japan-dominant*. Panel B then examines Korea's export capability for these specific 594 goods. Korea already exported 574 of them—out of its 3,554 exported goods—to the global market during the same period. The final row shows that Korea already exported 97 percent of these goods.

At the structural level, the analysis measures the overall similarity between Korea's and Japan's export structures. The Export Similarity Index (ESI), first introduced by Finger and Kreinin (1979), is a standard measure in international trade used to quantify the degree of overlap (competition) between two countries' export portfolios. The index ESI_{ij} between country i and country j is calculated as:

$$ESI_{ij} = \sum_k \min(s_{i,k}, s_{j,k}),$$

where $s_{i,k}$ is the share of product k in country i's total export value to the world.[47] The index ranges from 0 (no overlap) to 1 (identical export structures). Because the calculation requires the export data of Japan and other countries, which the Korea Customs Service data do not cover, it draws on the CEPII BACI database (Gaulier and Zignago (2010)); for consistency, Korea's exports are also taken from the same source. The database includes annual trade data for 5,199 goods at the HS-6 level under the 2012 classification across

[47] Formally, $s_{i,k} = \frac{\sum_{j \neq i} X_{ij,k}}{\sum_m \sum_{j \neq i} X_{ij,m}}$, where $X_{ij,k}$ is country i's export value of product k to country j.

226 countries, and the ESI is therefore calculated for the pre-event year 2018. Table A10 presents the results, listing the top five countries with the most similar export structures to Korea (Panel A) and Japan (Panel B). For Korea, Japan is the most similar counterpart. For Japan, Korea is the second-most similar counterpart, following only Germany.

Together, the two results in Tables A9 and A10 show that Korea's export capability was already in place, both in the *Japan-dominant* goods themselves and in the broader similarity of the two countries' export structures. This explains how the rapid expansion noted in Fact 5 was feasible, consistent with the production-side response in Table 2.

Table A9: Korea's Export Capability in Japan-dominant Goods

	Number of HS-6 Goods
A. Korea's Imports	
Japan-dominant Goods (a)	594
Total Goods	3,810
B. Korea's Exports	
Japan-dominant Goods (b)	574
Total Goods	3,554
Share (b/a)	0.97

Notes: Table A9 reports Korea's pre-existing export capability in Japan-dominant goods, which are defined as products for which Japan was the leading supplier to Korea during the pre-event period (July 2018–June 2019). The table covers intermediate and capital goods, consistent with the main empirical analysis in Section 3. Panel A presents the number of these goods relative to Korea's total imported goods. Panel B presents the number of these same Japan-dominant goods that Korea also exported, relative to Korea's total exported goods. The final row reports the share of Japan-dominant goods that Korea already had the capability to export.
Source: Korea Customs Service (2018–2019).

Table A10: Export Similarity Index (ESI) Ranking (2018)

Panel A: Korea		Panel B: Japan	
Country	ESI with Korea	Country	ESI with Japan
Japan	**0.518**	Germany	0.545
Taiwan	0.496	**Korea**	**0.518**
Germany	0.420	United States	0.459
United States	0.419	United Kingdom	0.432
Singapore	0.417	Austria	0.396

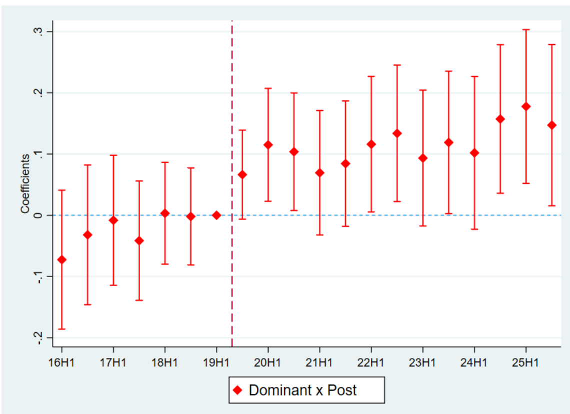
Notes: Table A10 lists the top five countries with the highest Export Similarity Index (ESI) relative to Korea (Panel A) and Japan (Panel B), with the Korea–Japan pair shown in bold. The index is calculated at the HS-6 product level using annual trade data for the pre-event year 2018. Trade data for Taiwan are proxied using the data for "Other Asia, not elsewhere specified" (ISO code 490), following the CEPII documentation, as this category almost exclusively contains trade data for Taiwan.
Source: CEPII BACI database (Gaulier and Zignago (2010)).

A.8 The Timing of Korea's Export Quantities and Prices

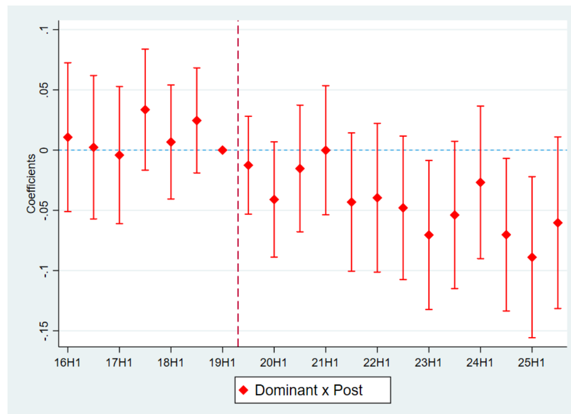
This appendix re-examines Fact 5—the rise in Korea's export quantities and the decline in export prices for *Japan-dominant* goods—at monthly frequency to trace the timing of the two responses. The specification is identical to that of Fact 5, except that the data are monthly (January 2016 to December 2025) and the event study uses half-year windows rather than annual ones. As in the main analysis, the sample is restricted to intermediate and capital goods.

The difference-in-differences estimates are consistent with the quarterly results in Fact 5. Relative to *Non-dominant* goods and the pre-event period, export quantities of *Japan-dominant* goods rise by approximately 14.5% ($p < 0.01$), while export prices fall by approximately 5.3% ($p < 0.01$).

Figure A5 reports the event-study estimates. Neither series exhibits a pre-trend. Following the event, export quantities rise first and remain elevated throughout the post-event period (Panel A). Export prices, by contrast, change little initially and decline only with a lag, turning statistically significant from 2023 and deepening toward the end of the sample (Panel B). This Q-first, P-lag dynamic is the signature of scale economies: the quantity expansion comes first, and the unit price declines only later, as the productivity gains accumulate with the scale of production. Such a gradual, lagged price decline is difficult to reconcile with markup cutting, which would lower prices immediately rather than progressively.



(A) Export Quantity



(B) Export Price

Figure A5: Event Study of Korea's Exports in Japan-dominant Goods

Notes: Figure A5 illustrates the event-study coefficients for Korea's export quantities (Panel A) and unit prices (Panel B) of Japan-dominant goods, using the specification of Fact 5 with half-year windows. The reference period is the first half of 2019, immediately preceding the event. The sample is restricted to intermediate and capital goods. All specifications are estimated by ordinary least squares (OLS) at the HS-6 product level, using monthly data for the period from January 2016 to December 2025.
Source: Korea Customs Service (2016–2025).

A.9 Variable Definitions for the Production Response

All variables are constructed from the Mining and Manufacturing Survey conducted annually by Statistics Korea, which covers establishments with 10 or more workers. The unit of observation is the 5-digit KSIC sector by year. The eight dependent variables in Table 2 are defined as follows, in the order of their columns.

- *output*: Gross output, defined as the value of product shipments, including goods produced under subcontract and free transfers to other plants within the same firm at producer prices.
- *machinery*: The book value of machinery and equipment as of year-end (December 31), one of the categories that make up tangible fixed assets.
- *inputs*: The total cost of major intermediate inputs directly used in the manufacturing process, defined as the sum of raw materials, fuel, electricity, water, subcontracted processing, and repairs. Indirect intermediate inputs, such as labor compensation and depreciation, are excluded.
- *emp*: The number of regular workers, defined as those employed under a contract of one year or more, or those who work as permanent staff without a fixed term of employment.
- *wagebill*: The annual wage bill of regular workers, inclusive of bonuses and allowances.
- *assets*: The book value of tangible fixed assets as of year-end (December 31), comprising land, buildings and structures, machinery and equipment, and other assets such as vehicles, tools, instruments, and fixtures.
- *invest*: The total value of investment in tangible fixed assets made during the year.
- *estab*: The number of establishments with 10 or more workers in the sector.

B Appendix to Section 5.1

B.1 Theoretical Foundation of the Instrumental Variable

This section details the theoretical mechanism through which a shock to importing costs from Japan ($\Delta \ln \tau_{JPN\,KOR,k}$) affects Korea's total sales ($\Delta \ln Y_{KOR,k}$). The total elasticity can be decomposed using the chain rule as follows:

$$\frac{\Delta \ln Y_{KOR,k}}{\Delta \ln \tau_{JPN\,KOR,k}} = \frac{\Delta \ln p_{JPN\,KOR,k}}{\Delta \ln \tau_{JPN\,KOR,k}} \frac{\Delta \ln P_{KOR,k}}{\Delta \ln p_{JPN\,KOR,k}} \frac{\Delta \ln X_{KOR\,KOR,k}}{\Delta \ln P_{KOR,k}} \frac{\Delta \ln Y_{KOR,k}}{\Delta \ln X_{KOR\,KOR,k}}.$$

Each term on the RHS is derived below.

The first term of the RHS is the elasticity of the price of Japanese goods in Korea with respect to its trade cost. From equation (3), holding all other variables constant, this elasticity is equal to one:

$$\frac{\Delta \ln p_{JPN\,KOR,k}}{\Delta \ln \tau_{JPN\,KOR,k}} = \frac{\Delta \ln \left(\frac{\tau_{JPN\,KOR,k} w_{JPN}^{1+\gamma_k}}{A_{JPN,k} Y_{JPN,k}^{\gamma_k}} \right)}{\Delta \ln \tau_{JPN\,KOR,k}} = 1.$$

The second term of the RHS is the elasticity of Korea's sectoral price index ($P_{KOR,k}$) with respect to the price of Japanese goods in Korea ($p_{JPN\,KOR,k}$). Holding all other prices constant, this elasticity is equivalent to Japan's market share in Korea for good k:

$$\frac{\Delta \ln P_{KOR,k}}{\Delta \ln p_{JPN\,KOR,k}} = \frac{\Delta \ln \left(\sum_i p_{i\,KOR,k}^{-\theta_k} \right)^{-\frac{1}{\theta_k}}}{\Delta \ln p_{JPN\,KOR,k}} = \frac{\left(p_{JPN\,KOR,k}\right)^{-\theta_k}}{\sum_i \left(p_{i\,KOR,k}\right)^{-\theta_k}} = \frac{X_{JPN\,KOR,k}}{\sum_i X_{i\,KOR,k}}.$$

The third term of the RHS is the elasticity of Korea's domestic sales ($X_{KOR\,KOR,k}$) with respect to its sectoral price index ($P_{KOR,k}$). From the bilateral trade equation (4), holding all other variables constant, this elasticity is equal to the trade elasticity (θ_k):

$$\frac{\Delta \ln X_{KOR\,KOR,k}}{\Delta \ln P_{KOR,k}} = \frac{\Delta \ln \left(\left(\frac{\tau_{KOR\,KOR,k} w_{KOR}^{1+\gamma_k}}{A_{KOR,k}} \right)^{-\theta_k} Y_{KOR,k}^{\gamma_k \theta_k} X_{KOR,k} P_{KOR,k}^{\theta_k} \right)}{\Delta \ln P_{KOR,k}} = \theta_k.$$

The fourth term of the RHS is the elasticity of Korea's total sales ($Y_{KOR,k}$) with respect to its domestic sales ($X_{KOR\,KOR,k}$). Holding sales to other destinations constant, this elasticity is equivalent to the share of domestic sales in Korea's total sales for good k:

$$\frac{\Delta \ln Y_{KOR,k}}{\Delta \ln X_{KOR\,KOR,k}} = \frac{X_{KOR\,KOR,k}}{Y_{KOR,k}} = \frac{X_{KOR\,KOR,k}}{\sum_j X_{KOR\,j,k}}.$$

Finally, combining these four terms, the total elasticity of Korea's total sales ($Y_{KOR,k}$) with respect to the shock to importing costs from Japan ($\tau_{JPN\,KOR,k}$) is given by:

$$\begin{aligned}\frac{\Delta \ln Y_{KOR,k}}{\Delta \ln \tau_{JPN\,KOR,k}} &= \frac{\Delta \ln p_{JPN\,KOR,k}}{\Delta \ln \tau_{JPN\,KOR,k}} \frac{\Delta \ln P_{KOR,k}}{\Delta \ln p_{JPN\,KOR,k}} \frac{\Delta \ln X_{KOR\,KOR,k}}{\Delta \ln P_{KOR,k}} \frac{\Delta \ln Y_{KOR,k}}{\Delta \ln X_{KOR\,KOR,k}} \\ &= 1 \times \frac{X_{JPN\,KOR,k}}{\sum_i X_{i\,KOR,k}} \times \theta_k \times \frac{X_{KOR\,KOR,k}}{\sum_j X_{KOR\,j,k}} \\ &= \theta_k \times \frac{X_{JPN\,KOR,k}}{\sum_i X_{i\,KOR,k}} \times \frac{X_{KOR\,KOR,k}}{\sum_j X_{KOR\,j,k}}.\end{aligned}$$

This total effect is therefore the product of three components: the trade elasticity, Japan's share of the Korean market, and the share of domestic sales in Korea's total sales.

In summary, the structural model provides a clear justification for the instrumental variable used in Section 5.1, $IV_{KOR,k} = \theta_k Share^{2018}_{JPN\,KOR,k}$. The derivation shows that it corresponds to the first two components of the total elasticity, the trade elasticity and Japan's share; because Korea's domestic sales are unobserved at the HS-6 level, it uses Japan's import share ($X_{JPN\,KOR,k} / \sum_{i \neq KOR} X_{i\,KOR,k}$) as the observable proxy for its market share and omits the domestic-sales-share component. It thereby preserves the cross-sectional variation in exposure that drives the first stage, establishing its relevance. Moreover, its demand-side nature ensures it is uncorrelated with the supply-side shocks, thus satisfying the exclusion restriction.

B.2 Estimation of the Output Elasticity

This appendix presents the estimation of the output elasticity, which allows for a direct comparison with the findings of Breinlich et al. (2026). The specification is identical to the main specification (9) with the sole exception that the dependent variable is not adjusted by the trade elasticity (θ_k). The estimating equation is therefore:

$$\Delta \ln X_{KOR\,j,k} = \beta \Delta \ln Y_{KOR,k} + \alpha_{KOR} + \alpha_{KOR\,j} + \alpha_{KOR,K(k)} + \varepsilon_{KOR\,j,k}. \tag{A1}$$

The data, sample, endogenous variable, instrumental variable, and fixed effects are the same as those used in the main analysis in Section 5.1.

In this specification, the coefficient of interest, β, captures the output elasticity. As derived in the bilateral trade equation (4), this is the total elasticity of bilateral trade ($X_{ij,k}$) with respect to total sales ($Y_{i,k}$). This output elasticity arises from two channels. First, an increase in total sales lowers prices via the scale elasticity channel (γ_k). Second, this price decline increases sales in destination j via the trade elasticity channel (θ_k). The product of these two effects ($\gamma_k \theta_k$) is the output elasticity; since the specification imposes a common coefficient across sectors, the estimate of β is interpreted as a weighted average of the sector-level output elasticities.

The 2SLS estimate for the output elasticity (β) is 0.806 and is statistically significant, as reported in Table A11. This estimate is in line with both the structural estimate of 0.74 reported by Breinlich et al. (2026) and the calibrated value of 0.821 used in their quantitative analysis. Furthermore, the estimated output elasticity is below one, consistent with the condition $\gamma_k \theta_k < 1$ that ensures a unique counterfactual equilibrium (Kucheryavyy et al. (2023)).

Table A11: Estimates of Output Elasticity (HS-6)

	$\Delta \ln X_{KOR\,j,k}$		
	2SLS	OLS	Reduced-form
	(1)	(2)	(3)
$\Delta \ln Y_{KOR,k}$	0.806^{***}	0.412^{***}	
	(0.191)	(0.008)	
$\theta_k Share^{2018}_{JPN\,KOR,k}$			0.029^{***}
			(0.007)
First-stage Coefficient	0.036		
First-stage F-statistic	422.47		
Destination Fixed Effects	Yes	Yes	Yes
Sector (KSIC-5) Fixed Effects	Yes	Yes	Yes
Number of Goods	4,634	4,634	4,634
Observations	220,763	220,763	220,763

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Table A11 reports the estimation results for the output elasticity (β) at the HS-6 level using specification (A1). Column (1) presents the main 2SLS estimate, with its corresponding first-stage coefficient and F-statistic reported below. For this 2SLS estimation, the log difference in total sales ($\Delta \ln Y_{KOR,k}$) is instrumented by the product of the trade elasticity and Japan's 2018 import share in Korea ($\theta_k Share^{2018}_{JPN\,KOR,k}$). Columns (2) and (3) show the OLS and reduced-form estimates, respectively. The dependent variable ($\Delta \ln X_{KOR\,j,k}$) is not adjusted by the trade elasticity (θ_k). The independent variable ($\Delta \ln Y_{KOR,k}$) is proxied by the log difference in total exports. The analysis focuses on manufacturing sectors, which correspond to KSIC-2 codes 10 to 33.
Source: CEPII BACI (Gaulier and Zignago (2010)) and Statistics Korea.

B.3 Robustness: Varying the Aggregation Level of Fixed Effects

Table A12: Estimates of Scale Elasticity (KSIC-2 Fixed Effects)

	$\frac{1}{\theta_k}\Delta \ln X_{KOR\,j,k}$		
	2SLS	OLS	Reduced-form
	(1)	(2)	(3)
$\Delta \ln Y_{KOR,k}$	0.177^{***}	0.115^{***}	
	(0.051)	(0.002)	
$\theta_k Share^{2018}_{JPN\,KOR,k}$			0.005^{***}
			(0.002)
First-stage Coefficient	0.031		
First-stage F-statistic	326.80		
Destination Fixed Effects	Yes	Yes	Yes
Sector (KSIC-2) Fixed Effects	Yes	Yes	Yes
Number of Goods	4,634	4,634	4,634
Observations	220,764	220,764	220,764

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Table A12 reports the estimation results from re-estimating the main specification (9). The sole difference from the main analysis is that the origin-sector fixed effects ($\alpha_{KOR,K(k)}$) are defined at the 2-digit rather than the 5-digit level. Column (1) presents the 2SLS estimate, with its corresponding first-stage coefficient and F-statistic reported below. For this 2SLS estimation, the log difference in total sales ($\Delta \ln Y_{KOR,k}$) is instrumented by the product of the trade elasticity and Japan's 2018 import share in Korea ($\theta_k Share^{2018}_{JPN\,KOR,k}$). Columns (2) and (3) show the OLS and reduced-form estimates, respectively. The independent variable ($\Delta \ln Y_{KOR,k}$) is proxied by the log difference in total exports. The trade elasticity (θ_k) is taken from Giri et al. (2021) following Bartelme et al. (2025), assuming a common elasticity for all HS-6 goods within the same KSIC-2 sector. The analysis focuses on manufacturing sectors, which correspond to KSIC-2 codes 10 to 33.
Source: CEPII BACI (Gaulier and Zignago (2010)) and Statistics Korea.

Table A13: Estimates of Scale Elasticity (KSIC-3 Fixed Effects)

	$\frac{1}{\theta_k}\Delta \ln X_{KOR\,j,k}$		
	2SLS	OLS	Reduced-form
	(1)	(2)	(3)
$\Delta \ln Y_{KOR,k}$	0.142**	0.111***	
	(0.057)	(0.002)	
$\theta_k Share^{2018}_{JPN\,KOR,k}$			0.004**
			(0.002)
First-stage Coefficient	0.028		
First-stage F-statistic	271.27		
Destination Fixed Effects	Yes	Yes	Yes
Sector (KSIC-3) Fixed Effects	Yes	Yes	Yes
Number of Goods	4,634	4,634	4,634
Observations	220,764	220,764	220,764

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Table A13 reports the estimation results from re-estimating the main specification (9). The sole difference from the main analysis is that the origin-sector fixed effects ($\alpha_{KOR,K(k)}$) are defined at the 3-digit rather than the 5-digit level. Column (1) presents the 2SLS estimate, with its corresponding first-stage coefficient and F-statistic reported below. For this 2SLS estimation, the log difference in total sales ($\Delta \ln Y_{KOR,k}$) is instrumented by the product of the trade elasticity and Japan's 2018 import share in Korea ($\theta_k Share^{2018}_{JPN\,KOR,k}$). Columns (2) and (3) show the OLS and reduced-form estimates, respectively. The independent variable ($\Delta \ln Y_{KOR,k}$) is proxied by the log difference in total exports. The trade elasticity (θ_k) is taken from Giri et al. (2021) following Bartelme et al. (2025), assuming a common elasticity for all HS-6 goods within the same KSIC-2 sector. The analysis focuses on manufacturing sectors, which correspond to KSIC-2 codes 10 to 33.
Source: CEPII BACI (Gaulier and Zignago (2010)) and Statistics Korea.

C Appendix to Section 5.2

C.1 Theoretical Foundation of the Instrumental Variable

The derivation for the cross-country instrumental variable is analogous to the one detailed for the Korea-only case in Appendix B.1. The logic of the chain rule and the derivation of each component are the same. The total elasticity of origin i's total sales ($Y_{i,k}$) with respect to the shock to Korea's importing costs from Japan ($\tau_{JPN\,KOR,k}$) simplifies to:

$$\begin{aligned}\frac{\Delta \ln Y_{i,k}}{\Delta \ln \tau_{JPN\,KOR,k}} &= \frac{\Delta \ln p_{JPN\,KOR,k}}{\Delta \ln \tau_{JPN\,KOR,k}} \frac{\Delta \ln P_{KOR,k}}{\Delta \ln p_{JPN\,KOR,k}} \frac{\Delta \ln X_{i\,KOR,k}}{\Delta \ln P_{KOR,k}} \frac{\Delta \ln Y_{i,k}}{\Delta \ln X_{i\,KOR,k}} \\ &= 1 \times \frac{X_{JPN\,KOR,k}}{\sum_\ell X_{\ell\,KOR,k}} \times \theta_k \times \frac{X_{i\,KOR,k}}{\sum_j X_{ij,k}} \\ &= \theta_k \times \frac{X_{JPN\,KOR,k}}{\sum_\ell X_{\ell\,KOR,k}} \times \frac{X_{i\,KOR,k}}{\sum_j X_{ij,k}}.\end{aligned}$$

This total effect is the product of three components: the trade elasticity, Japan's market share in Korea, and the share of sales to Korea in origin i's total sales. The instrument used in Section 5.2, $IV_{i,k} = \theta_k Share^{2018}_{JPN\,KOR,k} EXShare^{2018}_{i\,KOR,k}$, is the empirical counterpart of this total effect, with its three components—the trade elasticity, Japan's import share in Korea, and the share of sales to Korea in origin i's total exports—corresponding to the three terms above. This structural correspondence establishes the relevance of the instrument.[48] Moreover, its demand-side nature ensures it is uncorrelated with the supply-side shocks, thus satisfying the exclusion restriction.

[48]The empirical shares use total imports and exports in place of the theoretically ideal total sales, because origins' domestic sales are unobserved at the HS-6 level.

C.2 Robustness: Varying the Aggregation Level of Fixed Effects

Table A14: Cross-Country Estimates of Scale Elasticity (KSIC-2 Fixed Effects)

	$\frac{1}{\theta_k}\Delta \ln X_{ij,k}$		
	2SLS	OLS	Reduced-form
	(1)	(2)	(3)
$\Delta \ln Y_{i,k}$	0.220***	0.104***	
	(0.052)	(0.001)	
$\theta_k Share^{2018}_{JPN\,KOR,k} EXShare^{2018}_{i\,KOR,k}$			0.041***
			(0.008)
First-stage Coefficient	0.185		
First-stage F-statistic	11.96		
Origin Fixed Effects	Yes	Yes	Yes
Origin-Destination Fixed Effects	Yes	Yes	Yes
Destination-Good Fixed Effects	Yes	Yes	Yes
Origin-Sector (KSIC-2) Fixed Effects	Yes	Yes	Yes
Number of Goods	4,667	4,667	4,667
Number of Origins	20	20	20
Observations	4,479,133	4,479,133	4,479,133

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Table A14 reports the estimation results from re-estimating the main specification (11). The sole difference from the main analysis is that the origin-sector fixed effects ($\alpha_{i,K(k)}$) are defined at the 2-digit rather than the 5-digit level. Column (1) presents the 2SLS estimate, with its corresponding first-stage coefficient and F-statistic reported below. For this 2SLS estimation, the log difference in total sales ($\Delta \ln Y_{i,k}$) is instrumented by the product of the trade elasticity, Japan's 2018 import share in Korea, and origin i's 2018 export share to Korea ($\theta_k Share^{2018}_{JPN\,KOR,k} EXShare^{2018}_{i\,KOR,k}$). Columns (2) and (3) show the OLS and reduced-form estimates, respectively. The independent variable ($\Delta \ln Y_{i,k}$) is proxied by the log difference in total exports. The trade elasticity (θ_k) is taken from Giri et al. (2021) following Bartelme et al. (2025), assuming a common elasticity for all HS-6 goods within the same KSIC-2 sector. The analysis focuses on manufacturing sectors, which correspond to KSIC-2 codes 10 to 33, and on Korea's 20 largest import sources other than Japan. Standard errors are clustered at the origin-good level.
Source: CEPII BACI (Gaulier and Zignago (2010)) and Statistics Korea.

Table A15: Cross-Country Estimates of Scale Elasticity (KSIC-3 Fixed Effects)

	$\frac{1}{\theta_k}\Delta \ln X_{ij,k}$		
	2SLS (1)	OLS (2)	Reduced-form (3)
$\Delta \ln Y_{i,k}$	0.214***	0.102***	
	(0.051)	(0.001)	
$\theta_k Share^{2018}_{JPN\,KOR,k} EXShare^{2018}_{i\,KOR,k}$			0.039***
			(0.008)
First-stage Coefficient	0.184		
First-stage F-statistic	11.66		
Origin Fixed Effects	Yes	Yes	Yes
Origin-Destination Fixed Effects	Yes	Yes	Yes
Destination-Good Fixed Effects	Yes	Yes	Yes
Origin-Sector (KSIC-3) Fixed Effects	Yes	Yes	Yes
Number of Goods	4,667	4,667	4,667
Number of Origins	20	20	20
Observations	4,479,129	4,479,129	4,479,129

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Table A15 reports the estimation results from re-estimating the main specification (11). The sole difference from the main analysis is that the origin-sector fixed effects ($\alpha_{i,K(k)}$) are defined at the 3-digit rather than the 5-digit level. Column (1) presents the 2SLS estimate, with its corresponding first-stage coefficient and F-statistic reported below. For this 2SLS estimation, the log difference in total sales ($\Delta \ln Y_{i,k}$) is instrumented by the product of the trade elasticity, Japan's 2018 import share in Korea, and origin i's 2018 export share to Korea ($\theta_k Share^{2018}_{JPN\,KOR,k} EXShare^{2018}_{i\,KOR,k}$). Columns (2) and (3) show the OLS and reduced-form estimates, respectively. The independent variable ($\Delta \ln Y_{i,k}$) is proxied by the log difference in total exports. The trade elasticity (θ_k) is taken from Giri et al. (2021) following Bartelme et al. (2025), assuming a common elasticity for all HS-6 goods within the same KSIC-2 sector. The analysis focuses on manufacturing sectors, which correspond to KSIC-2 codes 10 to 33, and on Korea's 20 largest import sources other than Japan. Standard errors are clustered at the origin-good level.
Source: CEPII BACI (Gaulier and Zignago (2010)) and Statistics Korea.

C.3 Robustness: Varying the Set of Origin Countries

Table A16: Cross-Country Estimates of Scale Elasticity (10 Largest Import Sources)

	$\frac{1}{\theta_k}\Delta \ln X_{ij,k}$		
	2SLS	OLS	Reduced-form
	(1)	(2)	(3)
$\Delta \ln Y_{i,k}$	0.181***	0.104***	
	(0.042)	(0.001)	
$\theta_k Share^{2018}_{JPN\,KOR,k} EXShare^{2018}_{i\,KOR,k}$			0.041***
			(0.010)
First-stage Coefficient	0.227		
First-stage F-statistic	16.14		
Origin Fixed Effects	Yes	Yes	Yes
Origin-Destination Fixed Effects	Yes	Yes	Yes
Destination-Good Fixed Effects	Yes	Yes	Yes
Origin-Sector (KSIC-5) Fixed Effects	Yes	Yes	Yes
Number of Goods	4,667	4,667	4,667
Number of Origins	10	10	10
Observations	1,935,117	1,935,117	1,935,117

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Table A16 reports the estimation results from re-estimating the main specification (11). The sole difference from the main analysis is that the sample is restricted to Korea's 10 largest import sources, rather than the 20 largest. Column (1) presents the 2SLS estimate, with its corresponding first-stage coefficient and F-statistic reported below. For this 2SLS estimation, the log difference in total sales ($\Delta \ln Y_{i,k}$) is instrumented by the product of the trade elasticity, Japan's 2018 import share in Korea, and origin i's 2018 export share to Korea ($\theta_k Share^{2018}_{JPN\,KOR,k} EXShare^{2018}_{i\,KOR,k}$). Columns (2) and (3) show the OLS and reduced-form estimates, respectively. The independent variable ($\Delta \ln Y_{i,k}$) is proxied by the log difference in total exports. The trade elasticity (θ_k) is taken from Giri et al. (2021) following Bartelme et al. (2025), assuming a common elasticity for all HS-6 goods within the same KSIC-2 sector. The analysis focuses on manufacturing sectors, which correspond to KSIC-2 codes 10 to 33. Standard errors are clustered at the origin-good level.
Source: CEPII BACI (Gaulier and Zignago (2010)) and Statistics Korea.

Table A17: Cross-Country Estimates of Scale Elasticity (Major Competitors of Japan)

	$\frac{1}{\theta_k}\Delta \ln X_{ij,k}$		
	2SLS	OLS	Reduced-form
	(1)	(2)	(3)
$\Delta \ln Y_{i,k}$	0.172^{***}	0.098^{***}	
	(0.040)	(0.002)	
$\theta_k Share^{2018}_{JPN\,KOR,k} EXShare^{2018}_{i\,KOR,k}$			0.044^{***}
			(0.013)
First-stage Coefficient	0.254		
First-stage F-statistic	12.53		
Origin Fixed Effects	Yes	Yes	Yes
Origin-Destination Fixed Effects	Yes	Yes	Yes
Destination-Good Fixed Effects	Yes	Yes	Yes
Origin-Sector (KSIC-5) Fixed Effects	Yes	Yes	Yes
Number of Goods	4,667	4,667	4,667
Number of Origins	4	4	4
Observations	1,442,759	1,442,759	1,442,759

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Table A17 reports the estimation results from re-estimating the main specification (11). The sole difference from the main analysis is that the sample is restricted to the four major suppliers to Korea that compete with Japan: China, the United States, Germany, and Taiwan. Column (1) presents the 2SLS estimate, with its corresponding first-stage coefficient and F-statistic reported below. For this 2SLS estimation, the log difference in total sales ($\Delta \ln Y_{i,k}$) is instrumented by the product of the trade elasticity, Japan's 2018 import share in Korea, and origin i's 2018 export share to Korea ($\theta_k Share^{2018}_{JPN\,KOR,k} EXShare^{2018}_{i\,KOR,k}$). Columns (2) and (3) show the OLS and reduced-form estimates, respectively. The independent variable ($\Delta \ln Y_{i,k}$) is proxied by the log difference in total exports. The trade elasticity (θ_k) is taken from Giri et al. (2021) following Bartelme et al. (2025), assuming a common elasticity for all HS-6 goods within the same KSIC-2 sector. The analysis focuses on manufacturing sectors, which correspond to KSIC-2 codes 10 to 33. Standard errors are clustered at the origin-good level.
Source: CEPII BACI (Gaulier and Zignago (2010)) and Statistics Korea.

D Appendix to Section 6.1

D.1 Derivation of Equation (12)

The bilateral trade equation (4) can be expanded as follows:

$$
\begin{aligned}
X_{ij,k} &= \left(\frac{\tau_{ij,k} w_i^{1+\gamma_k}}{A_{i,k}}\right)^{-\theta_k} Y_{i,k}^{\gamma_k \theta_k} X_{j,k} P_{j,k}^{\theta_k} \\
&= \left(\frac{\tau_{ij,k} w_i^{1+\gamma_k}}{A_{i,k}}\right)^{-\theta_k} (w_i L_{i,k})^{\gamma_k \theta_k} P_{j,k}^{\theta_k} X_{j,k} \\
&= \left(\frac{\tau_{ij,k} w_i}{A_{i,k} L_{i,k}^{\gamma_k}}\right)^{-\theta_k} \frac{X_{j,k}}{\sum_\ell p_{\ell j,k}^{-\theta_k}} \\
&= \frac{p_{ij,k}^{-\theta_k}}{\sum_\ell p_{\ell j,k}^{-\theta_k}} \beta_{j,k} w_j L_j. \qquad \text{(A2)}
\end{aligned}
$$

The second line is obtained by substituting the zero-profit condition, $w_i L_{i,k} = Y_{i,k}$. The third line incorporates the definition of the CES price index, $P_{j,k} = (\sum_\ell p_{\ell j,k}^{-\theta_k})^{-1/\theta_k}$. The last line is derived by substituting total sectoral expenditure ($X_{j,k} = \beta_{j,k} w_j L_j$) and the pricing condition $p_{ij,k} = c_{ij,k}$ from equation (2).

The market clearing condition (5) can be expressed in terms of expenditure shares ($\lambda_{ij,k}$) by substituting equation (A2) as follows:

$$
\begin{aligned}
w_i L_i &= \sum_k \sum_j X_{ij,k} \\
&= \sum_k \sum_j \frac{p_{ij,k}^{-\theta_k}}{\sum_\ell p_{\ell j,k}^{-\theta_k}} \beta_{j,k} w_j L_j \\
&= \sum_k \sum_j \frac{X_{ij,k}}{X_{j,k}} \beta_{j,k} w_j L_j \\
&= \sum_k \sum_j \lambda_{ij,k} \beta_{j,k} w_j L_j. \qquad \text{(A3)}
\end{aligned}
$$

The third line is obtained by the standard property of CES demand systems, where a variety's expenditure share is equal to its relative price term. The last line applies the definition of $\lambda_{ij,k}$.

Denoting the proportional change in any variable x as $\hat{x}$ (the ratio of its counterfactual value to its initial value), equation (A3) can be re-written as follows:

$$
\begin{aligned}
w_i L_i \hat{w}_i \hat{L}_i &= \sum_k \sum_j \lambda_{ij,k} \beta_{j,k} w_j L_j \hat{\lambda}_{ij,k} \hat{\beta}_{j,k} \hat{w}_j \hat{L}_j, \\
\hat{w}_i &= \frac{1}{w_i L_i} \sum_k \sum_j \lambda_{ij,k} \beta_{j,k} w_j L_j \hat{\lambda}_{ij,k} \hat{w}_j \\
&= \frac{1}{Y_i} \sum_k \sum_j \lambda_{ij,k} X_{j,k} \hat{\lambda}_{ij,k} \hat{w}_j. \qquad \text{(A4)}
\end{aligned}
$$

The second line is derived from the first line by dividing by $w_i L_i$ and assuming both constant total labor ($\hat{L}_i = \hat{L}_j = 1$) and fixed preference shifters ($\hat{\beta}_{j,k} = 1$). The last line is obtained by substituting total income ($Y_i = w_i L_i$) and total sectoral expenditure ($X_{j,k} = \beta_{j,k} w_j L_j$). This completes the derivation of equation (12).

D.2 Derivation of Equation (13)

By applying the standard property of CES demand systems, the expenditure share of origin i in destination j's market for sector k ($\lambda_{ij,k}$) can be expressed as follows:

$$\lambda_{ij,k} = \frac{X_{ij,k}}{X_{j,k}} = \frac{p_{ij,k}^{-\theta_k}}{\sum_\ell p_{\ell j,k}^{-\theta_k}}.$$

Expressing this expenditure share in proportional changes ($\hat{x}$) yields the following derivation:

$$\begin{aligned}
\hat{\lambda}_{ij,k} &= \frac{p_{ij,k}^{-\theta_k}\hat{p}_{ij,k}^{-\theta_k}}{\lambda_{ij,k}\sum_\ell p_{\ell j,k}^{-\theta_k}\hat{p}_{\ell j,k}^{-\theta_k}} \\
&= \frac{\hat{p}_{ij,k}^{-\theta_k}}{\sum_\ell \frac{p_{\ell j,k}^{-\theta_k}}{\sum_m p_{mj,k}^{-\theta_k}}\hat{p}_{\ell j,k}^{-\theta_k}} \\
&= \frac{\hat{p}_{ij,k}^{-\theta_k}}{\sum_\ell \lambda_{\ell j,k}\hat{p}_{\ell j,k}^{-\theta_k}} \\
&= \frac{\left(\frac{\hat{\tau}_{ij,k}\hat{w}_i}{\hat{A}_{i,k}\hat{L}_{i,k}^{\gamma_k}}\right)^{-\theta_k}}{\sum_\ell \lambda_{\ell j,k}\left(\frac{\hat{\tau}_{\ell j,k}\hat{w}_\ell}{\hat{A}_{\ell,k}\hat{L}_{\ell,k}^{\gamma_k}}\right)^{-\theta_k}} \\
&= \frac{\hat{\tau}_{ij,k}^{-\theta_k}\hat{L}_{i,k}^{\gamma_k\theta_k}\hat{w}_i^{-\theta_k}}{\sum_\ell \lambda_{\ell j,k}\hat{\tau}_{\ell j,k}^{-\theta_k}\hat{L}_{\ell,k}^{\gamma_k\theta_k}\hat{w}_\ell^{-\theta_k}}. \qquad \text{(A5)}
\end{aligned}$$

The derivation begins by expressing the change in share ($\hat{\lambda}_{ij,k}$) as a function of the changes in prices ($\hat{p}_{ij,k}$). The pricing condition $p_{ij,k} = c_{ij,k}$ from equation (2), also in proportional changes, is then substituted for $\hat{p}_{ij,k}$ to express $\hat{\lambda}_{ij,k}$ in terms of the underlying cost shifters. Assuming that the exogenous productivity parameter is unchanged ($\hat{A}_{i,k} = 1$), this yields the final expression for the change in expenditure shares. This completes the derivation of equation (13).

D.3 Derivation of Equation (14)

The derivation for the change in sectoral labor allocation ($\hat{L}_{i,k}$) begins with the zero-profit condition and the definition of total sales for an origin i in sector k:

$$w_i L_{i,k} = Y_{i,k} = \sum_j X_{ij,k}.$$

Expressing this condition in proportional changes yields the following:

$$w_i L_{i,k} \hat{w}_i \hat{L}_{i,k} = Y_{i,k} \hat{Y}_{i,k} = \sum_j X_{ij,k} \hat{X}_{ij,k}.$$

The change in bilateral trade values ($\hat{X}_{ij,k}$) can be further expressed as $\hat{X}_{ij,k} = \hat{\lambda}_{ij,k} \hat{w}_j$. This expression is derived by combining the definition of the expenditure share ($X_{ij,k} = \lambda_{ij,k} X_{j,k}$) with total sectoral expenditure ($X_{j,k} = \beta_{j,k} w_j L_j$), under the assumptions of constant total labor ($\hat{L}_j = 1$) and fixed preference shifters ($\hat{\beta}_{j,k} = 1$). Substituting $\hat{X}_{ij,k} = \hat{\lambda}_{ij,k} \hat{w}_j$ into the equation above and solving for $\hat{L}_{i,k}$ gives the final expression:

$$\hat{L}_{i,k} = \frac{1}{Y_{i,k} \hat{w}_i} \sum_j X_{ij,k} \hat{\lambda}_{ij,k} \hat{w}_j.$$

This completes the derivation of equation (14).

E Appendix to Section 6.2

E.1 Data Imputation for Domestic Sales

The primary dataset (CEPII BACI; Gaulier and Zignago (2010)) lacks information on domestic sales ($X_{ii,k}$), which is a necessary component for the quantitative analysis. This appendix details the procedure used to impute this missing variable.

The procedure begins by sourcing aggregate sector-level data from the International Trade and Production Database for Simulation, Release 1.1 (ITPD-S; Borchert et al. (2024)). These data are first summed over the pre-event period (2015–2018). Using the concordance tables provided with the ITPD-S, the original industry classifications are then mapped to the SIC-2 level. From this prepared dataset, the domestic sales share for each country at the SIC-2 level is calculated. These shares are in turn assigned to the corresponding KSIC-2 sectors used in this paper. Finally, a proportionality assumption is applied to impute the domestic sales shares at the more granular HS-6 level. This assumption posits that the domestic sales share of a 6-digit sector is identical to that of its corresponding aggregate 2-digit sector. Formally, this is expressed as:

$$\frac{X_{ii,k}}{Y_{i,k}} = \frac{X_{ii,K(k)}}{Y_{i,K(k)}} \quad \text{for all } k,$$

where $K(k)$ denotes the aggregate KSIC-2 sector to which the HS-6 good k belongs.

E.2 Country Sample for Quantitative Analysis

This appendix details the construction of the 23 country units used in the quantitative analysis. The country sample consists of Korea, Japan, and the 20 major import sources from Section 5.2, namely the 20 largest sources of Korea's imports other than Japan, based on 2018 trade data from the CEPII BACI database. These 20 major import sources are:

> Australia, China, France, Germany, India, Indonesia, Iraq, Italy, Kuwait, Malaysia, Netherlands, Qatar, Russia, Saudi Arabia, Singapore, Taiwan, United Arab Emirates, United Kingdom, United States, and Vietnam.

For computational tractability, all remaining countries are consolidated into a single "Rest of the World" (RoW) unit, yielding 23 country units. Trade data for Taiwan are proxied using the data for "Other Asia, not elsewhere specified" (ISO code 490), following the CEPII documentation. This is a reliable proxy as, in practice, this category almost exclusively contains trade data for Taiwan.

E.3 Sector-Level Scale and Trade Elasticities from Bartelme et al. (2025)

This appendix details the sector-level scale and trade elasticities used in the quantitative analysis in Section 6; the trade elasticities are also used in the estimation in Section 5. The scale elasticities (γ_k) are the preferred IV estimates of Bartelme et al. (2025), reported in column (2) of their Table 1, covering 15 manufacturing sectors. The trade elasticities (θ_k) are taken from column (6) of the same table, which reports the estimates of Giri et al. (2021) used in their analysis.

The sector classification of Bartelme et al. (2025) is based on 2-digit SIC sectors, some of which are combined into single sectors. These 15 sectors are assigned to the 24 manufacturing sectors (KSIC-2 codes 10 to 33) used in this paper. A single estimate can therefore be assigned to multiple KSIC-2 sectors; for example, the estimate for Food, Beverages and Tobacco is assigned to KSIC-2 codes 10, 11, and 12. Table A18 reports the full assignment.

Two sectors, furniture (KSIC-2 code 32) and other manufacturing (KSIC-2 code 33), have no counterpart among the 15 sectors of Bartelme et al. (2025). The scale elasticity for these two sectors is conservatively set to zero. This treatment follows the practice of Bartelme et al. (2025), who assume no scale economies in the sectors outside their estimation sample. The trade elasticity for these two sectors is set to the estimate for furniture (4.47) from Giri et al. (2021), as they are commonly grouped into a single sector in aggregated industrial classifications.

Table A18: Sector-Level Scale and Trade Elasticities from Bartelme et al. (2025)

KSIC-2	Description	Sector in Bartelme et al. (2025)	γ_k	θ_k
10	Food Products	Food, Beverages and Tobacco	0.24	3.57
11	Beverages	Food, Beverages and Tobacco	0.24	3.57
12	Tobacco Products	Food, Beverages and Tobacco	0.24	3.57
13	Textiles, Except Apparel	Textiles	0.21	4.43
14	Wearing Apparel, Clothing Accessories and Fur Articles	Textiles	0.21	4.43
15	Leather, Luggage and Footwear	Textiles	0.21	4.43
16	Wood Products, Except Furniture	Wood Products	0.19	4.32
17	Pulp, Paper and Paper Products	Paper Products	0.31	2.97
18	Printing and Reproduction of Recorded Media	Paper Products	0.31	2.97
19	Coke and Refined Petroleum Products	Coke/Petroleum Products	0.11	8.94
20	Chemicals, Except Pharmaceuticals	Chemicals	0.22	3.75
21	Pharmaceuticals and Medicinal Chemicals	Chemicals	0.22	3.75
22	Rubber and Plastic Products	Rubber and Plastics	0.20	4.13
23	Other Non-metallic Mineral Products	Mineral Products	0.19	5.14
24	Basic Metals	Basic Metals	0.11	8.94
25	Fabricated Metal Products	Fabricated Metals	0.18	5.07
26	Electronic Components, Computer and Communication Equipment	Computers and Electronics	0.27	3.27
27	Medical, Precision and Optical Instruments	Computers and Electronics	0.27	3.27
28	Electrical Equipment	Electrical Machinery, NEC	0.25	3.27
29	Other Machinery and Equipment	Machinery and Equipment	0.26	3.27
30	Motor Vehicles and Trailers	Motor Vehicles	0.19	4.47
31	Other Transport Equipment	Other Transport Equipment	0.20	4.47
32	Furniture	–	0	4.47
33	Other Manufacturing	–	0	4.47

Notes: Table A18 reports the sector-level scale elasticities (γ_k) and trade elasticities (θ_k) used in the quantitative analysis, assigned to the manufacturing sectors of the Korean Standard Industrial Classification (KSIC-2 codes 10 to 33). The scale elasticities are the IV estimates reported in column (2) of Table 1 in Bartelme et al. (2025). The trade elasticities are from column (6) of the same table, sourced from Giri et al. (2021). Since the 15 sectors of Bartelme et al. (2025) are assigned to the 24 KSIC-2 sectors, a single estimate can be assigned to multiple KSIC-2 sectors. The sectors with KSIC-2 codes 32 and 33 have no counterpart in Bartelme et al. (2025); their scale elasticity is conservatively set to zero. Their trade elasticity is the estimate for furniture from Giri et al. (2021), as the two sectors are commonly grouped into a single sector in aggregated industrial classifications.
Source: Bartelme et al. (2025).

E.4 Derivation of Equation (15)

This appendix details the derivation of equation (15), which is used to calibrate the unobserved trade cost shock to Korea's imports from Japan ($\hat{\tau}_{JP\,KR,k}$). The derivation starts by totally differentiating the bilateral trade equation (4) in logs with respect to the trade cost. Using the chain rule, the resulting total effect can be decomposed into a partial equilibrium (PE) effect and a first-order general equilibrium (GE) effect. The PE effect captures the direct impact of the shock through the price of Japanese goods in Korea ($p_{JP\,KR,k}$), while the GE effect captures the indirect impact of the shock through Korea's sectoral price index ($P_{KR,k}$). This decomposition is expressed in elasticities as follows:

$$\begin{aligned}
\frac{\Delta \log X_{JP\,KR,k}}{\Delta \log \tau_{JP\,KR,k}} &= \frac{\Delta \log X_{JP\,KR,k}}{\Delta \log p_{JP\,KR,k}} \frac{\Delta \log p_{JP\,KR,k}}{\Delta \log \tau_{JP\,KR,k}} + \frac{\Delta \log X_{JP\,KR,k}}{\Delta \log P_{KR,k}} \frac{\Delta \log P_{KR,k}}{\Delta \log p_{JP\,KR,k}} \frac{\Delta \log p_{JP\,KR,k}}{\Delta \log \tau_{JP\,KR,k}} \\
&= -\theta_k + \theta_k \frac{X_{JP\,KR,k}}{\sum_i X_{i\,KR,k}} \\
&= -\theta_k \big(1 - \lambda_{JP\,KR,k}\big).
\end{aligned}$$

The second line is derived by substituting the elasticities implied by equations (3) and (4), together with the standard property of CES demand systems. The last line applies the definition of $\lambda_{ij,k}$. This equation is rearranged to solve for the change in trade cost ($\Delta \log \tau_{JP\,KR,k}$) as a function of the change in imports ($\Delta \log X_{JP\,KR,k}$). Finally, expressing this in "hat" notation ($\log \hat{x} = \Delta \log x$) yields equation (15).

E.5 Alternative Interpretation of the Shock

The baseline analysis interprets the shock from the 2019 Korea-Japan Trade Dispute as an increase in the bilateral trade cost on Korea's imports from Japan ($\tau_{JP\,KR,k}$), calibrated as the cost-equivalent of the threat (Section 6.2). In the Armington framework, an observationally equivalent alternative is a shift in Korea's preferences away from Japanese goods. This appendix formalizes the taste interpretation, shows that all equilibrium responses coincide exactly with those of the baseline counterfactual, and quantifies welfare under this interpretation. The main result is that the role of scale economies—mitigating the impact of the shock on Korea while exacerbating it for Japan—is unchanged; only the level of Korea's welfare change differs across the two interpretations.

Preferences with Taste Shifters. The lower-tier preferences in Section 4 are extended with a taste shifter $b_{ij,k} > 0$, which scales the utility contribution of a unit of good k produced in origin i and consumed in destination j:

$$Q_{j,k} = \Big(\sum_i b_{ij,k}^{\frac{1}{1+\theta_k}} \, q_{ij,k}^{\frac{\theta_k}{1+\theta_k}} \Big)^{\frac{1+\theta_k}{\theta_k}}.$$

The exponent on $b_{ij,k}$ is a normalization that makes the taste shifter enter linearly in the demand function below.[49] The model in Section 4 corresponds to the special case in which $b_{ij,k}$ is constant; since the model is solved in proportional changes, the level of $b_{ij,k}$ is absorbed into the initial expenditure shares, and the implicit assumption in the baseline analysis is that preferences do not change with the shock ($\hat{b}_{ij,k} = 1$).

Demand. The representative consumer in destination j allocates the sectoral expenditure $X_{j,k} = \beta_{j,k} w_j L_j$ across origins to maximize $Q_{j,k}$. Then, the demand in destination j for good k produced in origin i is:

$$q_{ij,k} = b_{ij,k} \, p_{ij,k}^{-1-\theta_k} \, P_{j,k}^{\theta_k} \, \beta_{j,k} w_j L_j, \tag{A6}$$

where $P_{j,k}$ is the price index for sector k in destination j given by $P_{j,k} = \big(\sum_i b_{ij,k} \, p_{ij,k}^{-\theta_k} \big)^{-\frac{1}{\theta_k}}$. Setting $b_{ij,k} = 1$ recovers the demand function (1) and the price index in Section 4.

[49] Any alternative exponent is equivalent up to a relabeling of $b_{ij,k}$. The parameterization follows the convention of Feenstra (1994).

Observational Equivalence. The same derivation as in Section 4.2—substituting the price equation (3) into the demand function (A6)—yields the taste-augmented counterpart of the bilateral trade equation (4):

$$X_{ij,k} = \left(b_{ij,k}\,\tau_{ij,k}^{-\theta_k}\right)\left(\frac{w_i^{1+\gamma_k}}{A_{i,k}}\right)^{-\theta_k} Y_{i,k}^{\gamma_k\theta_k}\,X_{j,k}\,P_{j,k}^{\theta_k}.$$

Relative to equation (4), the single difference is that the trade cost now enters together with the taste shifter, through the composite term $b_{ij,k}\,\tau_{ij,k}^{-\theta_k}$. Bilateral trade flows therefore identify only this composite: an increase in $\tau_{JP\,KR,k}$ and a decline in $b_{JP\,KR,k}$ satisfying $\hat{b}_{JP\,KR,k} = \hat{\tau}_{JP\,KR,k}^{-\theta_k}$ generate identical data, and the two interpretations cannot be distinguished from trade flows alone.

Equilibrium Responses. Applying the hat algebra of Section 6.1 to the taste-augmented model leaves equations (12) and (14) unchanged: the taste shifter enters both equations solely through the expenditure shares, and the upper-tier expenditure shares $\beta_{j,k}$ are unaffected by the lower-tier taste shifter. The modification is therefore confined to the expenditure-share equation (13), where $\hat{b}_{ij,k}$ enters alongside $\hat{\tau}_{ij,k}^{-\theta_k}$:

$$\hat{\lambda}_{ij,k} = \frac{\hat{b}_{ij,k}\,\hat{\tau}_{ij,k}^{-\theta_k}\,\hat{L}_{i,k}^{\gamma_k\theta_k}\,\hat{w}_i^{-\theta_k}}{\sum_\ell \lambda_{\ell j,k}\,\hat{b}_{\ell j,k}\,\hat{\tau}_{\ell j,k}^{-\theta_k}\,\hat{L}_{\ell,k}^{\gamma_k\theta_k}\,\hat{w}_\ell^{-\theta_k}}. \tag{A7}$$

The taste counterfactual sets $\hat{\tau}_{ij,k} = 1$, $\hat{b}_{JP\,KR,k} = \hat{\tau}_{JP\,KR,k}^{*\,-\theta_k}$, and $\hat{b}_{ij,k} = 1$ for all other pairs, where $\hat{\tau}_{ij,k}^*$ denotes the trade cost shock calibrated under Shock Calibration in Section 6.2. Because $\hat{b}_{ij,k}$ and $\hat{\tau}_{ij,k}^{-\theta_k}$ enter equation (A7) only through their product, the system of equilibrium conditions is numerically identical to that of the baseline counterfactual. The solutions for wages, expenditure shares, and sectoral labor allocations ($\hat{w}_i$, $\hat{\lambda}_{ij,k}$, $\hat{L}_{i,k}$)—and hence all trade, production, and scale responses—coincide exactly with those of the baseline. The two interpretations differ only in the evaluation of welfare.

Welfare under the Taste Shift. Under the trade cost interpretation, preferences are unchanged and the welfare effect on country j is $\hat{U}_j = \hat{w}_j/\hat{P}_j$, which is uniquely defined. Under the taste interpretation, the utility function itself changes, and the welfare evaluation requires a choice of reference preferences (Fisher and Shell (1972)): the cost of living can be measured with either the pre-shock or the post-shock preferences, and the two measures can differ.[50]

This taste counterfactual evaluates welfare under the *post-shock* preferences. This choice takes the preference shift as genuine and is the most favorable case for the taste interpretation.[51] The shift in tastes is then not itself a welfare loss, because the reallocation of expenditure reflects what consumers now demand. Accordingly, the cost of living in both periods is evaluated with the same post-shock taste weights, so that only prices differ, and the change in the sectoral price index becomes:

$$\hat{P}_{j,k} = \left(\frac{\sum_{\ell} \lambda_{\ell j,k} \, \hat{b}_{\ell j,k} \, \hat{p}_{\ell j,k}^{-\theta_k}}{\sum_{\ell} \lambda_{\ell j,k} \, \hat{b}_{\ell j,k}} \right)^{-\frac{1}{\theta_k}}. \tag{A8}$$

Without the denominator, equation (A8) would reduce to the unadjusted price index, $\left(\sum_{\ell} \lambda_{\ell j,k} \hat{b}_{\ell j,k} \hat{p}_{\ell j,k}^{-\theta_k}\right)^{-1/\theta_k}$. This index rises mechanically with the decline in $b_{JP\,KR,k}$ alone; no price needs to change. The denominator removes exactly this mechanical increase, so that only actual price changes, including the scale-induced ones, affect measured welfare. This choice of reference matters only for Korea: the taste shift is in Korea's preferences over Japanese goods, while Japan's utility function is unchanged, so Japan's welfare remains uniquely defined.

Welfare Results. Table A19 confirms that the main result of the quantitative analysis is robust to the interpretation of the shock: the presence of scale economies mitigates the welfare loss in Korea, while exacerbating it in Japan. Specifically, Columns (1) and (2) show that, under the taste interpretation, scale economies improve the welfare outcome for Korea, from 0.083% to 0.140%, and worsen it for Japan, from −0.051% to −0.057%. These directions are the same as under the trade cost interpretation (Columns (3) and (4)), which correspond to Columns (1) and (2) of Table 7 in Section 6.4; only the level of Korea's welfare change differs.[52]

Other Interpretations. The main result—that the presence of scale economies mitigates the welfare loss in Korea and exacerbates it in Japan—is also unchanged under other in-

[50] See Redding and Weinstein (2020) for a treatment of price index measurement under CES taste shocks.

[51] An evaluation under the pre-shock preferences yields approximately the same result as the post-shock evaluation, as discussed at the end of this appendix.

[52] The difference lies in the price index. Under the trade cost interpretation, the higher cost of sourcing from Japan raises Korea's cost of living. Under the taste interpretation, the shock is a shift in preferences, so equation (A8) does not count it as a rise in the cost of living; only actual price changes enter the index. The wage gain, by contrast, is identical across the two interpretations and exceeds the remaining price increase, turning Korea's welfare change positive. For Japan, the welfare levels are identical, as the taste shift is in Korea's preferences and its own utility function is unchanged.

Table A19: Welfare Changes under Alternative Interpretations of the Shock

	Taste Shift		Trade Cost	
	No Scale	Scale	No Scale	Scale
(% change)	(1)	(2)	(3)	(4)
Korea	0.083	0.140	−0.282	−0.226
Japan	−0.051	−0.057	−0.051	−0.057

Notes: Table A19 reports the percentage change in welfare for Korea and Japan from the counterfactual analysis of the 2019 Korea-Japan Trade Dispute, under the two interpretations of the shock. Columns (1) and (2) represent the shock as a decline in the taste shifter for Japanese goods in Korea ($\hat{b}_{JP\,KR,k} = \hat{\tau}^{*\,-\theta_k}_{JP\,KR,k}$), with welfare evaluated under the post-shock preferences using equation (A8). Columns (3) and (4) represent the shock as an increase in the bilateral trade cost, as in Section 6.2; they correspond to Columns (1) and (2) of Table 7, respectively. In the No Scale columns, scale economies are absent ($\gamma_k = 0$); in the Scale columns, the scale elasticity is set to the sector-level estimates of Bartelme et al. (2025), as in Section 6.2.

terpretations of the shock. Within the taste interpretation, the welfare evaluation requires a choice of reference preferences, and the analysis above adopted the post-shock preferences. An evaluation under the pre-shock preferences instead yields approximately the same result as the post-shock evaluation: the two references average the same equilibrium price changes and differ only in the taste weights placed on Japanese goods, whose prices change little.

Alternatively, the shock can be interpreted as a deterioration in the effective quality (reliability) of Japanese goods, a natural reading for buyers of intermediate and capital goods. In the model, this reading corresponds to measuring consumption in effective units, $\tilde{q}_{ij,k} = b^{1/\theta_k}_{ij,k} q_{ij,k}$, and leaves preferences over these units unchanged. A lower $b_{ij,k}$ then implies that each unit of the good delivers fewer effective units, which is exactly what an increase in the trade cost represents. Under the iceberg trade cost, more units must be shipped for one unit to arrive, so both readings raise the cost of an effective unit in the same way. The welfare calculation therefore coincides with that of the trade cost interpretation in Section 6.

F Appendix to Section 6.3

F.1 Calculation of Untargeted Moments

This appendix details how the untargeted moments used for the model validation in Section 6.3 are computed ex post from the model's equilibrium solution. These moments are the changes in Korea's total sales ($\hat{Y}_{KOR,k}$), export prices ($\hat{p}_{KOR\,j,k}$), and export quantities ($\hat{q}_{KOR\,j,k}$). The equilibrium changes in the endogenous variables ($\hat{w}_i$, $\hat{\lambda}_{ij,k}$, $\hat{L}_{i,k}$) are obtained by solving the system of equations (12), (13), and (14). The untargeted moments are then derived from these solutions as follows.

The change in Korea's total sales ($\hat{Y}_{KOR,k}$) is derived from the zero-profit condition, $Y_{KOR,k} = w_{KOR} L_{KOR,k}$. Expressing this in proportional changes gives:

$$\hat{Y}_{KOR,k} = \hat{w}_{KOR} \hat{L}_{KOR,k}.$$

This value is computed using the solved equilibrium values for the change in Korea's wage ($\hat{w}_{KOR}$) and the change in its sectoral labor allocation ($\hat{L}_{KOR,k}$).

The change in Korea's export price to destination j ($\hat{p}_{KOR\,j,k}$) is derived from the pricing condition $p_{ij,k} = c_{ij,k}$ in equation (2). Expressing this in proportional changes gives:

$$\hat{p}_{KOR\,j,k} = \frac{\hat{\tau}_{KOR\,j,k} \hat{w}_{KOR}}{\hat{A}_{KOR,k} \hat{L}_{KOR,k}^{\gamma_k}}.$$

Assuming that the exogenous productivity parameter is unchanged ($\hat{A}_{KOR,k} = 1$) and that the trade costs on Korea's exports remain unchanged ($\hat{\tau}_{KOR\,j,k} = 1$), this value is computed using the solved equilibrium values for $\hat{w}_{KOR}$ and $\hat{L}_{KOR,k}$.

The change in export quantity to destination j ($\hat{q}_{KOR\,j,k}$) is derived from its definition, $q_{KOR\,j,k} = X_{KOR\,j,k} / p_{KOR\,j,k}$. The price change ($\hat{p}_{KOR\,j,k}$) is taken from the previous step. The change in bilateral export values ($\hat{X}_{KOR\,j,k}$) is derived by combining the definition of the expenditure share ($X_{KOR\,j,k} = \lambda_{KOR\,j,k} X_{j,k}$) with total sectoral expenditure ($X_{j,k} = \beta_{j,k} w_j L_j$). Expressing this in proportional changes gives:

$$\hat{X}_{KOR\,j,k} = \hat{\lambda}_{KOR\,j,k} \hat{\beta}_{j,k} \hat{w}_j \hat{L}_j.$$

Assuming fixed preference shifters ($\hat{\beta}_{j,k} = 1$) and constant total labor in destination j ($\hat{L}_j = 1$), this simplifies to $\hat{X}_{KOR\,j,k} = \hat{\lambda}_{KOR\,j,k} \hat{w}_j$. This value is computed using the solved equilibrium values for $\hat{\lambda}_{KOR\,j,k}$ and $\hat{w}_j$. Finally, the change in export quantities is computed by combining these results:

$$\hat{q}_{KOR\,j,k} = \frac{\hat{X}_{KOR\,j,k}}{\hat{p}_{KOR\,j,k}} = \frac{\hat{\lambda}_{KOR\,j,k}\hat{w}_j}{\hat{p}_{KOR\,j,k}}.$$

F.2 Statistical Tests of Untargeted Moments

This appendix provides the detailed statistical results for the model validation exercise in Section 6.3. The validation tests whether the model's predicted untargeted moments—the changes in Korea's total sales ($\hat{Y}_{KOR,k}$), export prices ($\hat{p}_{KOR\,j,k}$), and export quantities ($\hat{q}_{KOR\,j,k}$)—are consistent with the empirical findings documented in Section 3. Two separate analyses are conducted.

First, a group comparison (Analysis 1) tests whether the moments differ between *Japan-dominant* and *Non-dominant* sectors. This is estimated using the following specification:

$$M_{KOR,k} = \alpha_0 + \alpha_1 \times \left(Dominant_k\right) + \varepsilon_{KOR,k}, \tag{A9}$$

where $M_{KOR,k}$ denotes one of the three untargeted moments ($\hat{Y}_{KOR,k}$, $\hat{p}_{KOR\,j,k}$, or $\hat{q}_{KOR\,j,k}$)[53] and $Dominant_k$ is an indicator variable for *Japan-dominant* sectors, defined as sectors for which Japan was the leading supplier to Korea during the pre-event period (2015–2018).

Second, a correlation analysis (Analysis 2) tests whether the moments are systematically correlated with Japan's pre-event share in Korea. This is estimated using the following specification:

$$M_{KOR,k} = \beta_0 + \beta_1 \times Share^{Pre}_{JPN\,KOR,k} + \varepsilon_{KOR,k}, \tag{A10}$$

where $Share^{Pre}_{JPN\,KOR,k}$ is Japan's pre-event import share in Korea for sector k.[54]

The validation consists of examining the sign and statistical significance of the coefficients α_1 and β_1. The full results for the scenario with scale economies (Panel A) and the scenario without scale economies (Panel B) are presented in Table A20. Panel A shows that the model's predictions for the untargeted moments are fully consistent with the empirical findings. First, the results from Analysis 1 show that the estimate of α_1 is positive and statistically significant for both $\hat{Y}_{KOR,k}$ (Column (1)) and $\hat{q}_{KOR\,j,k}$ (Column (3)). This indicates that *Japan-dominant* sectors experience significantly higher growth than *Non-dominant* sectors, both in total sales and in export quantities. Similarly, the results from Analysis 2 confirm this pattern: Japan's pre-event share is positively and significantly correlated with growth in total sales and export quantities. Second, and more

[53] The unit of observation is the sector k for $\hat{Y}_{KOR,k}$ and $\hat{p}_{KOR\,j,k}$, and the sector-destination pair (j,k) for $\hat{q}_{KOR\,j,k}$. Although indexed by destination, $\hat{p}_{KOR\,j,k}$ does not vary across destinations: with $\hat{A}_{KOR,k} = 1$ and $\hat{\tau}_{KOR\,j,k} = 1$ as in the previous subsection, it reduces to $\hat{p}_{KOR\,j,k} = \hat{w}_{KOR} / \hat{L}^{\gamma_k}_{KOR,k}$, which does not depend on j.

[54] The share is defined as $Share^{Pre}_{JPN\,KOR,k} = \frac{X^{Pre}_{JPN\,KOR,k}}{\sum_{i \neq KOR} X^{Pre}_{i\,KOR,k}}$, where trade values are summed over the pre-event period (2015–2018).

importantly, the coefficients for the change in export prices (Column (2)) are negative and significant in both Analysis 1 and Analysis 2. This indicates that the predicted values for $\hat{p}_{KOR\,j,k}$ are significantly lower for *Japan-dominant* sectors than for *Non-dominant* sectors, and that they are negatively and significantly correlated with Japan's pre-event share.

By contrast, Panel B of Table A20 shows that the scenario without scale economies fails this validation test. This scenario still generates a relative increase in total sales in *Japan-dominant* sectors and a positive correlation with Japan's pre-event share (Column (1)). However, it cannot generate the decrease in export prices and the simultaneous increase in export quantities. As expected, when $\gamma_k = 0$, export prices are identical across all sectors, resulting in no variation to test (Column (2)).[55] In terms of export quantities (Column (3)), the coefficients are positive and statistically significant. This, however, does not indicate export growth: the group comparison captures only the relative difference between the two groups. Given the overall decrease in export quantities shown in Panel B of Table 6, the positive coefficient merely reflects a smaller decrease in *Japan-dominant* sectors. This failure to generate an increase in export quantities is inconsistent with Fact 5. This comparison reconfirms that scale economies are the key mechanism behind the full set of empirical findings in Section 3.

[55]With $\gamma_k = 0$, the expression for the export price change derived in the previous subsection reduces to $\hat{p}_{KOR\,j,k} = \hat{w}_{KOR}$, which is identical across all sectors.

Table A20: Statistical Tests of Untargeted Moments

	(1) $\hat{Y}_{KOR,k}$	(2) $\hat{p}_{KOR\,j,k}$	(3) $\hat{q}_{KOR\,j,k}$
A. Scale Economies ($\gamma_k > 0$)			
Analysis 1			
$Dominant_k$	0.232***	−0.026***	0.258***
	(0.011)	(0.001)	(0.033)
Analysis 2			
$Share^{Pre}_{JPN\,KOR,k}$	0.583***	−0.061***	0.647***
	(0.019)	(0.002)	(0.086)
B. No Scale Economies ($\gamma_k = 0$)			
Analysis 1			
$Dominant_k$	0.164***	-	0.015***
	(0.015)		(0.003)
Analysis 2			
$Share^{Pre}_{JPN\,KOR,k}$	0.465***	-	0.041***
	(0.027)		(0.009)
No. of Dominant Sectors	611	611	611
No. of Sectors	4,667	4,667	4,667
Observations	4,667	4,667	107,341

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Table A20 reports the detailed OLS regression results for the model validation exercise in Section 6.3. Each entry is from a separate regression where the dependent variable is one of the model's predicted untargeted moments: the change in total sales ($\hat{Y}_{KOR,k}$) in Column (1), the change in export prices ($\hat{p}_{KOR\,j,k}$) in Column (2), and the change in export quantities ($\hat{q}_{KOR\,j,k}$) in Column (3). Panel A presents the results from the scenario with scale economies ($\gamma_k > 0$), using the same calibration as the baseline in Table 7. Panel B presents the results from the scenario without scale economies ($\gamma_k = 0$). Analysis 1 reports the coefficient (α_1) on $Dominant_k$ in equation (A9). Analysis 2 reports the coefficient (β_1) on $Share^{Pre}_{JPN\,KOR,k}$ in equation (A10). The unit of observation is the sector (k) in Columns (1) and (2) and the sector-destination pair (j,k) in Column (3); standard errors in Column (3) are clustered at the sector level. In Panel B, the dash in Column (2) indicates that no regression is estimated, as export prices are identical across all sectors when $\gamma_k = 0$.

G Appendix to Section 6.4

This appendix details the derivation of the welfare change equation used in the quantitative analysis. The change in welfare for country j ($\hat{U}_j$) is defined as the change in its real income. This is the change in the nominal income ($\hat{w}_j \hat{L}_j$) divided by the change in the aggregate price index ($\hat{P}_j$):

$$\hat{U}_j = \frac{\hat{w}_j \hat{L}_j}{\hat{P}_j}. \tag{A11}$$

The change in the aggregate price index ($\hat{P}_j$) is a Cobb-Douglas combination of the change in the sectoral price index ($\hat{P}_{j,k}$):

$$\hat{P}_j = \prod_k \hat{P}_{j,k}^{\beta_{j,k}}.$$

The change in the sectoral price index ($\hat{P}_{j,k}$) is a CES aggregate of the price changes of individual varieties from each origin i ($\hat{p}_{ij,k}$), weighted by their initial expenditure shares ($\lambda_{ij,k}$):

$$\hat{P}_{j,k} = \left(\sum_i \lambda_{ij,k} \hat{p}_{ij,k}^{-\theta_k} \right)^{-\frac{1}{\theta_k}}.$$

The price change of an individual variety ($\hat{p}_{ij,k}$) is derived from the pricing condition $p_{ij,k} = c_{ij,k}$ in equation (2), expressed as a function of the changes in trade costs ($\hat{\tau}_{ij,k}$), wages ($\hat{w}_i$), exogenous productivity ($\hat{A}_{i,k}$), and sectoral labor allocation ($\hat{L}_{i,k}$):

$$\hat{p}_{ij,k} = \frac{\hat{\tau}_{ij,k} \hat{w}_i}{\hat{A}_{i,k} \hat{L}_{i,k}^{\gamma_k}}.$$

Substituting each of these components into the welfare definition in equation (A11) yields the full expression used for the quantitative analysis:

$$\hat{U}_j = \frac{\hat{w}_j \hat{L}_j}{\prod_k \left[\left(\sum_i \lambda_{ij,k} \left(\frac{\hat{\tau}_{ij,k} \hat{w}_i}{\hat{A}_{i,k} \hat{L}_{i,k}^{\gamma_k}} \right)^{-\theta_k} \right)^{-\frac{1}{\theta_k}} \right]^{\beta_{j,k}}}.$$